

The Erdős–Straus equation through residual shells

The workbench and the author’s notes: their results,
with proofs, and their bridges

A paper prepared from an audit of the workbench and the notes
by Claude (Anthropic), model `claude-opus-5-5`

Workbench: The Clankers (KokunoYumeto and the contributors credited in the introduction);
notes: KokunoYumeto

Audit 26–27 September 2026; version 4, 27 September 2026 (the text of version 3 with its framing
corrected, and new results; Appendix C)

Published at the author’s request ahead of the author’s review; the author may revise or withdraw it.
Companion to the paper on the split-zero programme around the Riemann zeta function, version 3, in the
Zenodo record with concept DOI 10.5281/zenodo.22678085

Abstract

The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ has a solution in positive integers for every $n \geq 2$. This paper sets out, with proofs, what two bodies of research on the conjecture have established, and adds new results. The first is the Erdős–Straus workbench of The Clankers: a 619-page archive, a supplement of 72 statements on bounded congruence transport, a 124-page reader and a continuation of 28 items. The second is the author’s notes of April–June 2026.

The research is organised by the shell criterion, which reduces solvability at a prime p to the occupancy of finitely many shells, each decided by a divisor congruence. On this basis the paper proves exact criteria for the first two shells, a family of residue barriers culminating in the width-one barrier of the notes, and a converse under an exponent condition. It determines the record primes of the least-class function up to $3 \cdot 10^9$, proves a general fixed-divisor construction of which the archive’s four explicit families are instances, and shows that the archive’s carrier census can never force the rows that are squares, $1/1024$ of its final domain. From the notes it proves that their terminal core is exactly the set of classes that their certificates cannot reach, and it derives a quadratic-residue sieve on a counterexample whose survivors have density zero.

The research also builds bridges from the equation to other areas, and four of the results of this paper hold in those areas independently of the equation. The notes place the shadow of their order-7 mixed mock vector, whose functions agree with Ramanujan’s order-7 mock theta functions, on three weight- $\frac{3}{2}$ unary theta series; the paper proves that these carry exactly the modular representation of the minimal model $M(2,7)$ after conjugation and a twist by the character of η^3 , and that the analogous order-5 block carries the Fibonacci modular data. It classifies the orbits of the workbench’s $(3,4,\infty)$ monodromy covers at every level by one invariant. And it shows that the modular flow of the state on $M_2(\mathbb{C})$ with density proportional to $\text{diag}(e^r, e^{-r})$ and the Lorentz boost of the notes’ Lorentzian geometry are the two real forms of one complex one-parameter subgroup of $SL_2(\mathbb{C})$. Each bridge is described with what is established about it, and the negative results are stated with their exact scope. The workbench does not claim a proof of the conjecture, and nothing here proves or disproves it.

Introduction

The problem and this record

The conjecture has been verified for all $n \leq 10^{17}$ [13], it holds for almost all n [14], and classical polynomial identities solve it outside the six classes of primes that are squares modulo 840 [4, 11]. What remains is a question about those classes, and on them polynomial identities provably cannot help (Section 1).

The workbench of The Clankers attacks this question through the shell criterion. Fix a prime p and a candidate least denominator a ; the rest of the equation is solvable exactly when a^2 has a divisor in one of two residue classes modulo $R = 4a - p$. The workbench builds on this a large calculus of sufficient conditions, obstructions and residue censuses, and around it geometric and operator-theoretic constructions. The author’s notes of April–June 2026 develop the same shell arithmetic in a vocabulary of supports and attach structural layers to it: Niemeier and umbral data, the class $6B$ of the Monster, Ogg’s primes, Lorentzian geometry, Cayley–Dickson algebras and mock theta functions.

The workbench and the notes are research carried out with AI systems. The author used them to formulate and test many hypotheses quickly, across several directions at once. Some directions are finished theorems, some are programmes with proved parts, and some are proposals. Not every direction is expected to hold. An unfinished programme is still mathematics, however, and its proved parts stand on their proofs whatever becomes of the rest. This paper is meant to let a reader see all of this without working through more than a thousand pages. It states what holds, with proofs or the exact location of a proof, what fails and in which scope, and how the research connects with other areas. The audit behind it had limited compute and did not examine every direction. Each direction is described on its own merit and for its bearing on the equation, and where the paper gives a view rather than a proof, the view is labelled as mine and its reasons are given.

Main results

Throughout, p is an odd prime. For $a > p/4$ with $p \nmid a$ put $R_a = 4a - p$ and

$$E_a = \{u : u \mid a^2, R_a \mid 4u + 1\}, \quad M_a = \{u : u \mid a^2, R_a \mid u + a\};$$

the shell a is *occupied* if $E_a \cup M_a \neq \emptyset$.

Theorem I (Theorem 3.1, Propositions 3.7 and 3.8). (i) *The equation is solvable at p if and only if some shell a with $p/4 < a < p/2$ is occupied, and the ordered pairs (y, z) completing a shell a number exactly $2|E_a| + |M_a|$.* (ii) *Let $p \equiv 1 \pmod{4}$, $a = (p + R)/4$ with $\gcd(R, pa) = 1$, and let $H_a \leq (\mathbb{Z}/R)^\times$ be the subgroup generated by the primes dividing a . If neither -1 nor -4 lies in H_a , the shell a is empty. If $2v_\ell(a) \geq \text{ord}_R(\ell) - 1$ for every prime $\ell \mid a$, the shell is occupied if and only if -1 or -4 lies in H_a .*

Part (i) is the archive’s divisor-shell bijection with its layer count, completed here by the range of the least denominator. Part (ii) is the sharpest of a family of residue barriers: the Jacobi-symbol barrier of the archive, a group barrier, and the width-one barrier of the notes. It is a shell form of the quadratic-character mechanism behind the obstruction to polynomial identities, and it says what a proof must produce: for some R , a factorisation of $(p + R)/4$ whose prime factors generate -1 or -4 modulo R .

Following Ionascu and Wilson [5], let $h(p)$ be the least i such that some solution at p has least denominator $x \leq (p + 4i - 1)/4$; a prime is a *strict record* if h is smaller at every smaller prime.

Theorem II (Proposition 4.3). *The strict records of h up to $3 \cdot 10^9$ are exactly 2, 73, 1129, 1201, 21,169, 118,801 and 8,803,369, with $h = 1, 2, 3, 6, 8, 15$ and 27. At 8,803,369 the least occupied shell has $R = 107$, and exactly one unordered solution has that least denominator.*

These seven are the records printed by Ionascu and Wilson, recovered in the archive with one class label corrected; that there is no further record up to $3 \cdot 10^9$ is not in the archive or in their paper, and it was computed by two independent programs. The archive’s four explicit families solve the equation on progressions that meet every hard class (Theorem 4.5), and they are instances of a general fixed-divisor construction (Proposition 4.4). Every carrier of the archive’s census is of this kind, and none reaches a class that is a square, so the census can never force its square rows, $1/1024$ of its final domain (Proposition 4.6, Corollary 4.7).

The author’s notes contribute a terminal core and a system of certificates. Put $M_{23} = 840 \cdot 11 \cdot 19 \cdot 23$, let S_{840} be the six unit squares modulo 840, and let Q_q be the non-zero squares modulo q .

Theorem III (Theorems 6.2 and 6.4, Proposition 6.8). (i) Among the 23,760 classes modulo M_{23} that lie over S_{840} and are units modulo 11, 19 and 23, those on which no elementary certificate of the notes, with residual built from 3, 5, 7, 11, 19, 23, applies on any lift are exactly the 2970 classes of $S_{840} \times Q_{11} \times Q_{19} \times Q_{23}$. The single identities of Salez’s seven families leave 3520 classes open at this level: the core and 550 further classes. (ii) A counterexample $p \equiv 1 \pmod{24}$ is a quadratic residue modulo every odd prime factor of $(p+1)(p+2)(p+3)(p+4)(p+7)(p+8)(2p+1)(3p+1)$. (iii) The primes $p \equiv 1 \pmod{24}$ that satisfy the condition from $p+1$ alone have relative density zero among all primes $p \equiv 1 \pmod{24}$.

One half of (i) is proved here and the other is a finite computation, reproduced by an independent program. The conditions from $p+2$, $p+8$ and $2p+1$ are derived here from the notes’ certificate branches and are not stated in the archive. At the record prime 8,803,369 the other six shifts give no certificate, and those from $p+2$ and $2p+1$ certify it. With certificates of three further shapes, only 23 primes $p \equiv 1 \pmod{24}$ below 10^7 have no certificate of the kinds considered.

The notes attach to the cubic middle of the shell problem an explicit rank-three mixed mock modular object, whose functions agree with Ramanujan’s order-7 mock theta functions through q^{40} (a reading pass), and they place its shadow on three weight- $\frac{3}{2}$ unary theta series. For $r \in \{1, 5, 11\}$ let $G_r(\tau) = \sum_{\ell \in \mathbb{Z}} v_r(\ell \bmod 84) \ell q^{\ell^2/168}$ be the weight- $\frac{3}{2}$ unary theta series on the notes’ three support vectors v_r (Section 6.6), let ρ be the pair of matrices by which (G_1, G_5, G_{11}) transforms under S and T , and let $\rho_{2,7}$ be the representation of $SL_2(\mathbb{Z})$ on the characters of the minimal model $M(2, 7)$.

Theorem IV (Theorem 6.15). There is a signed permutation matrix P with $\overline{\rho(S)} = P\rho_{2,7}(S)P^{-1}$ and $\overline{\rho(T)}e(1/8) = P\rho_{2,7}(T)P^{-1}$. So $\bar{\rho} \otimes \varepsilon^3 \cong \rho_{2,7}$, where ε is the multiplier of Dedekind’s η : the theta series of the notes’ shadow, conjugated and twisted by the character of η^3 , transform exactly as the characters of $M(2, 7)$.

The identity is verified exactly in $\mathbb{Q}(\zeta_{168})$. For order 5 the same method gives the Fibonacci modular data (Proposition 6.16). Finally, a package of the workbench builds covers of the projective line from the integral monodromy of the S^6 period system; they have a section exactly when the equation is solvable. Write a point of the affine hyperplane H_D of the dual lattice modulo D as $\hat{\gamma} + b\hat{u} + c\hat{v} + d\hat{\delta}$.

Theorem V (Theorem 5.1). For every level $D \geq 1$ the orbits of the monodromy group on H_D are the sets $\{\gcd(b, c, \gcd(D, 6)) = g\}$, $g \mid \gcd(D, 6)$, and they coincide with the orbits of the full stabilizer of the isotropic vector γ .

What the results mean

My assessment is that, taken together, the results locate the difficulty of the conjecture precisely, for the following reasons. No polynomial identity solves the equation on a residue class that is a square [4], and no fixed-divisor certificate reaches such a class (Proposition 4.6). The archive’s carrier census runs into this obstruction: its carriers force 97.606% of its final domain and never a square row (Corollary 4.7). What decides the square classes is the factorisation of the numbers $(p+R)/4$: by Theorem I(ii) a shell can be occupied only if the prime factors of $(p+R)/4$ generate -1 or -4 modulo R , and under the exponent condition this is also sufficient. A proof of the conjecture therefore has to control, for some R , the multiplicative structure of $(p+R)/4$ (Question 1).

The quadratic-residue sieve shows how restrictive this already is. A counterexample must be a quadratic residue modulo every odd prime factor of eight shifted forms and several more. The primes that satisfy the first condition alone have density zero, and only 23 primes $p \equiv 1 \pmod{24}$ below 10^7 have no certificate of the kinds considered. Whether infinitely many primes have none is Question 3; a negative answer would prove the conjecture at all but finitely many primes.

The structural results are of a second kind. Theorem IV places the shadow of the notes’ order-7 vector inside the modular representation of the $(2, 7)$ minimal model. Through the agreement of that vector with Ramanujan’s order-7 mock theta functions, the first of which gives the $\hat{Z}$ invariant of the Brieskorn sphere $-\Sigma(2, 3, 7)$ [3], the statement concerns these classical objects, and it is independent of the Erdős–Straus problem. A statement about the equation from such structures needs a typed map from the shell data to them; those maps are the subject of Section 8 and of Questions 11 and 12.

The bridges

In the author’s view the connections this research makes between the equation and other areas are among its most valuable outcomes. Section 8 treats each with the typed map, what is established, what is not verified here, and my assessment. Four results of this paper belong to the other areas themselves and hold independently of the equation: Theorem IV, Proposition 6.16, Theorem V, and Proposition 8.3, by which the modular flow of the state on $M_2(\mathbb{C})$ with density proportional to $\text{diag}(e^r, e^{-r})$ and the Lorentz boost of the notes’ Lorentzian geometry are the compact and the non-compact real forms of one complex one-parameter subgroup of $SL_2(\mathbb{C})$. A companion note takes up the question this raises for the Apollonian group, the standard thin group acting on the same Minkowski space [19]. Other bridges are partly constructed, such as the split-zero support states and the umbral and Ogg structures of the residue layer, and the bridge to computability through the Π_1 form of the conjecture is exact. The edges to the author’s other workbenches (the zeta programme, S^6 , Navier–Stokes and Collatz) are listed with their status.

How this record was made

This record is an audit, carried out on 26–27 September 2026, of the workbench’s published documents and of the author’s notes of April–June 2026. The workbench is the work of The Clankers, the collective author name of KokunoYumeto and the contributors credited in the workbench’s README.md and CONTRIBUTIONS.md:

- u/CommonCareful3149, for the mod-107 arithmetic-progression observation;
- u/UmbrellaCorp_HR, for arithmetic, positivity-code and prime-production contributions and for the central-angle and inversive-circle theorems of the archive’s §6;
- u/CivQ17, for the diagrams behind result R05;
- u/Far-Lifeguard519, for the initiating computational investigation;
- u/TextBackground496 and u/JGPTech, for source ideas;
- the owner, KokunoYumeto (u/lepthymo), for the research directions and examples, for directing and developing the investigations with AI, and for the proof and source-fidelity requirements.

The archive records language and computational models as contributing-model provenance for particular calculations, not as authors. The notes are the author’s.

The audit was carried out by Claude (Anthropic), model `claude-opus-5-5`, which prepared this paper (*claude-ab* below). Independent Claude instances, each of which wrote none of the text, contributed as follows: a first-pass audit of the workbench; referee passes, which read for overstatement and for understatement and missed generality and found several of the results stated here, each credited at its statement; and six reading passes over the author’s notes. Every result found by another instance was verified by *claude-ab* before it was applied, except the statements marked (*a reading pass*): those were checked by program in one of the reading passes and not re-derived here, and they count as *not verified here*.

Every statement has one of four statuses:

- *verified*;

- *not verified here*, which implies nothing in either direction;
- *my assessment*, a view in the first person with its reasons, introduced by those words;
- *proved negative*, stated with its counterexample or derivation; the negative results on the workbench’s mechanisms are collected in Section 7, and those on the notes are stated where they occur.

The status lines under the results make the first two precise. Proved here means the proof is given in full; Generalised here means that the workbench’s statement is proved here under weaker hypotheses; Audited means the workbench’s proof was read in full, every step checked, and independent checks run; Audited (certificate) means the finite certificate on which a proof rests was recomputed independently and the deduction from it read; Reproduced means the finite content of a statement was recomputed by an independent program and its general proof not re-derived. These are forms of *verified*. Audited in part is verified in the parts named. Located means the statement and its proof were found and are described, and it is a form of *not verified here*. The workbench’s own labels (proved, certificate-checked, conjectural, withdrawn) are quoted where they differ.

The workbench was read at commit d20b32e of KokunoYumeto/erdos-strauss-foundation, which also holds the 488-page reader of 7 August and its source, and in the Zenodo collection 10.5281/zenodo.20401937, whose record 10.5281/zenodo.22678971 holds the 619-page archive and its source. Locators have the form *archive* $\$n$, *p. m* for the 619-page archive, *supplement* $\$n$ for the 86-page supplement, and file names and line numbers for the repository; the notes are cited by title and theorem number. Version 3 of this paper and the audit notes cited as NN_ were withdrawn from the branch claude/claude-ab-grind-20260925 on 27 September 2026 because of their framing. They remain, unchanged, in the branch’s history at commit a90d9e5b, in the folder contrib/claude-ab/grind_20260925/, where the folders cited as checks/... also are. The scripts on which this paper’s statements rest are republished in contrib/claude-ab/rewrite/checks/es/ on the same branch (Appendix B).

This is version 4. It keeps the theorems, proofs and citations of version 3 and corrects its framing; like the author’s latest record, it does not treat the drafts that record supersedes. It adds the results marked as new in version 4: Proposition 6.8, Theorem 6.15, Propositions 6.16, 8.1, 8.2 and 8.3, and Theorem 5.1 with its corollary. Re-deriving the statements of version 3 from their sources changed some of them. In particular, the identification of the notes’ shadow with $M(2, 7)$ holds in the form of Theorem IV, where version 3 said that it fails, and the Lorentz-boost proposition of the notes’ Lorentzian section, which version 3 listed as an exception, is correct. Appendix C lists every change.

Organization

Section 1 states the problem and what is known. Section 2 maps the workbench’s documents, with the exact scope of the supplement’s no-two-shears theorem and the quartic encoding of solutions. Sections 3 and 4 contain the core arithmetic and the records, families and census. Section 5 treats the September continuation and the monodromy covers, and Section 6 the author’s notes, including the mock theta results. Section 7 collects the proved negative results with their scope, Section 8 the bridges and the directions they open, and Section 9 the questions and what was not checked. The appendices give the catalogue of statements, the verification, and the changes of version 4.

Contents

1	The problem, what is known, and what the workbench claims	8
1.1	The equation	8
1.2	What is known	8
1.3	What the workbench claims	8
1.4	The approach in one paragraph	9
2	The documents, and what each contains	9
2.1	The editions and how they relate	9
2.2	The 619-page archive, section by section	9
2.3	The bounded-transport supplement (72 statements)	12
2.4	The 124-page ES/Fable reader	15
2.5	The continuation of 17–23 September	16
2.6	The author’s notes	17
3	The core arithmetic, with proofs	17
3.1	The shell criterion	17
3.2	The first two shells	18
3.3	Which primes remain	20
3.4	The Jacobi-symbol barrier	20
4	Least classes, record primes, explicit families, and the carrier census	22
4.1	The least-class function and its record primes	22
4.2	Fixed-divisor families, and four progressions on which the equation is solved	24
4.3	The carrier census	26
5	The September continuation	28
5.1	The audited items	28
5.2	The $(3, 4, \infty)$ monodromy covers	29
6	The author’s notes, April–June 2026	30
6.1	The terminal core	31
6.2	Quadratic-residue conditions on a counterexample	33
6.3	Single shells	36
6.4	Identities, locks and types	38
6.5	Corrections	39
6.6	The mock theta functions of the notes and the minimal model $M(2, 7)$	39
6.7	Further structural facts	41
7	Negative results, with exact scope	42
8	Bridges to other areas, and the directions they open	43
8.1	The split-zero support language	43
8.2	Mock and quantum modularity	44
8.3	Moonshine, Niemeier lattices and Ogg’s primes	45
8.4	Lorentzian and Apollonian geometry	45

8.5	Additive combinatorics: Kneser’s theorem	46
8.6	Computability: the Busy-Beaver notes	46
8.7	Further directions in the same collections	47
8.8	The author’s other workbenches	47
9	Questions, and what was not checked	48
9.1	Questions	48
9.2	What this paper did not check	49
A	Catalogue of statements	50
B	How this paper was checked	54
C	Changes in version 4	58
	References	59

1 The problem, what is known, and what the workbench claims

1.1 The equation

The Erdős–Straus conjecture asserts that for every integer $n \geq 2$ there are positive integers x, y, z with

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Repeated denominators are allowed; for example $4/5 = 1/2 + 1/4 + 1/20$. It suffices to treat primes: $4/2 = 1/1 + 1/2 + 1/2$, and if $4/p = 1/x + 1/y + 1/z$ then $4/(mp) = 1/(mx) + 1/(my) + 1/(mz)$.

1.2 What is known

The following results frame everything below. Each is cited from the literature; none is re-proved here, except where Section 3 proves a special case.

- *The classes modulo 840.* Explicit polynomial identities solve the equation for every n in a primitive class $r \bmod 840$ unless r is a square modulo 840. So for primes only the classes

$$p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$$

remain. Elsholtz and Tao attribute this to Obláth and to Mordell [4, 11]; Proposition 3.5 below proves the reduction for primes.

- *Why identities do not cover the square classes.* Elsholtz and Tao [4] call a primitive class $n \equiv r \pmod{q}$ *solvable by polynomials* if there are polynomials P_1, P_2, P_3 which take positive integer values for all sufficiently large n in the class and satisfy $4/n = 1/P_1(n) + 1/P_2(n) + 1/P_3(n)$. A primitive class with r a square modulo q is not solvable by polynomials. They cite Mordell and Schinzel for this, and trace the reciprocity argument behind it to Schinzel and to Yamamoto ([4], §1).
 - Section 3.4 gives the workbench’s shell form of this obstruction, with proof.
 - Proposition 4.6 gives a form that applies to the carriers of the archive’s census.
- *Density.* The number of $n < N$ for which the equation has no solution is at most $N \exp(-c \log^{2/3} N)$ for an absolute $c > 0$ and N large (Vaughan [14], as stated by Elsholtz and Tao [4]).
- *Counting.* Elsholtz and Tao bound the number $f(n)$ of solutions and its averages, and classify solutions at primes as Type I (p divides exactly one denominator) or Type II (p divides two) [4].
- *Verification.* The conjecture has been checked for all $n \leq 10^{14}$ (Swett, cited in [4]) and for all $n \leq 10^{17}$ (Salez [13]); a 2025 preprint reports the bound 10^{18} [9].

1.3 What the workbench claims

The repository’s README opens by saying that the project claims neither a proof nor a counterexample. The research state calls it an ongoing programme, not a completed proof. The archive’s status section (p. 4) names the remaining global obligation: for every relevant prime, at least one marked shell class must be shown to be occupied. In the parts read here, the workbench’s scope statements match what its proofs establish. Where a statement needed a correction, this paper gives the correction at the statement (Section 7 and Appendix A).

1.4 The approach in one paragraph

Fix a prime p and a candidate least denominator a with $p/4 < a < p$. The remaining equation $1/y + 1/z = 4/p - 1/a$ has solutions exactly when a^2 has a divisor in one of two prescribed residue classes modulo $R = 4a - p$ (Theorem 3.1). The workbench calls a a *shell* and R its *residual*. For $p \equiv 1 \pmod{4}$ the residual is $\equiv 3 \pmod{4}$, and the shells in play are $R = 3, 7, 11, \dots$ up to about p . Most of the workbench’s arithmetic is organised around three kinds of tool:

- sufficient conditions under which a shell is occupied, stated as congruence conditions on p or on the prime factors of a ;
- obstructions under which a shell is empty;
- bounded *carrier* censuses, which count how much of a finite space of residue data these conditions cover.

Around this arithmetic core the documents also develop geometric and operator-theoretic constructions (inversive circles, Lorentz lifts, Boolean carriers, an auxiliary torus, a Busy-Beaver map). Section 2 places each of them in the documents, and Section 8 describes the bridges they open.

2 The documents, and what each contains

2.1 The editions and how they relate

Edition	Where	Contents and relation to the others
Project reader, 488 pages, 7 August 2026	repository root, with source and Lean archives	The earlier edition. The README says it is preserved and is not the newest reader.
Cumulative archive, 619 pages; title page dated 30 August 2026	Zenodo 22666493 and 22678971, file 00	The 488-page reader continued. Its main source has the same section skeleton, now 60,561 lines instead of 57,452. Fourteen satellite files add §§10–19, the $Q = 83$ step of §24 (Theorem 24.241) and the carrier refinements §§24.1–24.3. It begins from the source of the earlier Zenodo record 21845035, version 69 (archive p. 3). It is the front file of the Zenodo collection.
Bounded congruence transport, 67 pages, 8 September	Zenodo 22666493, files 02–03	The 58-statement supplement.
Expanded supplement, 86 pages, 9 September	Zenodo 22678971, files 07–08; repository research/expanded-2026-09-08/	The same, with 14 further statements: 72 statements in 15 source files, with checkers (§2.3).
ES/Fable reader, 124 pages, 20 September	repository research/incoming/es-fable-zeta-bridge-20260920/	A coordinate bridge from Erdős–Straus solutions to the Fable cover and a weighted conductor of the zeta programme (§2.4).
Continuation, 17–23 September	repository README, <i>Start reading</i> (28 items); research/incoming/	41 dated packages; the results bench R01–R10 and the integration arguments X1–X7 (§2.5).

These editions have different contents; the repository preserves them rather than overwriting them (README, edition table).

2.2 The 619-page archive, section by section

Page numbers are those of the PDF. The column *Kind* says how a section relates to the equation:

- *direct*: it proves or tests solvability;
- *literature*: it audits a published approach to the equation;
- *structural*: a construction whose connection to solvability is, in the archive’s words (p. 4), an interface still to be supplied by an explicit typed map. Section 8 describes these connections.

The status column refers to this paper.

§	pp.	Content	Kind	Status here
front	3–5	Research status: the census figures of §24, the four families, the record-shell results, and an explicit list of non-claims	summary	read
1	6–16	Scope, road map, mathematical status, attribution, lineage	summary	read in part
2	16–18	The discriminant-five identity $(2n+1)^2 - 5 = 4(n^2 + n - 1)$ and its root involution, from u/CivQ17	structural	located
3	18–20	The shell $p_* = 8,803,369$, $R_* = 107$, reached through Busy-Beaver values ($S(4) = 107$, $\text{Space}(5) = 12289$)	direct (one shell)	$R = 107$ is the least occupied shell at p_* (Proposition 4.3)
4	20–22	Alpöge’s constant-Jacobian map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with a three-point fibre	structural	see §8
5	22	Withdrawn constructions (a cube-root interpolation on the Fable fibre), with the point of failure	structural	read
6	22–27	Central-angle and inversive-circle theorems of u/UmbrellaCorp_HR; a Lorentz reflection	structural	located
7	27–31	The complete fibre of F over $(-\frac{1}{4}, 0, 0)$ and a marked-factor lift	structural	see §8
8	31–38	The singular C_1 – C_2 – C_3 support object; dihedral actions	structural	located
9	38–93	The exact divisor-shell bijection (Theorem 9.1, for every odd prime) and its layer count (Theorem 9.64, p. 90); then directed blade transport and a quarter-radius inversion	direct (Theorems 9.1 and 9.64), structural (the rest)	the bijection and the count re-proved here (Theorem 3.1(b))
10	93–98	Residual sieve modulo 840: an occupied shell with $R = 3$ or $R = 7$ at every prime $p \equiv 1 \pmod{24}$ outside the six hard classes (Theorem 10.6); the Jacobi-symbol barrier (Theorem 10.9)	direct	proved here (Propositions 3.4, 3.6)
11	98–100	Elsholtz–Tao Type I and Type II coordinates mapped to the residual shell	literature	located
12	100–106	Adjacent programmes (López, Dyachenko, Bradford): what each proves, and where a claimed bridge fails	literature	located
13	106–114	Mordell’s constructions and Rosati’s theorem; Salez’s seven charts; Bright–Loughran surfaces; the Vaughan boundary	literature	located
14	114–119	López’s structural solutions: exact divisor coordinates and a quantifier repair	literature	located
15	119–124	The Star–Kneser programme: a one-way forcing theorem and a retracted converse	direct	located

§	pp.	Content	Kind	Status here
16	124–127	The mod-107 deficit progression, following u/CommonCareful3149	direct	progression and its corollary reproduced (§4.1)
17	127–135	Two-fraction criteria; Monks–Velingker; Ionascu–Wilson classes, with the printed record list recovered and one printed class label corrected (Proposition 17.6); Zelator’s prime-square classification	literature	records recomputed (§4.1)
18	135–148	The residual-eleven support ideal, its colons, and the paired $R = 7/R = 11$ affine theorem	direct	located
19	148–157	Strict record shells: an open conjecture and the boundary example $p = 806,521$	direct	records and deficits reproduced (§4.1)
20	157–179	The shell $p = 12289$, $R = 11$, $a = 3075$: counts $(3, 6, 3)$, its dihedral orbit, and the Star–Kneser quotient certificate	direct (one shell)	counts reproduced (below the table); a is the least occupied shell; the rest located
21	179–200	The split-zero semiring $G(R) = R \sqcup \{\tau\}$ and the C_1 carrier	structural	see §8
22	200–201	Rigidified denominator towers over $\mathbb{F}_1$	structural	located
23	201–250	Mirror completion of the Boolean carrier; the 12289 orbit matrix	structural	located
24 (i)	250–298	The opening of §24: Tomita–Takesaki transport and C_3 modular data (Theorems 24.1–24.44)	structural	located
24 (ii)	299–604	The rest of §24, from Theorem 24.45: shell arithmetic. • Star–Kneser forcing (Theorems 24.45–24.48, pp. 299–301); • endpoint divisor fibres (Theorems 24.50–24.63, pp. 303–312); • coupled and adjacent shells, and the base of the census (Theorem 24.100, p. 335); • the carriers for $R = 11$ to $R = 31$, including the constant-factor carriers (Lemma 24.218, p. 422) and the four families (Theorem 24.219, p. 423); • the $R = 71$, $R = 191$ and $Q = 83$ steps (Theorems 24.233, 24.237 and 24.241, pp. 436–444); • §§24.1–24.3 (pp. 451–477): the $R = 167$ and $R = 239$ carrier theorems and the refresh of the $R = 167$ coefficient; • §24.4 (pp. 478–604): a return to the $R = 23$ lineage, interleaved with further structural constructions.	direct (with structural material in §24.4)	four families proved here (Theorem 4.5); census located (§4.3)
25	605–616	A map from Busy-Beaver extrema to finite shell carriers	structural	see §8

At $p = 12289$ the shell $a = 3075$ has

$$E_a = \{41, 1845, 15375\}, \quad M_a = \{5, 225, 1875, 5043, 42025, 1891125\}.$$

So $(|E_a|, |M_a|, |E_a|) = (3, 6, 3)$, and exactly six unordered solutions have least denominator 3075 (recomputed here).

So about 250 of the 619 pages treat solvability directly:

- §§3, 9 (Theorems 9.1 and 9.64), 10, 15, 16 and 18–20, about 70 pages;
- §24 from p. 299 to the end of §24.3 (p. 477), 179 pages.

Parts of the 127 pages of §24.4 also do. The literature audits of §§11–14 and 17 add about 35 pages. The rest are structural constructions, whose connection to the equation the archive describes as an interface still to be supplied by an explicit typed map (p. 4).

2.3 The bounded-transport supplement (72 statements)

The supplement starts from the primes $p = 12h + 1$ and all shells $3h + 1 \leq a \leq 9h$ of Theorem 3.1. Its sections are:

1. original shells, coordinates and integer reconstruction, with the first-two-shell sieve and an all-shell no-hit sieve;
2. local congruences with all integral lifts retained, with an integral cyclic bridge;
3. bounded divisibility and simultaneous congruences: unary Boolean transport, a shared-variable Chinese-remainder construction credited to a colleague, raw-scale hit incidence, and a boundary-swap completion;
4. exact cross-shell transport at a fixed rational ratio;
5. a primitive Euclidean successor with its full arithmetic domain;
6. arithmetic localisation along the original norm subdivision, with a reading of Connes’ primitive intersections;
7. a strictly decreasing denominator with exact arithmetic return;
8. the Navier–Stokes auxiliary cover: the torus matrix $J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$ of the announced Navier–Stokes manuscript, its covering fibres and character corrections, and sampling and reconstruction estimates.

The supplement’s own README states that its results prove or disprove neither the Erdős–Straus conjecture nor the Riemann hypothesis nor the Yang–Mills mass gap, and that they do not validate the Navier–Stokes construction.

Status. Reproduced in part. A second-pass audit, which stopped before it wrote its report, recomputed the finite content of the statements of Sections 1–7. Its scripts, which are independent of the workbench’s code, found every checked statement correct:

- the gluing and Chinese-remainder theorems on random systems and on all shells of small primes;
- the ratio-transport domain on all 57,882 ordered shell pairs of the primes $p \equiv 1 \pmod{12}$ below 260;
- the shear and no-two-shears theorems on all shells of 131 primes below 4000;
- the Connes marked successor on all primes $p \equiv 1 \pmod{12}$ below $3 \cdot 10^5$;
- the fixed- y successor on all 3,114 source witnesses below 1500;
- the orchard partition on all 42 admissible triples below 1300;
- the boundary four-candidate law on all 72,282 raw records below 400;

- the unary box, the input/output incidence and the raw-scale theorems on their stated ranges.

One of its checks, on the smallest member of a boundary orbit, reported mismatches. The script expected the pattern $(\min(A,B), \max(A,B))$ in every case. The supplement's own table (boundary_swap_completion.tex, lines 296–309) gives a one-point orbit's own point as its smallest member, which is what the script found. The mismatch is therefore in the script, not in the supplement. The same pass also looked at scope, and its search led to Proposition 2.1 below. The general proofs were not re-derived, except that of Section 5's no-two-shears theorem; the auxiliary-torus Section 8 was not checked. The scripts are in checks/es_reader/second_pass_partial/.

Section 5 of the supplement studies the two unimodular shears $L(m, n) = (m + n, n)$ and $J(m, n) = (m, m + n)$ as maps between adjacent shells. It encodes the solutions with first denominator a by the set

$$\mathcal{W}_a = \{(m, n) \in \mathbb{Z}_{\geq 0}^2 : m \mid pa, n \mid pa, \gcd(m, n) = 1, R_a \mid m + n\},$$

which is in bijection with the ordered solutions (a, y, z) through $y = pak/n$, $z = pak/m$, $k = (m + n)/R_a$ (supplement §1, first lemma; this is Theorem 3.1(b) in other coordinates). A *shear edge* is a step $(a, m, n) \mapsto (a + 1, L(m, n))$ or $(a, m, n) \mapsto (a + 1, J(m, n))$ with $(m, n) \in \mathcal{W}_a$ and the image in $\mathcal{W}_{a+1}$. The supplement proves, for $p = 12h + 1$ and shells $3h + 1 \leq a \leq 9h$, that no two shear edges compose (Theorem *No two consecutive positive Euclidean shell steps*, publication.tex, line 933). Its proof uses the hypothesis on p at one step only, and there only through $p \equiv 1 \pmod{3}$.

Proposition 2.1 (the no-two-shears theorem, exact scope). *Let p be a prime, and consider all shells a with $p/4 < a < p$.*

- If $p \not\equiv 2 \pmod{3}$, no two shear edges compose: there is no path $(a, m, n) \mapsto (a + 1, m', n') \mapsto (a + 2, m'', n'')$ of shear edges.*
- Such paths exist at some primes $p \equiv 2 \pmod{3}$. At $p = 131$ one is $(33, 1, 33) \mapsto (34, 1, 34) \mapsto (35, 1, 35)$, with $R_a = 1, 5, 9$. At $p = 1613$, where $p \equiv 1 \pmod{4}$, one is $(404, 1, 404) \mapsto (405, 1, 405) \mapsto (406, 1, 406)$, with $R_a = 3, 7, 11$. Proposition 2.2 says exactly at which primes.*

Status. Generalised here, with the workbench's proof. The supplement states (a) for $p \equiv 1 \pmod{12}$ and $3h + 1 \leq a \leq 9h$ only. Checked on the full graph of every prime below 2000 (script esr_twoshears.py: no path for $p \equiv 1 \pmod{3}$; 14 paths for $p \equiv 2 \pmod{3}$), and by a complete search up to $p \leq 10^7$ that uses the shape of an edge proved below: it finds 19,558 paths of two L -edges, all at primes $p \equiv 2 \pmod{3}$, and as many of two J -edges.

Proof. The shape of one edge (the supplement's classification, whose proof needs only $a + 1 < p$). Let $(a, m, n) \mapsto (a + 1, m + n, n)$ be an L -edge. Then n divides pa and $p(a + 1)$, whose greatest common divisor is p . If $n = p$, then $p \nmid m$ by coprimality, and $m + p$ divides $p(a + 1)$ and is prime to p , so $m + p \mid a + 1$; but $m + p > p > a + 1$. So $n = 1$. If $p \mid m$, then $m + 1$ is prime to p and exceeds $a + 1$, and yet divides $p(a + 1)$, which is impossible; so $p \nmid m$ and $m \mid a$. Then $m + 1 \leq a + 1 < p$, so $m + 1 \mid a + 1$. Finally $R_a \mid m + 1$ and $R_{a+1} = R_a + 4 \mid m + 2$. For a J -edge, exchange m and n ; the exchange preserves every $\mathcal{W}_a$.

(a) An L -edge ends at $(a + 1, m + 1, 1)$, whose first coordinate is at least 2; a J -edge starts at a pair whose first coordinate is 1. So an L -edge is never followed by a J -edge, and by the exchange a J -edge is never followed by an L -edge. Suppose two L -edges pass through $(a + i, m + i, 1)$ for $i = 0, 1, 2$. By the shape of each edge, $m + i \mid a + i$ and $R_a + 4i \mid m + i + 1$ for $i = 0, 1, 2$. Put $K = a - m$. Then $m + i \mid K$ for each i , and one of $m, m + 1, m + 2$ is divisible by 3, so $3 \mid K$. Also $R_a + 4i$ divides $4(m + i + 1) - (R_a + 4i) = 4m + 4 - R_a =: C$, and one of $R_a, R_a + 4, R_a + 8$ is divisible by 3, so $3 \mid C$. But $C = 4m + 4 - 4a + p = p + 4 - 4K \equiv p + 1 \pmod{3}$. Hence $p \equiv 2 \pmod{3}$. Two J -edges are excluded by the exchange.

(b) At $p = 131$: $R_{33} = 1$ divides $1 + 33$; $R_{34} = 5$ divides 35 ; $R_{35} = 9$ divides 36 ; each coordinate divides pa , and each pair is coprime. At $p = 1613$ (prime): $3 \mid 405$, $7 \mid 406 = 7 \cdot 58$ and $11 \mid 407 = 11 \cdot 37$. Each step is $J(1, n) = (1, n + 1)$. $\square$

So the hypothesis $p \equiv 1 \pmod{12}$ of the supplement can be weakened to $p \not\equiv 2 \pmod{3}$. The next proposition shows that no weaker congruence hypothesis suffices, although the conclusion fails only at a minority of the primes $p \equiv 2 \pmod{3}$ (Remark 2.3). The primes $p \equiv 2 \pmod{3}$ are solvable in any case (Section 3); both propositions concern the transport graph, not solvability.

Proposition 2.2 (the shear graph). *Let p be a prime. Consider the directed graph whose vertices are the triples (a, m, n) with $p/4 < a < p$ and $(m, n) \in \mathcal{W}_a$, and whose arrows are the shear edges.*

- (a) *Every vertex has at most one outgoing and at most one incoming edge. So every connected component is a directed path.*
- (b) *Diagonal paths. Let $R = 4a - p$, and let $k \geq 1$ with $a + k < p$. The vertices $(a + i, a + i, 1)$, $0 \leq i \leq k$, form a path of L -edges if and only if $R + 4i$ divides $p + 4$ for every $0 \leq i \leq k$.*
- (c) *If $(a + i, m + i, 1)$, $0 \leq i \leq k$, is a path of L -edges and $R = 4a - p$, then $R + 4i$ divides $p + 4$ for every $0 \leq i \leq k - 1$. In particular a path of two edges forces $R(R + 4) \mid p + 4$.*
- (d) *Every prime $p \equiv 131 \pmod{180}$ has a path of two edges. More generally, let $q \geq 1$ and let c be prime to q , with $c \equiv 2 \pmod{3}$ if $3 \mid q$ (these are exactly the classes that contain primes $\equiv 2 \pmod{3}$). Then infinitely many primes $p \equiv c \pmod{q}$ have a path of two edges. So if Proposition 2.1(a) holds at all but finitely many primes of a union of classes modulo q , then none of those classes contains primes $\equiv 2 \pmod{3}$.*
- (e) *For every $k \geq 1$, infinitely many primes have a path of length k .*

Status. Proved here. Found by the referee pass of version 1; each part checked here. Part (a) was confirmed on the full graph of every prime below 1200. A complete enumeration below 10^7 through (c) and the shape of an edge finds:

- 19,558 two-edge paths at 17,425 primes;
- $R(R + 4) \mid p + 4$ on every one of those paths;
- all 13,822 primes $p \equiv 131 \pmod{180}$ among the primes with a path.

The least primes with a path of length 1, 2, 3, 4 and 5 are 11, 131, 3461, 109,391 and 1,183,451. The referee gave 3,533,141 for length 5; it has such a path, but it is not the least.

Proof. (a) By the shape of an edge, an L -edge starts at a vertex with $n = 1$ and a J -edge at one with $m = 1$. So the only possible successors of (a, m, n) are $(a + 1, m + 1, 1)$, when $n = 1$, and $(a + 1, 1, n + 1)$, when $m = 1$. Both are possible only if $m = n = 1$. Then $(1, 1) \in \mathcal{W}_a$ forces $R_a \mid 2$, so $R_a = 1$. Either successor would then need $R_{a+1} = 5$ to divide 3, which is false. The only possible predecessors of $(a + 1, m', n')$ are $(a, m' - 1, 1)$, by an L -edge when $n' = 1$, and $(a, 1, n' - 1)$, by a J -edge when $m' = 1$. Both are possible only if $m' = n' = 1$, and then $m' - 1 = 0$ is not a positive coordinate. Finally, a increases along edges, so there are no cycles.

(b) The triple $(a + i, a + i, 1)$ satisfies the divisibility and coprimality conditions of $\mathcal{W}_{a+i}$. It lies in $\mathcal{W}_{a+i}$ exactly when $R + 4i \mid a + i + 1$. As $R + 4i$ is odd and $4(a + i + 1) = (R + 4i) + (p + 4)$, this holds exactly when $R + 4i \mid p + 4$. Consecutive vertices are joined by an L -edge, since $L(a + i, 1) = (a + i + 1, 1)$.

(c) Let $0 \leq i \leq k - 1$ and put $K = a - m$. The edge from the i -th vertex gives $m + i + 1 \mid a + i + 1$, hence $m + i + 1 \mid K$. The i -th vertex lies in $\mathcal{W}_{a+i}$, so $R + 4i \mid m + i + 1$. Hence $R + 4i \mid K$. Also $R + 4i$ divides $4(m + i + 1) - (R + 4i) = 4m + 4 - R$. Adding $4K$, it divides $4a + 4 - R = p + 4$. For two edges, R and $R + 4$ are odd with difference 4, hence coprime, so $R(R + 4) \mid p + 4$.

(d) If $p \equiv 131 \pmod{180}$, then $p \equiv 3 \pmod{4}$, so $R = 1$ occurs, at $a = (p + 1)/4$. Also $45 \mid p + 4$, so 1, 5 and 9 divide $p + 4$, and (b) applies with $k = 2$. In general, choose an odd R with the following properties:

- $R \equiv -c \pmod{4}$ if $4 \mid q$;
- no prime $\ell \geq 5$ dividing q divides $R(R + 4)(R + 8)$;
- the one of $R, R + 4, R + 8$ that is divisible by 3 is not divisible by 9.

Such an R exists by the Chinese remainder theorem: each prime $\ell \geq 5$ excludes three classes, and modulo 9 the classes 0, 1 and 5 are excluded. Put $m' = R(R + 4)(R + 8)$. Its three factors are odd with pairwise differences 4 and 8, hence pairwise coprime, and $\gcd(m', q) \mid 3$. The congruences $p \equiv c \pmod{q}$, $p \equiv -4 \pmod{m'}$ and $p \equiv -R \pmod{4}$ are compatible: modulo 3 both of the first two give $p \equiv 2$, and modulo 4 the first and the third agree by the choice of R . Together they define a class prime to its modulus, which contains infinitely many primes by Dirichlet's theorem. Each such prime $p > R + 8$ has the diagonal path of (b) with $a = (p + R)/4$ and $k = 2$. If $3 \nmid q$, every class prime to q contains primes $\equiv 2 \pmod{3}$ (by the Chinese remainder theorem and Dirichlet's theorem). If $3 \mid q$, a class contains such primes exactly when $c \equiv 2 \pmod{3}$.

(e) Take $R = 1$. The modulus $\text{lcm}(1, 5, 9, \dots, 4k + 1)$ is odd, so the congruences $p \equiv 3 \pmod{4}$ and $p \equiv -4$ modulo it define a class prime to its modulus. Each prime $p > (4k + 1)/3$ in it has, by (b), a diagonal path of length k from $a = (p + 1)/4$. $\square$

Remark 2.3 (how often the theorem fails). Among the primes $p \equiv 2 \pmod{3}$, those with a path of two edges have upper relative density at most 0.335. So the conclusion of Proposition 2.1(a) holds on a set of primes $p \equiv 2 \pmod{3}$ of lower relative density at least 0.665. Below 10^7 it fails at 17,425 of the 332,384 primes $p \equiv 2 \pmod{3}$ (5.24%); the smallest primes $p \equiv 2 \pmod{3}$ at which it holds are 2, 5, 11, 17, 23, 29, 41, 47, 53, 59, 71, 83, ...

Proof. By Proposition 2.1(a) and Proposition 2.2(c), a prime with a path of two edges satisfies:

- $p \equiv 2 \pmod{3}$;
- for some odd R with $R(R + 4) \leq p + 4$, both $p \equiv -4 \pmod{R(R + 4)}$ and $p \equiv -R \pmod{4}$.

Fix R and put $m = R(R + 4)$. By Dirichlet's theorem, the second condition picks out a set of relative density $\delta_R = (1 + [3 \mid m])/(2\varphi(m))$ among the primes $p \equiv 2 \pmod{3}$. For large R the bound must be uniform in x :

- For $4m \leq x^{1/2}$. The Brun–Titchmarsh inequality $\pi(x; Q, b) \leq 2x/(\varphi(Q) \log(x/Q))$ for $Q < x$ [10], with $Q = 4m$, bounds the number of such primes up to x by $2x/(\varphi(m) \log x)$. This is $(4 + o(1))/\varphi(m)$ times the number of primes $p \equiv 2 \pmod{3}$ up to x .
- For $4m > x^{1/2}$. Each class has at most $x/m + 1$ members up to x , and $R \leq \sqrt{x + 4}$. The total over these R is $O(x^{3/4})$.

Since $\sum_R 1/\varphi(R(R + 4))$ converges, letting $x \rightarrow \infty$ and then the cut between small and large R tend to infinity bounds the upper relative density by $\sum_{R \text{ odd}} \delta_R$. Numerically, the partial sum over $R < 2 \cdot 10^5$ is 0.334275. The rest is less than $2 \cdot 10^{-4}$, by $n/\varphi(n) < e^\gamma \log \log n + 3/\log \log n$ for $n \geq 3$ [12]. So the sum is less than 0.335. $\square$

2.4 The 124-page ES/Fable reader

It encodes a solution of $4/p = 1/x + 1/y + 1/z$ as the binary quartic

$$H_{\text{ES}}(U, V) = -S^{-1}(U - pV)(U - xV)(U - yV)(U - zV), \quad S = p + x + y + z,$$

with normalised coefficients $u_0 = -1/S$, $u_1 = 1$, $u_2 = -(p(x+y+z) + xy + yz + zx)/S$, $u_3 = (p(xy+yz+zx) + xyz)/S$ (which equals $5xyz/S$ on solutions) and $u_4 = -pxyz/S$. It then transports the quartic to the Fable cover and to a weighted conductor of the zeta programme.

Proposition 2.4. (a) On solutions, $p = -5u_4/u_3$ and

$$625u_0u_4^3 - 125u_3u_4^2 + 25u_2u_3^2u_4 - 4u_3^4 = 0.$$

(b) Conversely, let u_0, u_2, u_3, u_4 be complex numbers with $u_0u_3u_4 \neq 0$ that satisfy this relation. Put

$$S = -1/u_0, \quad p = -5u_4/u_3, \quad P = u_3S/5, \quad \sigma_1 = S - p, \quad \sigma_2 = -u_2S - p\sigma_1.$$

Then the roots x, y, z of $T^3 - \sigma_1T^2 + \sigma_2T - P$ are non-zero and satisfy $4/p = 1/x + 1/y + 1/z$. The quartic $-S^{-1}(U - pV)(U - xV)(U - yV)(U - zV)$ has the coefficients $u_0, 1, u_2, u_3, u_4$.

Status. Proved here. The README of the package states (a), and 21_ (Proposition 21.6) also checked the relation. Part (b) was found by the referee pass of version 1 and is proved here.

Proof. (a) Write $\sigma_1 = x + y + z$, $\sigma_2 = xy + yz + zx$ and $P = xyz$. The equation is $4P = p\sigma_2$, so the third elementary symmetric function of p, x, y, z is $p\sigma_2 + P = 5P$; this gives $u_3 = 5P/S$, and $p = -5u_4/u_3$ follows. Substituting and multiplying by $S^4/625$, the left side becomes $P^3(p^3 - p^2S + p(p\sigma_1 + \sigma_2) - 4P) = P^3(p\sigma_2 - 4P) = 0$.

(b) With the stated definitions, $u_0 = -1/S$, $u_4 = -p\sigma_1/5$, $u_2 = -(p\sigma_1 + \sigma_2)/S$ and $u_3 = 5P/S$. Substituting, and using $S = p + \sigma_1$, gives

$$625u_0u_4^3 - 125u_3u_4^2 + 25u_2u_3^2u_4 - 4u_3^4 = \frac{u_3^3}{S}(5p^3 - 5p^2S + 5p(p\sigma_1 + \sigma_2) - 20P) = \frac{5u_3^3}{S}(p\sigma_2 - 4P).$$

So the relation gives $p\sigma_2 = 4P$. Now $xyz = P \neq 0$, since $u_3 \neq 0$, and $p \neq 0$, since $u_4 \neq 0$. Hence $1/x + 1/y + 1/z = \sigma_2/P = 4/p$. For the quartic, the coefficients of U^4 , U^3V , U^2V^2 , UV^3 and V^4 in $-S^{-1}(U - pV)(U - xV)(U - yV)(U - zV)$ are:

- $-1/S = u_0$;
- $(p + \sigma_1)/S = 1$;
- $-(p\sigma_1 + \sigma_2)/S = u_2$;
- $(p\sigma_2 + P)/S = 5P/S = u_3$;
- $-pP/S = u_4$.

□

So the relation is the equation itself, written in these coordinates: by (a) and (b), it cuts out exactly the quartics that come from triples, possibly complex, with $4/p = 1/x + 1/y + 1/z$. The Fable and conductor maps built on it are located; their comparison with the zeta programme is in the zeta audit (47_ overlap OV-13).

2.5 The continuation of 17–23 September

The README's *Start reading* list has 28 items with proof packages in `research/incoming/`. The results bench R01–R10 and the integration arguments X1–X7 accompany them.

- The first-pass audit 43a_ lists them all.
- Section 5 states the eleven items audited there.
- Section 7 gives their negative results.

2.6 The author's notes

The author's notes of April–June 2026 are separate from the workbench's documents, and Section 6 treats them. Their arithmetic is the shell calculus of Section 3 together with the classical identity families. Their structural layers are Niemeier and umbral data, the class 6B, Ogg's primes, Lorentzian geometry, Cayley–Dickson algebras and mock theta functions. The archive's §25 is drawn from one of them.

3 The core arithmetic, with proofs

Throughout, p is an odd prime and a *solution* is a triple of positive integers with $4/p = 1/x + 1/y + 1/z$. The workbench mostly works with $p = 12h + 1$ ($h \geq 1$). The results of this section, except Propositions 3.6, 3.7 and 3.8, were first stated in the reader of four workbenches [17], whose Erdős–Straus section was refereed three times (Appendix B); the audit note is 43_ and the first-pass report 43a_.

3.1 The shell criterion

For an integer $a > p/4$ not divisible by p put $R_a = 4a - p$, $S_a = pa$ and

$$E_a = \{u : u \mid a^2, R_a \mid 4u + 1\}, \quad M_a = \{u : u \mid a^2, R_a \mid u + a\}.$$

A shell a is *occupied* if $E_a \cup M_a \neq \emptyset$.

Theorem 3.1 (shell criterion). (a) If $x \leq y \leq z$ is a solution, then $p/4 < x \leq 3p/4$. More precisely, $x < p/2$ and $x < y$, unless $p \equiv 3 \pmod{4}$ and $(x, y, z) = (\frac{p+1}{2}, \frac{p+1}{2}, \frac{p(p+1)}{4})$. For $p = 12h + 1$ this gives $3h + 1 \leq x \leq 6h$.

(b) For each integer $a > p/4$ not divisible by p (in particular each a with $p/4 < a < p$), the ordered pairs (y, z) of positive integers with $1/a + 1/y + 1/z = 4/p$ number exactly $2|E_a| + |M_a|$. A pair with $y = z$ occurs if and only if $R_a = 1$ (possible only for $p \equiv 3 \pmod{4}$ and $a = (p+1)/4$); otherwise $|M_a|$ is even and the unordered pairs number $|E_a| + \frac{1}{2}|M_a|$. For $p \equiv 1 \pmod{4}$ this is always the case.

(c) Hence the equation is solvable at p if and only if some shell a with $p/4 < a < p/2$ is occupied.

Status. Proved here. Where each part is in the workbench:

- The bijection of (b). Archive Theorem 9.1 (Exact divisor-shell bijection, p. 38) states it for every odd prime p and every $R \equiv -p \pmod{4}$ with $\gcd(R, pa) = 1$. That is exactly the generality of (b), since $\gcd(R, pa) = 1$ holds if and only if $p \nmid a$.
- The count $2|E_a| + |M_a|$. Archive Theorem 9.64 (p. 90) splits the divisors by their p -adic layer and proves by an involution that the two outer layers have equal size. The targets $-4a^2$, $-a$ and -4^{-1} are the archive's table (394) (§10, pp. 93–94, stated there for $R \equiv 3 \pmod{4}$).
- The version for $p = 12h + 1$. The supplement's ES-01/ES-02 (§1), audited in the first pass, state the criterion for $p = 12h + 1$ and $a \in [3h + 1, 9h]$.

This paper adds (a), with its exceptional solution at $p \equiv 3 \pmod{4}$; the analysis of the swap in (b), that is, when $y = z$ occurs, the parity of $|M_a|$ and the unordered count; and (c). The halved range at $p \equiv 1 \pmod{12}$, with x unique, is already a reduction step of the workbench's item 22, checked by reading in the first pass (43a_ §2.22), and items 6 and 7 work in it; the workbench's later packages attribute these coordinates to Elsholtz and Tao [4], §2. Checked against a brute-force count of the ordered pairs on all 4,726 pairs (p, a) with $p < 260$ an odd prime and $p/4 < a < p$ (a pair with $y = z$ exactly in the 29 cases with $R_a = 1$) and on all 4,303 pairs with $p < 120$, $p/4 < a \leq 3p$ and $p \nmid a$; on all 4,398 shells of the primes $p \equiv 1 \pmod{12}$ below 700, and in the first pass on all 8,124 shells below 1000; (a) against all 2,259 solutions at the odd primes below 400; every prime $p \equiv 1 \pmod{12}$ below 30,000 has an occupied shell.

Proof. (a) The other two terms are positive, so $1/x < 4/p$; and x is the least denominator, so $4/p \leq 3/x$. Hence $p/4 < x \leq 3p/4$. Suppose $x > p/2$, so $x \geq (p+1)/2$ as p is odd, and $x < p$. Then $2/y \geq 1/y + 1/z = 4/p - 1/x > 2/p$, so $y < p$. From $4xyz = p(xy + yz + zx)$ the prime p divides xyz , and as $0 < x \leq y < p$ it divides $z = pz'$; then $xy(4z' - 1) = pz'(x + y)$ with $p \nmid xy$, so $4z' - 1 = pk$ with $k \geq 1$. Put $N = pk + 1 = 4z'$; then $1/x + 1/y = 4k/N$, that is, $(4kx - N)(4ky - N) = N^2$ with both factors positive. But $4kx - N \geq 2k(p+1) - N = N + 2(k-1)$, and likewise for y ; so $k = 1$ and $x = y = N/2 = (p+1)/2$, with $z = p(p+1)/4$, which needs $p \equiv 3 \pmod{4}$. Finally, if $x < p/2$ and $y = x$, then $1/x + 1/y > 4/p$, which is impossible. For $p = 12h + 1$ the integers in $(p/4, p/2)$ are $3h + 1, \dots, 6h$.

(b) Write $R = R_a$, $S = S_a$. Then $R \geq 1$ is odd, because p is. Since $p \nmid a$, $\gcd(a, R) = \gcd(a, p) = 1$ and $\gcd(p, R) = \gcd(p, 4a) = 1$, so S is a unit modulo R . The equation $1/y + 1/z = R/S$ is equivalent to $(Ry - S)(Rz - S) = S^2$, and both factors are positive because $1/y$ and $1/z$ are smaller than R/S . So the ordered pairs correspond bijectively to ordered factorisations $S^2 = dd'$ into positive divisors with $d \equiv d' \equiv -S \pmod{R}$, through $y = (d + S)/R$, $z = (d' + S)/R$. If $d \mid S^2$ and $d \equiv -S \pmod{R}$, then d is a unit and $d' = S^2/d \equiv S^2(-S)^{-1} = -S$; so the pairs are counted by $\#\{d \mid S^2 : d \equiv -S \pmod{R}\}$. Since $\gcd(p, a) = 1$, each divisor of $S^2 = p^2 a^2$ is uniquely $d = p^i v$ with $i \in \{0, 1, 2\}$ and $v \mid a^2$. Using $4a \equiv p \pmod{R}$: for $i = 2$, $p^2 v \equiv -pa \iff pv \equiv -a \iff 4pv \equiv -p \iff R \mid 4v + 1$, so these v form E_a ; for $i = 1$, $pv \equiv -pa \iff R \mid v + a$, so these v form M_a ; and $d \mapsto S^2/d$ maps the layer $i = 2$ bijectively onto the layer $i = 0$ while preserving the congruence. The total is $2|E_a| + |M_a|$. The swap $(y, z) \mapsto (z, y)$ is $d \mapsto S^2/d$; it maps each layer i to the layer $2 - i$, so a fixed point lies in the layer $i = 1$, where it is $v \mapsto a^2/v$ with only possible fixed point $v = a$. That needs $R \mid 2a$, i.e. $R \mid 2$ (as $\gcd(R, a) = 1$), i.e. $R = 1$. If $R \neq 1$ the swap has no fixed point, $|M_a|$ is even, $y \neq z$, and the unordered pairs number $|E_a| + \frac{1}{2}|M_a|$. For $p \equiv 1 \pmod{4}$, $R \equiv -p \equiv 3 \pmod{4}$, so $R \neq 1$.

(c) For $p \equiv 1 \pmod{4}$ it follows from (a) and (b). For $p \equiv 3 \pmod{4}$ both sides hold: the shell $a = (p+1)/4$ lies in $(p/4, p/2)$ and has $R_a = 1$, so $u = a$ lies in M_a . $\square$

The workbench also gives the exact bookkeeping around this criterion:

- an explicit bijection between a tagged set and the ordered solutions, with its inverse read off from the p -adic valuation of $Ry - S$ (archive (526), p. 151; supplement §1);
- the global statement that a counterexample prime would satisfy the finite, computable condition $E_a = M_a = \emptyset$ for every a .

Both were re-derived in the first pass (43a_ §1.2).

3.2 The first two shells

Proposition 3.2 (the residual-3 shell). *Let $a = 3h + 1 = (p+3)/4$, so $R_a = 3$. Let $P_1 = \prod (2v_\ell(a) + 1)$ over the primes $\ell \equiv 1 \pmod{3}$ dividing a , and P_2 the same product over the primes $\ell \equiv 2 \pmod{3}$. Then $E_a = M_a = \{u \mid a^2 : u \equiv 2 \pmod{3}\}$ and $|E_a| = |M_a| = P_1(P_2 - 1)/2$. So a solution with least denominator a exists if and only if a has a prime factor $\equiv 2 \pmod{3}$, and there are then exactly $3P_1(P_2 - 1)/4$ unordered solutions with least denominator a .*

Status. Proved here; the workbench's residual-3 sieve (supplement §1, `first_two_shell_sieve.tex`; archive Theorem 10.3, *Exact $R = 3$ factor criterion*, p. 94, for $p \equiv 1 \pmod{24}$), audited. Checked on all 4,472 primes $p \equiv 1 \pmod{12}$ below $2 \cdot 10^5$, and in the first pass on all 19,564 below 10^6 .

Proof. Since $a \equiv 1 \pmod{3}$, $3 \nmid a$. Modulo 3 both $4u + 1$ and $u + a$ are $\equiv u + 1$, so both conditions say $u \equiv 2 \pmod{3}$. A divisor $u = \prod \ell^{e_\ell}$ of a^2 has $u \equiv (-1)^{\sum e_\ell} \pmod{3}$, the sum over the primes $\ell \equiv 2 \pmod{3}$. Their exponent vectors $(0 \leq e_\ell \leq 2v_\ell(a))$ number P_2 , and the difference between the numbers with even and with odd sum is $\prod_\ell \sum_{e=0}^{2v_\ell} (-1)^e = 1$; so $(P_2 - 1)/2$ have odd sum, and the other primes contribute the factor P_1 . By Theorem 3.1(b) the unordered solutions containing

a number $\frac{3}{2}P_1(P_2 - 1)/2$, and $a = 3h + 1$ is the least denominator of each of them because it is the smallest possible one. Finally $P_2 > 1$ exactly when a has a prime factor $\equiv 2 \pmod{3}$. (The count is an integer: $a \equiv 1 \pmod{3}$ makes the total exponent of those primes even, so $P_2 \equiv 1 \pmod{4}$.) $\square$

Proposition 3.3 (the residual-7 shell). *Let $a = 3h + 2 = (p + 7)/4$, so $R_a = 7$ and $7 \nmid a$. Let n_3 be the total exponent in a of the primes $\equiv 3 \pmod{7}$ and n_{24} that of the primes $\equiv 2$ or $4 \pmod{7}$. The shell is occupied if and only if a has a prime factor $\equiv 5$ or $6 \pmod{7}$, or $n_3 \geq 3$, or $n_3 \geq 1$ and $n_{24} \geq 1$.*

Status. Proved here; the workbench’s residual-7 sieve, audited, together with its count as coefficients of $\prod_{\ell} \sum_{e=0}^{2v_{\ell}(a)} T^{\lambda(\ell)e}$ in $\mathbb{Z}[T]/(T^6 - 1)$. Checked on all 4,472 primes $p \equiv 1 \pmod{12}$ below $2 \cdot 10^5$ (and below 10^6 in the first pass).

Proof. The exterior condition $7 \mid 4u + 1$ is $u \equiv -4^{-1} \equiv 5 \equiv 3^5$, and the middle condition is $u/a \equiv -1 \equiv 3^3 \pmod{7}$; in base 3 the discrete logarithms of $1, \dots, 6$ are $0, 2, 1, 4, 5, 3$. *Sufficiency.* A prime $\ell \equiv 5$ dividing a gives $u = \ell$; a prime $\ell \equiv 6$ gives $u = a\ell$. If $n_3 \geq 3$, a product of primes $\equiv 3$ dividing a^2 with total exponent 5 gives $u \equiv 3^5$. If primes $t \equiv 3$ and $\ell \in \{2, 4\} \pmod{7}$ divide a , then $u = alt$ (when $\ell \equiv 2 \equiv 3^2$) or $u = a\ell/t$ (when $\ell \equiv 4 \equiv 3^4$) divides a^2 and has $u/a \equiv 3^3$. *Necessity.* If none of the conditions holds, every prime factor of a is $\equiv 1, 2, 3$ or 4 , and either $n_3 = 0$ or $(1 \leq n_3 \leq 2$ and $n_{24} = 0)$. If $n_3 = 0$, a and all divisors of a^2 lie in the subgroup $\{1, 2, 4\}$ of squares, which contains neither 5 nor -1 . Otherwise $a \equiv 3^{n_3}$ and a divisor of a^2 is $\equiv 3^e$ with $0 \leq e \leq 2n_3 \leq 4$; the exterior target needs $e \equiv 5 \pmod{6}$ and the middle one $e - n_3 \equiv 3 \pmod{6}$ with $|e - n_3| \leq 2$, both impossible. $\square$

The shells with $R = 3$ and $R = 7$ exist exactly for the primes $p \equiv 1 \pmod{4}$, with $a = (p + 3)/4$ and $a = (p + 7)/4$ (Generalised here; checked on all 1,611 such primes below $3 \cdot 10^4$). Proposition 3.3 holds for all of them, since its proof never uses $p \equiv 1 \pmod{3}$. Proposition 3.2 holds as stated for $p \equiv 1 \pmod{12}$ only. For $p \equiv 5 \pmod{12}$, $a = (p + 3)/4 \equiv 2 \pmod{3}$, so modulo 3 the middle condition reads $u \equiv 1$; then $E_a = \{u \mid a^2 : u \equiv 2\}$ and $M_a = \{u \mid a^2 : u \equiv 1\}$, with $|E_a| = P_1(P_2 - 1)/2$ and $|M_a| = P_1(P_2 + 1)/2$ by the parity count of the proof above, and there are exactly $|E_a| + \frac{1}{2}|M_a| = P_1(3P_2 - 1)/4$ unordered solutions with least denominator a (an integer: the total exponent in a of the primes $\equiv 2 \pmod{3}$ is odd, so $P_2 \equiv 3 \pmod{4}$). That shell is always occupied ($1 \in M_a$), and its occupancy criterion holds trivially, since $a \equiv 2 \pmod{3}$ has a prime factor $\equiv 2 \pmod{3}$; those primes are solvable anyway, since $p \equiv 2 \pmod{3}$.

Proposition 3.4 (the two-shell survivors). *Let $p \equiv 1 \pmod{12}$ be prime. Both shells $a = (p + 3)/4$ and $a = (p + 7)/4$ are unoccupied if and only if the following three conditions hold:*

- $p \equiv 1 \pmod{24}$;
- every prime factor of $(p + 3)/4$ is $\equiv 1 \pmod{3}$;
- every prime factor of $(p + 7)/4$ is $\equiv 1, 2$ or $4 \pmod{7}$.

Every such p is a square modulo 7 and satisfies $p \equiv \pm 1 \pmod{5}$. So it lies in one of the six hard classes $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$.

Status. Proved here. The conclusion modulo 840 is archive Corollary 10.7 (p. 95), the contrapositive of its Theorem 10.6; the first pass stated the congruence mod 24 as an observation on its computation. Checked on all primes $p \equiv 1 \pmod{12}$ below 10^6 . Of these, exactly 989 have both shells unoccupied. All 989 lie in the six hard classes, and each class occurs; the least in the classes 1, 121, 169, 289, 361, 529 are 2521, 12721, 2689, 1129, 1201 and 3049.

Proof. If $p \equiv 13 \pmod{24}$, then $(p + 3)/4$ is even and $2 \equiv 2 \pmod{3}$, so the first shell is occupied (Proposition 3.2). If $p \equiv 1 \pmod{24}$, the first shell is unoccupied exactly when every prime factor

of $(p+3)/4 \equiv 1 \pmod{3}$. Also $(p+7)/4$ is even, so $n_{24} \geq 1$, and Proposition 3.3 says that the second shell is occupied exactly when $(p+7)/4$ has a prime factor $\equiv 3, 5$ or $6 \pmod{7}$, a non-residue. If it has none, $(p+7)/4$ is a square modulo 7, and so is $p \equiv 4 \cdot \frac{p+7}{4}$. For the congruence modulo 5:

- if $p \equiv 2 \pmod{5}$, then $5 \mid (p+3)/4$ and $5 \equiv 2 \pmod{3}$, so the first shell is occupied;
- if $p \equiv 3 \pmod{5}$, then $5 \mid (p+7)/4$ and 5 is a non-residue modulo 7, so the second shell is occupied;
- $p = 5$ is not $\equiv 1 \pmod{24}$.

So $p \equiv \pm 1 \pmod{5}$, and $p \equiv 1, 2, 4 \pmod{7}$, and $p \equiv 1 \pmod{24}$. These are the unit squares modulo 840. $\square$

3.3 Which primes remain

The workbench proves solvability for $p \equiv 2 \pmod{3}$, $p \equiv 3 \pmod{4}$ and $p \equiv 13 \pmod{24}$ (scope identities that it does not advertise as new) by the families $(p, \frac{p+1}{3}, \frac{p(p+1)}{3})$, $(\frac{p+1}{4}, \frac{p(p+1)}{2}, \frac{p(p+1)}{2})$ and $(a, \frac{pa}{2}, pa)$ with $a = \frac{p+3}{4}$ (even when $p \equiv 13 \pmod{24}$); so its own reduction leaves the primes $p \equiv 1 \pmod{24}$ (archive Proposition 10.2). Its later packages restrict to the *hard* primes $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$, which are exactly the unit squares modulo 840 (a group of order $1 \cdot 1 \cdot 2 \cdot 3 = 6$). The reduction to them is classical (Obláth and Mordell; see Elsholtz–Tao [4] and Mordell [11]). The 22 September packages use it without proof.

The archive proves a stronger form (Theorem 10.6, *Non-square modulo 840 theorem*, p. 95). For every prime $p \equiv 1 \pmod{24}$ outside the six hard classes, the shell with $R = 3$ or the shell with $R = 7$ is occupied. Its proof uses explicit divisors in these two shells; Proposition 3.4 above is the same statement. The classical proof uses shells up to $R = 15$ and five identities. It is given here for comparison.

Proposition 3.5 (the reduction modulo 840). *Let $p \equiv 1 \pmod{24}$ be prime, and let (A, B, R) be $(1, 2, 7)$, $(2, 1, 7)$, $(1, 1, 7)$, $(1, 2, 15)$ or $(2, 1, 15)$ according as $p \equiv 3, 5, 6 \pmod{7}$ or $p \equiv 2, 3 \pmod{5}$. Then $D = (p+R)/(4AB)$ and $C = (A+pB)/R$ are positive integers and*

$$\frac{4}{p} = \frac{1}{ABD} + \frac{1}{ACD} + \frac{1}{pBCD}.$$

So the equation is solvable at every prime $p \equiv 1 \pmod{24}$ that is not a square modulo 840.

Status. Proved here (the reduction is classical); the first pass covered the 18 non-square unit classes $\equiv 1 \pmod{24}$ by the same kind of families. The identities hold for all integers in the stated classes, not only for primes. Checked on all primes $p \equiv 1 \pmod{24}$ below $3 \cdot 10^5$.

Proof. With $RC = A+pB$ and $4ABD = p+R$: $\frac{1}{ABD} + \frac{1}{ACD} + \frac{1}{pBCD} = \frac{pC+pB+A}{pABCD} = \frac{C(p+R)}{pABCD} = \frac{p+R}{pABD} = \frac{4}{p}$. Since $p \equiv 1 \pmod{8}$, 8 divides $p+7$ and $p+15$, so $4AB \mid p+R$; $R \mid A+pB$ is the stated congruence modulo 7 or 5, together with $A+pB \equiv A+B \equiv 0 \pmod{3}$ when $R = 15$. The primes $p \equiv 1 \pmod{24}$ left over have $p \equiv 1, 2, 4 \pmod{7}$ and $p \equiv \pm 1 \pmod{5}$, which are exactly the unit squares modulo 840. $\square$

3.4 The Jacobi-symbol barrier

Proposition 3.6 (the Jacobi-symbol barrier). *Let p be an odd prime, $R \equiv 3 \pmod{4}$, $a = (p+R)/4$ an integer, and $\gcd(R, pa) = 1$. If the Jacobi symbol (q/R) equals 1 for every prime $q \mid a$, then the shell a is unoccupied. The converse is false: at $p = 5$, $R = 7$, $a = 3$ the only prime factor of a is a non-residue modulo 7, and yet the shell is unoccupied.*

Status. Proved here; archive Theorem 10.9 (*Jacobi-symbol barrier*, p. 96) and Proposition 10.10 (*The Jacobi barrier has no converse*, pp. 96–97), whose proofs are correct as read.

Proof. The hypothesis $\gcd(R, pa) = 1$ gives $p \nmid a$: otherwise p would divide $4a - p = R$. So, by the proof of Theorem 3.1(b), the shell is occupied exactly when some divisor d of $S^2 = p^2a^2$ satisfies $d \equiv -pa \pmod{R}$. Let $\chi = (\cdot/R)$ be the Jacobi symbol, a character on the units modulo R . The hypothesis gives $\chi(a) = 1$, and $p \equiv 4a \pmod{R}$ gives $\chi(p) = \chi(4)\chi(a) = 1$. So $\chi(d) = 1$ for every divisor d of p^2a^2 , while $\chi(-pa) = \chi(-1) = (-1)^{(R-1)/2} = -1$ because $R \equiv 3 \pmod{4}$. So no divisor lies in the target class. For the converse: the squares modulo 7 are 1, 2, 4, so $(3/7) = -1$; the divisors of $15^2 = 225$ are $\equiv 1, 2, 3, 4, 5 \pmod{7}$, and the target $-15 \equiv 6$ is not among them. $\square$

So at a prime $p \equiv 1 \pmod{4}$, a shell with residual R can be occupied only if $a = (p + R)/4$ has a prime factor that is a non-residue modulo R in the Jacobi sense. The argument is a quadratic-character argument: every divisor inherits the residue condition. This is the kind of mechanism behind the obstruction to polynomial identities recalled in Section 1. The barrier explains why the workbench’s occupancy criteria, such as Propositions 3.2 and 3.3, are phrased through non-residue prime factors of a .

The quadratic character can be replaced by the whole group that the divisors generate. This gives a sharper barrier, and a converse when the exponents of a are large enough. Both were found by the referee pass of version 1 of this paper; the proofs below were checked here. Part (c) of Proposition 3.7, added in version 3, is from the author’s notes and the archive (see its status).

Proposition 3.7 (the group barrier). *Let p be an odd prime, $R \geq 3$ with $R \equiv -p \pmod{4}$, $a = (p+R)/4$ an integer, and $\gcd(R, pa) = 1$. Let $G_a \leq (\mathbb{Z}/R)^\times$ be the subgroup generated by 4 and the primes dividing a .*

- (a) *If $-1 \notin G_a$, the shell a is unoccupied.*
- (b) *This contains Proposition 3.6. For prime $R \equiv 3 \pmod{4}$ the two hypotheses are equivalent. For composite R the barrier (a) is strictly stronger: at $p = 53$, $a = 26$, $R = 51$ one has $G_a = \langle 2 \rangle = \{1, 2, 4, 8, 13, 16, 26, 32\}$, which does not contain -1 , while $(2/51) = -1$.*
- (c) *(The width-one barrier.) Let $H_a \leq G_a$ be the subgroup generated by the primes dividing a . If neither -1 nor -4 lies in H_a , the shell a is unoccupied. This contains (a), and it is strictly stronger. At $p = 433$, $R = 91$, $a = 131$ (a prime), $H_a = \langle 40 \rangle = \{1, 27, 40, 53, 66, 79\}$ contains neither $90 = -1$ nor $87 = -4$. Yet $-1 \in G_a$, and $(131/91) = -1$, so neither (a) nor Proposition 3.6 applies; the shell is empty. When every prime factor of R is $\equiv 3 \pmod{4}$, (a) and (c) are equivalent.*

Proof. (a) Every divisor u of a^2 , and a itself, lie in G_a . The two targets -4^{-1} and $-a$ lie in G_a exactly when -1 does, because $4, a \in G_a$. So if $-1 \notin G_a$, no divisor hits a target. (b) If $(q/R) = 1$ for every prime $q \mid a$, then G_a lies in the kernel of the Jacobi character, which does not contain -1 when $R \equiv 3 \pmod{4}$. For prime $R \equiv 3 \pmod{4}$ the group $(\mathbb{Z}/R)^\times$ is cyclic of order $2 \cdot \frac{R-1}{2}$ with $\frac{R-1}{2}$ odd. Its element -1 is the only element of order 2. So $-1 \notin G_a$ exactly when G_a has odd order, that is, when G_a lies in the subgroup of squares, which has odd order $\frac{R-1}{2}$. That is the hypothesis of Proposition 3.6. At $R = 51$: 2 has order 2 modulo 3 and 8 modulo 17, hence order 8 modulo 51, with the powers listed. Also $13 = 2^6$ and $4 = 2^2$ modulo 51, so $G_{26} = \langle 2 \rangle$. Finally $(2/51) = (2/3)(2/17) = -1$. (c) Every divisor of a^2 , and a , lie in H_a . The targets -4^{-1} and $-a$ lie in H_a exactly when -4 , respectively -1 , does. If $-1 \notin G_a = \langle 4, H_a \rangle$, then $-1 \notin H_a$, and $-4 \notin H_a$ since otherwise $-1 = -4 \cdot 4^{-1} \in G_a$; so (c) contains (a).

Now let every prime factor of R be $\equiv 3 \pmod{4}$. Then each factor $(\mathbb{Z}/q^k)^\times$ is cyclic of order $\equiv 2 \pmod{4}$. So $(\mathbb{Z}/R)^\times = P \times O$, where the 2-part P is elementary abelian and O has odd order $|O|$.

- The element 4 is a square, so it lies in O . Write $-1 = (\varepsilon, 1)$.

- If $-1 = 4^j h$ with $h \in H_a$, then $h = (\varepsilon, w)$ for some $w \in O$.
- So $h^{[O]} = (\varepsilon, 1) = -1$ lies in H_a .

Hence $-1 \in G_a$ implies $-1 \in H_a$, and (a) and (c) are equivalent. The example was computed directly; there $91 = 7 \cdot 13$ has the factor $13 \equiv 1 \pmod{4}$. $\square$

Status. Part (c) is Theorem 3.1 of the author’s note *Normalized divisor supports and the residual pre-Niemeier datum* (Section 6), and the *reachable targets* of archive Theorem 15.5. It was added to this paper in version 3. Among the shells $R < 3p$ of the primes $p \equiv 1 \pmod{24}$ below 3000, it kills 124 empty shells on which (a) is silent; every one of these R has a prime factor $\equiv 1 \pmod{4}$. The equivalence for R with all prime factors $\equiv 3 \pmod{4}$ was found by the referee of version 3.

Proposition 3.8 (a converse under an exponent condition). *In the setting of Proposition 3.7, let H_a be the subgroup generated by the primes dividing a . Suppose that $2v_\ell(a) \geq \text{ord}_R(\ell) - 1$ for every prime $\ell \mid a$.*

- (a) *The residues modulo R of the divisors of a^2 are exactly the elements of H_a .*
- (b) *The shell a is occupied if and only if $-1 \in H_a$ or $-4 \in H_a$ (equivalently $-4^{-1} \in H_a$, since H_a is a group).*
- (c) *If moreover R is a prime $\equiv 3 \pmod{4}$, the shell is occupied if and only if some prime factor of a is a non-residue modulo R .*

Proof. (a) A divisor of a^2 is $\prod \ell^{e_\ell}$ with $0 \leq e_\ell \leq 2v_\ell(a)$. Under the hypothesis, e_ℓ runs through a complete set of residues modulo $\text{ord}_R(\ell)$, so ℓ^{e_ℓ} runs through all of $\langle \ell \rangle$. The products then fill the subgroup generated by these cyclic subgroups, which is H_a . (b) The targets are -4^{-1} and $-a$, and $-a \in H_a$ exactly when $-1 \in H_a$, since $a \in H_a$. (c) If every prime factor of a is a residue, Proposition 3.6 applies. Suppose some prime factor is a non-residue. The group $(\mathbb{Z}/R)^\times$ is cyclic of order $R-1 = 2 \cdot \frac{R-1}{2}$ with $\frac{R-1}{2}$ odd. So every subgroup of odd order lies in its unique subgroup of order $\frac{R-1}{2}$, which is the subgroup of squares. Since H_a contains a non-square, its order is even. It therefore contains the unique element of order 2, which is -1 , and the shell is occupied by (b). $\square$

Status. Both proved here. Checked on all 123,538 shells with $R \geq 3$ of the primes $5 \leq p < 1500$: the group barrier holds on every one of them. On 9,462 of them, all with composite $R \equiv 3 \pmod{4}$, it applies where the Jacobi barrier does not. The criterion of Proposition 3.8(b) is correct on all 677 shells that satisfy the exponent condition. For prime $R \equiv 3 \pmod{4}$ the two barriers agree on all 18,157 shells of the primes $p \equiv 1 \pmod{4}$ below 1500. The counterexample $p = 5, R = 7, a = 3$ to the converse of Proposition 3.6 fails the exponent condition: $\text{ord}_7(3) = 6 > 2v_3(3) + 1 = 3$.

4 Least classes, record primes, explicit families, and the carrier census

4.1 The least-class function and its record primes

The archive (§17.4 and §19, pp. 131 and 148–157) puts, for $i \geq 1$, following Ionascu and Wilson [5],

$$C_i = \left\{ n \geq 2 : \frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \text{ with } x \leq \frac{n+4i-1}{4} \right\}, \quad h(n) = \min\{i : n \in C_i\}.$$

An odd prime p is a *strict record* if $h(q) < h(p)$ for every prime $q < p$.

Lemma 4.1. *Let $p \equiv 1 \pmod{4}$. The least denominator occurring in any solution at p is the least occupied shell a . With $R = 4a - p$ one has $h(p) = (R+1)/4$. For $p \equiv 3 \pmod{4}$, $h(p) = 1$, and $h(2) = 1$.*

Status. Proved here.

Proof. If a is occupied, some solution has a as a denominator, so the least denominator of that solution is at most a . Conversely, the least denominator x of a solution is itself an occupied shell: it exceeds $p/4$ (Theorem 3.1(a)) and is completed by the other two denominators. So the least occupied shell equals the least denominator over all solutions. A solution lies in C_i exactly when $4x - p \leq 4i - 1$. For $p \equiv 1 \pmod{4}$ the residual $4x - p \equiv 3 \pmod{4}$, so the least i is $(R + 1)/4$. For $p \equiv 3 \pmod{4}$ the shell $(p + 1)/4$ is occupied (Theorem 3.1(c)). For $p = 2$, take $4/2 = 1/1 + 1/2 + 1/2$ with $x = 1 \leq 5/4$. $\square$

Corollary 4.2. *If a prime p does not lie in one of the six hard classes 1, 121, 169, 289, 361, 529 modulo 840, then $h(p) \leq 2$. Hence every strict record after 73 lies in a hard class.*

Status. Proved here; found by the referee pass of version 1. It is the archive’s Theorem 10.6 and Proposition 10.2 read through Lemma 4.1. An independent computation of $h(p)$ for all 144,449,537 primes below $3 \cdot 10^9$ finds no prime with $h(p) \geq 3$ outside the hard classes.

Proof. For $p = 2$ and $p \equiv 3 \pmod{4}$, $h(p) = 1$ by Lemma 4.1. Let $p \equiv 1 \pmod{4}$.

- If $p \equiv 5 \pmod{12}$, the shell $(p + 3)/4$ is occupied (Section 3, after Proposition 3.3).
- If $p \equiv 13 \pmod{24}$, the same shell is occupied (Proposition 3.4, first line of the proof).
- If $p \equiv 1 \pmod{24}$ is not hard, Proposition 3.4 shows that the shell with $R = 3$ or the shell with $R = 7$ is occupied.

In each case $R \leq 7$, so $h(p) \leq 2$. A strict record after 73, where $h = 2$, has $h \geq 3$. $\square$

Proposition 4.3 (the record primes). *The strict records $p \leq 8,803,369$, with $h(p)$ and the residual R of the least occupied shell, are exactly*

$(2, 1, 2), (73, 2, 7), (1129, 3, 11), (1201, 6, 23), (21169, 8, 31), (118801, 15, 59), (8803369, 27, 107)$.

For the odd records $R = 4h(p) - 1$; for $p = 2$ the least denominator is 1, so $R = 4 \cdot 1 - 2 = 2$. Moreover $h(806521) = h(118801) = 15$, so 806,521 is not a strict record. There is no strict record between 8,803,369 and $3 \cdot 10^9$. Below $3 \cdot 10^9$, no prime other than 8,803,369 has $h \geq 22$. The primes with $h = 19, 20$ or 21 are

- $h = 21$: 287,567,281, 651,412,441, 2,188,875,529;
- $h = 20$: 778,103,881, 1,749,233,641, 2,535,613,921, 2,809,257,889;
- $h = 19$: 230,089,729, 793,592,809.

Status. Reproduced here: every prime up to 8,803,369 was computed by Lemma 4.1 and Theorem 3.1, using the residues of the divisors of a^2 modulo R (script `esr_records.py`, under three seconds).

- *Source of the list.* Ionascu and Wilson print the record list [5]. The archive (Proposition 17.6, pp. 131–132) recovers every printed record prime from the definition. It corrects one printed class label: $1201 \in C_6 \setminus C_5$, where the source prints C_4 . Its table (IW-RC) (p. 133) gives $R = 2$ for $p = 2$.
- *The range to $3 \cdot 10^9$.* It was found by the referee pass of version 1 with its own program, and recomputed here by an independent segmented program. The two agree on every record, on every prime with $h \geq 19$, and on the histogram of h .
- *The shell at p_* .* At $p_* = 8,803,369$ the least occupied shell has $R = 107$, the shell of archive §3.

For the five records whose residual R is a safe prime ($R = 2m + 1$ with m prime), and whose shell $a = (p + R)/4$ has exactly one non-residue prime factor modulo R , to exponent one, the archive defines a deficit set $D \subset \mathbb{Z}/m$ from the discrete logarithms of the prime factors of a (§19.1). It states the *strict-record deficit-box conjecture* (§19.2): at such a record with non-empty deficit, D is a difference-proper generalised arithmetic progression of rank at most two, and two induced odd sheets cover $\mathbb{Z}/m$. The deficits of the first three rows are empty. The two non-empty ones

are proved to be a rank-two rectangle $\{11, 14, 15, 18\} \subset \mathbb{Z}/29$ at $(118801, 59)$ and the progression $\{23 + 42j : 0 \leq j \leq 9\} \subset \mathbb{Z}/53$ at $(8803369, 107)$. The boundary example $p = 806,521$ satisfies every hypothesis except strict recordness, and both conclusions fail there. So the hypothesis cannot be dropped.

Status. The conjecture is open, as the archive says. Its finite evidence is reproduced here from the definitions (script `esr_deficits.py`, with the primitive root $g = 2$ that the archive uses for $R = 59$ and $R = 107$):

- the deficits of the rows $p = 73, 1129$ and 1201 are empty, and the row $p = 21169$ is outside the domain because $m = 15$ is composite;
- at $p = 118,801$: $a = 29715 = 3 \cdot 5 \cdot 7 \cdot 283$, $D = \{11, 14, 15, 18\}$, $D - D = \{3r + 4s : |r|, |s| \leq 1\}$ with nine elements, offsets $u = 11$ and $v = 17$, and the two odd sheets cover $\mathbb{Z}/29$;
- at $p = 8,803,369$: $a = 2,200,869 = 3^2 \cdot 11^2 \cdot 43 \cdot 47$, $D = \{23 + 42j : 0 \leq j \leq 9\}$, $D - D = \{42k : |k| \leq 9\}$ with nineteen elements, offsets 29 and 23, and the sheets cover $\mathbb{Z}/53$; also the corollary of archive §16, that the middle target -1 is reached by exactly two exponent vectors and the exterior target not at all;
- at $p = 806,521$: $a = 201,645 = 3^2 \cdot 5 \cdot 4481$, a deficit of 14 points with $D - D = \mathbb{Z}/29$, and the odd sheets miss exactly $\{1, 4, 8, 12, 16, 20, 24, 27\}$.

Another primitive root multiplies D by a unit of $\mathbb{Z}/m$ (archive §19.1). The script confirms that the sizes of D and $D - D$, and the number of missed points, do not change. The failure of conclusion (i) at $806,521$ follows from $D - D = \mathbb{Z}/29$ by the archive's counting argument. A difference-proper progression of rank one with 14 terms has exactly 27 differences. A proper one of rank two has side lengths $\{2, 7\}$, so a difference-proper one would have $3 \cdot 13 = 39 > 29$ distinct differences. That argument was read and is correct. By Theorem 3.1(b), the counts at $p_* = 8,803,369$ ($|E_a| = 0$, $|M_a| = 2$) mean that exactly one unordered solution has least denominator 2,200,869, namely the archive's (405) (p. 97), from the middle divisor $u = 11^2$:

$$\frac{4}{8803369} = \frac{1}{2200869} + \frac{1}{181085300330} + \frac{1}{3293760527702370}.$$

The archive's Lean module covers the $R = 107$ row only, as the archive states.

4.2 Fixed-divisor families, and four progressions on which the equation is solved

Theorem 3.1(b) turns a single divisor into a solution. If the divisor is a fixed multiple of the cofactor $h = a/c$, the solution is given by polynomials in n on a whole residue class. The archive states this for primes, as its constant-factor carriers (Lemma 24.218, p. 422). The form below holds for all n . It was found by the referee pass of version 1 and checked here.

Proposition 4.4 (fixed-divisor families). *Let $R \geq 3$ be odd, and let $c, s \geq 1$ with $\gcd(c, R) = 1$ and $s \mid c^2$. Let $n \geq 1$ with $n \equiv -R \pmod{4c}$, and put $a = (n + R)/4$ and $h = a/c$, both integers.*

(i) Exterior. *If $n \equiv -cs^{-1} \pmod{R}$, then $4/n = 1/a + 1/y + 1/z$ with the positive integers*

$$y = \frac{n(nsh + a)}{R}, \quad z = \frac{c^2h/s + na}{R}.$$

(ii) Middle. *If $n \equiv -4s \pmod{R}$, then $4/n = 1/a + 1/y + 1/z$ with the positive integers*

$$y = \frac{n(a + s)}{R}, \quad z = \frac{n(a + a^2/s)}{R}.$$

Each case is one residue class modulo $4cR$. If $\gcd(cR, 35) = 1$, that class meets all six hard classes modulo 840 or none of them. It meets all six exactly when it is $\equiv 1 \pmod{\gcd(4cR, 24)}$.

Proof. Put $S = na$. In both cases n is a unit modulo R , because $\gcd(cs, R) = 1$. So is a , because $4a \equiv n$. Hence S is a unit modulo R . The proof of Theorem 3.1(b) uses only this. A positive divisor d of S^2 with $d \equiv -S \pmod{R}$ therefore gives positive integers $y = (d + S)/R$ and $z = (S^2/d + S)/R$ with $1/y + 1/z = R/S$. Then $1/a + 1/y + 1/z = (n + R)/(na) = 4/n$.

- In case (i), take $d = n^2u$ with $u = sh$. Here $u \mid c^2h^2 = a^2$, and $d \equiv -S$ means $nu \equiv -a$, that is, $4u \equiv -1 \pmod{R}$. That holds because $c(4sh + 1) = s(n + R) + c \equiv sn + c \equiv 0 \pmod{R}$. Then $S^2/d = a^2/u = c^2h/s$.
- In case (ii), take $d = nu$ with $u = s$. Here $s \mid c^2 \mid a^2$, and $d \equiv -S$ means $s \equiv -a$, that is, $4s \equiv -n \pmod{R}$. Then $S^2/d = na^2/s$.

Since R is odd and prime to c , the congruences modulo $4c$ and modulo R define one class modulo $4cR$ (Chinese remainder theorem). If $\gcd(cR, 35) = 1$, then $\gcd(4cR, 840) = \gcd(4cR, 24)$. A class r modulo $4cR$ meets a class t modulo 840 exactly when $r \equiv t \pmod{\gcd(4cR, 840)}$. The six hard classes are all $\equiv 1 \pmod{24}$. $\square$

Status. Proved here. For primes it is archive Lemma 24.218; the archive’s proof of its Theorem 24.219 identifies the four families below as these middle and exterior channel maps.

Theorem 4.5 (four unconditional families). *For $k \geq 0$ put $g_1 = 31k+30, g_2 = 31k+15, m_3 = 23k+15, m_4 = 23k+19, w_3 = 1116k+727$ and $w_4 = 4464k+3683$. Then $4/n_i = 1/a_i + 1/b_i + 1/c_i$ with positive integers*

i	n_i	a_i	b_i	c_i
1	$248k + 209$	$2g_1$	$2n_1(k+1)$	$2n_1g_1(k+1)$
2	$248k + 89$	$2g_2$	$n_2(2k+1)$	$2n_2g_2(2k+1)$
3	$17112k + 11137$	$186m_3$	$12n_3m_3w_3$	$124m_3w_3$
4	$17112k + 14113$	$186m_4$	$6n_4m_4w_4$	$31m_4w_4$

No primality of n_i is needed.

Status. Proved here by direct expansion: $4a_ib_ic_i - n_i(b_ic_i + a_ic_i + a_ib_i)$ is the zero polynomial in k for each i (sympy), and the entries are positive for $k \geq 0$; exact fractions agree for $k \leq 200$ (script `esr_families.py`). Archive Theorem 24.219, p. 423.

- *As fixed-divisor families.* The four families are instances of Proposition 4.4. With $(R, c, s) = (31, 2, 2)$ and $(31, 1, 1)$ in case (ii), and $(23, 186, 18)$ and $(23, 186, 36)$ in case (i), where $h = m_3$ and $h = m_4$, its formulas give exactly the table (checked by expansion). Two of the families are halves of larger instances (archive Proposition 24.222(ii)–(iii), p. 425):
 - family 2 is half of the class $n \equiv 89 \pmod{124}$, which is $(31, 1, 1)$ itself;
 - family 3 is half of the class $n \equiv 2581 \pmod{8556}$, which is $(23, 93, 9)$ in case (i).
- *Rosati.* The archive shows that these triples are specialisations of Rosati’s two 1954 parametrisations. It claims no independence from the classical parametrisations (archive abstract, p. 2, and §13.1). That comparison is located, not checked.

Proof. All displayed linear factors have positive constant terms, so every entry is a positive integer for $k \geq 0$. The identity $4a_ib_ic_i = n_i(b_ic_i + a_ic_i + a_ib_i)$ is a polynomial identity in k , verified by expanding both sides. Dividing by $n_ia_ib_ic_i$ gives the decomposition. $\square$

Each progression meets all six hard classes modulo 840. This is the last part of Proposition 4.4, and directly: since $\gcd(248, 840) = 8$ and $\gcd(17112, 840) = 24$, the numbers n_1, n_2 run through every class $\equiv 1 \pmod{8}$ modulo 840, and n_3, n_4 through every class $\equiv 1 \pmod{24}$. Every hard class is $\equiv 1 \pmod{24}$, being the square of a unit. Each $n_i(0)$ is prime to the step of its progression, so by Dirichlet’s theorem every progression contains infinitely many primes in each hard class. Examples are the strict records $1201 = n_1(4) \equiv 361$ and $21169 = n_2(85) \equiv 169 \pmod{840}$. So the families solve the equation at infinitely many primes in every class that the reduction modulo 840 leaves open (script `esr_families.py` lists the least prime for each family and class). This does

not conflict with the obstruction to polynomial identities recalled in Section 1. That obstruction is modulo q , and it applies only when r is a square modulo q : here $11137 \equiv 5 \pmod{23}$, and 5 is not a square modulo 23. The same holds for the other rows, since $14113 \equiv 14 \pmod{23}$, $209 \equiv 23 \pmod{31}$ and $89 \equiv 27 \pmod{31}$ are non-squares. Proposition 4.6 below shows that this is forced: no fixed-divisor family covers a square class.

4.3 The carrier census

The largest part of the archive’s arithmetic is a sequence of *sufficient-carrier censuses*. It occupies §18 and §24 from the base of the census (Theorem 24.100, p. 335) to the end of §24.3 (p. 477).

- *Domain*. The census concerns only the primes $p \equiv 25 \pmod{72}$ whose shells $R = 7$ and $R = 11$ are both empty (archive Corollary 18.17 and Theorem 24.100). Such a prime lies in one of 30 classes modulo $11088 = 16 \cdot 9 \cdot 7 \cdot 11$, the base rows $B_1 \cup B_{25}$. The primes $p \equiv 1$ and $p \equiv 49 \pmod{72}$ with these two shells empty are outside the census. Below $3 \cdot 10^6$ there are 2,316 primes $p \equiv 1 \pmod{24}$ with both shells empty; 829 are $\equiv 1$, 682 are $\equiv 25$ and 805 are $\equiv 49 \pmod{72}$ (computed here).
- *Rows*. The archive considers residue data of p modulo a growing product of moduli. We call each such datum a *row*. The moduli are 11088, 5, 19, 23, 31 and 47, then 71, 191, 83, 167 and 239.
- *Forced rows*. A row is *forced* when an explicit construction shows that, for every prime p in the branch with that row, one of a finite list of shells is occupied. An example is the implication (2272) on p. 431: for a prime $p \equiv 25 \pmod{72}$ whose shells $R = 7$ and $R = 11$ are empty and whose row lies in the domain but outside the survivor set, some shell with $R \in \{15, 19, 23, 31, 47\}$ is occupied.
- *Survivors*. The remaining rows are *survivors*. Nothing is concluded for them.

The archive’s front section (pp. 3–5) reports the following chain.

Step	Domain (rows)	Forced	Survivors
six shells after the unit lift mod 71	4,590,432,000	4,212,008,640	378,423,360
$R = 191$	71,900,438,400	24,762,018,720	47,138,419,680
$Q = 83$	3,865,350,413,760	934,648,428,742	2,930,701,985,018
$R = 167$	486,496,529,512,988	115,851,891,806,170	370,644,637,706,818
$R = 239$	88,213,423,774,222,684	20,524,419,041,579,843	67,689,004,732,642,841
refresh of the $R = 167$ coefficient		39,025,685,947,842	67,649,979,046,694,999

Cumulatively 2,757,919,929,517,785,001 of the 2,825,569,908,564,480,000 rows of the final hard domain are forced, a proportion of 0.97606 The table is internally consistent (checked here):

- each domain is the previous survivor count times $q-1$, for the next modulus $q = 191, 83, 167, 239$;
- the final domain is $4,590,432,000 \cdot 190 \cdot 82 \cdot 166 \cdot 238$;
- forced plus survivors equals the domain in every row;
- the fractions $69643/75900$ and $6257/75900$ of the first row are exact.

Status. Located. The arithmetic consistency of the reported numbers was checked here; the carrier theorems and the enumerations behind them were not re-run. The archive states the following about its own checks.

- The enumerations are replayed by independent Python programs.
- Bounded Lean modules check the literal census rows, totals, fractions, target residues and candidate orders, but not the affine and gcd reconstructions, carrier membership or the positive-denominator witnesses (pp. 3–5).

- The ranking that selected 239 after 167 was corrected: on the full ten-coordinate object, 59 ranks first and 239 second. The $R = 239$ step is kept as a valid non-greedy continuation, not as an optimal one (p. 4).

The archive calls these *bounded sufficient-carrier censuses* and states that they are not shell-exhaustion, density, endpoint or optimality theorems (p. 2). A survivor row is only outside the carriers' images; it is not shown to be shell-empty.

What the census reaches. Every carrier of the census is a fixed-divisor family in the sense of Proposition 4.4. On its row it fixes the residual R , a divisor c of a that the row determines, and a divisor s of c^2 , and uses $u = sh$ or $u = s$:

- for $R = 15, 19, 23, 31$ and for the factor-31 carrier at $R = 23$, the maps listed in the proof of archive Theorem 24.228 (pp. 432–433), with $c = 2, 1, 6$ or $12, 2$ and 186;
- for $R = 47$ and later, the sets $E_q(c) = \{-cs^{-1}\}$ and $M_q(c) = \{-4s\}$ over the divisors $s \mid c^2$, with $c = \gcd(a, N)$ read off the row (archive (2268)–(2269), p. 431, and the coefficients c_Q , pp. 462–463).

In each case $4cR$ divides the modulus of the row. The obstruction of Schinzel and Yamamoto therefore applies, in the following form.

Proposition 4.6 (no fixed-divisor family covers a square class). *In the setting of Proposition 4.4, the class modulo $4cR$ on which (i) or (ii) applies contains no square modulo $4cR$. So it contains no class that is a square modulo a multiple of $4cR$.*

Proof. Suppose n lies in the class and is a square modulo $4cR$.

- *The prime 2.* n is odd, so $n \equiv 1 \pmod{4}$ and $R \equiv -n \equiv 3 \pmod{4}$. If $2 \mid c$, then $8 \mid 4c$, so $n \equiv 1 \pmod{8}$, $R \equiv 7 \pmod{8}$ and $(2/R) = 1$.
- *Odd primes $\ell \mid c$.* $-R \equiv n$ is a non-zero square modulo ℓ , so $(-R/\ell) = 1$. By reciprocity, $(\ell/R) = (R/\ell)(-1)^{(\ell-1)/2} = (-R/\ell) = 1$, since $(R-1)/2$ is odd.

Hence $(c/R) = 1$, and $(s/R) = 1$ because $s \mid c^2$. But n is a unit square modulo R , so $(n/R) = 1$. In case (ii), $n \equiv -4s$ gives $(n/R) = (-1/R)(s/R) = -1$. In case (i), $n \equiv -cs^{-1}$ gives $(n/R) = (-1/R)(c/R)(s/R) = -1$. Either way this is a contradiction. $\square$

Status. Proved here; found by the referee pass of version 1. It agrees with the archive's own residue sets (checked here):

- at $R = 47$ the forced residues F_{36} are exactly the 23 non-residues modulo 47, and F_{18} consists of 21 of them;
- the unforced sets contain every quadratic residue of their modulus: $S_5 = \{1, 4\}$, $A_{25} = Q_{23}$, and S_{19} and $S_{31}^\#$, whose excluded residues 15, 18 mod 19 and 15, 23, 27, 29, 30 mod 31 are all non-residues.

Corollary 4.7 (the square rows of the census). *No row that is a square modulo the modulus of the census is ever forced by its carriers. All 30 base rows are squares modulo 11088. So the rows whose coordinates at the ten odd primes 5, 19, 23, 31, 47, 71, 191, 83, 167, 239 are all non-zero squares are never forced. They number*

$$\frac{2,825,569,908,564,480,000}{2^{10}} = 2,759,345,613,832,500,$$

that is, 1/1024 of the final domain and 4.08% of its 67,649,979,046,694,999 survivors. Each further odd prime modulus added to the census halves this share, but no finite chain of such carriers removes it.

Proof. A forced row lies in a class of Proposition 4.4 modulo $4cR$, and $4cR$ divides the modulus of the row. Proposition 4.6 applies. The base rows were checked directly: each is a square of a

unit modulo 11088 (script `esr_referee_checks.py`). The domain is $30 \cdot \prod (q-1)$ over the ten odd primes, and exactly half of the units modulo each odd prime q are squares. $\square$

The survivors make up 2.39% of the final domain. Each of the four lifts forced between 23% and 35% of its domain ($R = 191$: 34.4%; $Q = 83$: 24.2%; $R = 167$: 23.8%; $R = 239$: 23.3%). By Corollary 4.7, a residue census built from such carriers does not reach the square rows. Carriers that reach them are not fixed-divisor families; one kind of candidate uses the prime factors of $(p+R)/4$ (compare Proposition 3.6 and Theorem I). The archive itself names the removal of the survivors as the open global obligation (p. 4).

5 The September continuation

The continuation of 17–23 September is organised as the 28-item *Start reading* list of the README, with proof packages in `research/incoming/`. The workbench describes the results of its 22 September release as written proofs with Python replays, without a Lean build or independent human review (43a_§2, citing DAILY_RESULTS_20260922).

5.1 The audited items

The 28 items of the README’s *Start reading* list (17–23 September) are listed in 43a_§2, with their packages. Those audited:

- *Item 6* (audited). With $L_a = 8\tau(a)^2/\varphi(R_a) - \mathcal{E}_a$, where $\mathcal{E}_a$ is the residue-collision energy of the $2\tau(a)$ divisors of pa modulo R_a : for $p \equiv 1 \pmod{4}$, $p \geq 2^{20}$, $\sum_{a \in (p/4, p/2)} L_a \leq -\frac{1}{24}p \log p$. So the lower bound that keeps only the principal character, summed over the shells, is negative for large p , and that bound alone does not show occupancy.
- *Item 7* (audited). For $p \equiv 1 \pmod{8}$, bounds in terms of the least quadratic nonresidue; at the hard primes a middle state (a, u) on a shell $p/4 < a < p/2$ has $a \leq 3(p+3)/11$, and an exterior state satisfies $22(p+1) \leq 23(p-2a+1)^2$. The workbench says this localises solutions and does not prove the region occupied.
- *Item 14* (audited). For $h' = 4j - 1 > t$ the divisors $W \mid j^2$ with $W \equiv t \pmod{h'}$ are canonical, companion or finite templates; the finite ones satisfy the sharp bound $h' \leq t^2 - 3t + 1$, with equality exactly in one parametrised family, attained at infinitely many primes $p \equiv 1 \pmod{840}$. For every $t \geq 4$ infinitely many hard primes have an exterior state whose same-grade middle box is empty.
- *Item 21* (audited). For a proper rational trace source the complete set of same-word residual returns is $\{k \mid R : d \mid k, k \equiv 1 \pmod{4K(u)}\}$; a uniform return exists exactly for the nine words $u \mid 36$.
- *Item 22* (audited, certificate). For every solution at $p \equiv 1 \pmod{12}$ an explicit symmetric polynomial $\mathfrak{N}(p, x, y, z)$ of degree 8 satisfies $\mathfrak{N} < -\frac{125873811}{262144}p^8$ ($\approx -480.17p^8$), and $\mathfrak{N} < -6960.19p^8$ when $p \equiv 1 \pmod{24}$; the proof reduces to finite polynomial positivity certificates, recomputed independently. It is a statement about existing solutions, not about their existence.
- *Item 26* (audited in part: the least example and the deduction from it; its prime-square trace theorem at $p \equiv 1 \pmod{24}$ and its terminality statement read and correct as read, with the identity $16(4q^3 + 1) - (4q + 1)(16q^2 - 4q + 1) = 15$ checked; the factor-sum, Pell and fixed-tail-sum sections not checked). $p = 67369$ is the least prime $p \equiv 1 \pmod{24}$ whose first *trace shell* ($a = 16849, R = 27$) precedes its first full shell ($a = 16850, R = 31$); so no universal map from rational traces to solutions avoids increasing a .
- *Item 4* (conditional on a computer enumeration of 1,381,117,764 rooted profiles, 5,922,259 of them range-compatible, run by two implementations of the workbench and not re-run here).

At every hard prime, from any nonresidue prime, factoring at most 25 numbers $(p + \sigma q t^2)/4$ with $t \in \{1, 3, 5, 7, 9\}$ yields at least six distinct nonresidue primes. A direct test of all 108 hard primes below 40,000 agrees.

Items 2 and 3 are method counterexamples, and their certificates are audited: the seven-cycle sink of item 2 at $p = 2521$ and the closed triangle of item 3; their closure and escape theorems are located. For item 8, the middle identity $p = RQ - (Q + 1)^2/(4A)$ and its two bounds were reproduced on all 1,613 oriented middle states at primes $p \equiv 1 \pmod{8}$ below 4000, and its atlas is located. The 22-digit prime 7510085481569082811681 of item 3 is proved prime here by a Pocklington chain ($p - 1 = 2^5 \cdot 3 \cdot 5 \cdot q_1$, $q_1 - 1 = 2 \cdot 3 \cdot 5 \cdot 351707 \cdot q_2$, $q_2 - 1 = 2 \cdot 3 \cdot 17^2 \cdot 5051 \cdot 169307$, witnesses below 12 at every level, the leaves by trial division). Item 23 is audited in part (its certificates; the deduction read in part); items 11–13, 15 and 20 use imported material (Section 8); the remaining 17 items, including these five, are located. The results bench R01–R10 and the arguments X1–X7 are checked in 43a_ §3; one source passage error (X5: $289^3 \equiv 169$, not -1 , modulo 840) is recorded by the workbench itself.

5.2 The $(3, 4, \infty)$ monodromy covers

A package of the workbench (ES-09) imports three integer matrices A_1, A_2, A_∞ with $A_1^3 = A_2^4 = A_1 A_2 A_\infty = I$ from the S^6 manuscript [1], written out in the package's `src/core.tex`, and nothing else from it. They are the duals $A_j = (T_j^{-1})^T$ of the local monodromies T_1, T_2 of the S^6 period system, acting on the dual lattice $\Lambda = \text{Hom}(V, \mathbb{Z})$ with dual basis $(\hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta})$ [18]. The package uses the induced affine action on the hyperplane

$$H_D = \{\lambda \in \Lambda/D\Lambda : \text{the } \hat{\gamma}\text{-coordinate of } \lambda \text{ is } 1\} \cong (\mathbb{Z}/D)^3,$$

and builds from its orbits covers of the projective line branched at three points. It presents the cover as a complete failure object: the equation fails at p if and only if a denominator-labelled cover has no section. That equivalence is correct, and it is Theorem 3.1 in cover-theoretic form: a section exists exactly when the tail denominator $R_a/\gcd(R_a, 4u + 1)$ or $R_a/\gcd(R_a, u + a)$ equals 1.

For odd levels D , which is the package's standing hypothesis and all that the application to the equation uses (there D divides the odd number R_a), the package's orbit and genus statements are correct. The action is transitive when $3 \nmid D$, and the cover is then connected of genus $(5D^3 - 12D^2 - 17D + 24)/24$. When $3 \mid D$ it has two orbits, of sizes $D^3/9$ and $8D^3/9$, which give components of genera $(5D^3 - 12D^2 - 81D + 216)/216$ and $(5D^3 - 12D^2 - 9D + 27)/27$. At even levels these statements do not hold as stated: at $D = 2$ the orbits have sizes 2 and 6 (found by the referee of the audit notes).

Status. Audited: the proof read; the relations, the orbits and the genera recomputed from cycle counts for $D = 1, 3, \dots, 15$. The equivalence with solvability is a re-encoding of Theorem 3.1.

The orbits at every level are determined by one invariant. Write $\lambda = \hat{\gamma} + b\hat{u} + c\hat{w} + d\hat{\delta} \in H_D$ and $e = \gcd(D, 6)$.

Theorem 5.1 (the orbits at every level). *Let the monodromy group $G = \langle T_1, T_2 \rangle$ act on H_D dually, through A_1, A_2 . For every $D \geq 1$ its orbits coincide with those of the full stabilizer $G_{\mathbb{Z}}$ of the isotropic vector γ in the symplectic group of the period lattice, acting in the same way. The orbit of λ is*

$$\{\lambda' \in H_D : \gcd(b', c', e) = \gcd(b, c, e)\},$$

of size $D^3 \cdot \#\{v \in (\mathbb{Z}/e)^2 : \gcd(v, e) = \gcd(b, c, e)\}/e^2$.

Status. New in version 4. Proved in the companion paper on the S^6 period system [18, Theorem 6.1], which describes G as the kernel of a character Φ of order 12 on $G_{\mathbb{Z}}$; the orbit argument is repeated below. Checked

as full orbit partitions for every $D \leq 14$ by the referee script `ver52_referee_claims.py`, for every $D \leq 24$ by the referee’s own scripts, and for $D = 2^k, k \leq 5$, by `es09_even_levels.py`.

Proof. We use from [18, §§2–3]: every element of $G_{\mathbb{Z}}$ is uniquely $L_M h(\alpha, \beta, s)$ with $M \in SL_2(\mathbb{Z})$ and $h(\alpha, \beta, s)$ in a Heisenberg group of integer triples; G is the kernel of the homomorphism $\Phi(L_M h(\alpha, \beta, s)) = \text{ab}(M) + s - 6(\alpha^2 + \alpha\beta + \beta^2) \pmod{12}$, where ab is the abelianisation of $SL_2(\mathbb{Z})$ onto $\mathbb{Z}/12$ with $\text{ab}\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) = 1$; and $G_{\mathbb{Z}} = G\langle t_{\gamma} \rangle$, where the transvection $t_{\gamma} = h(0, 0, 1)$ is central with $\Phi(t_{\gamma}) = 1$.

The dual action. The dual action of $h(\alpha, \beta, s)$ is $(b, c, d) \mapsto (b + 6\beta, c - 6\alpha, d - s - \alpha b - \beta c)$, and L_M acts on (b, c) by $M^{-\top}$ and fixes d .

The orbits of $G_{\mathbb{Z}}$. The coordinate d can be moved freely, by s , and (b, c) matters only modulo $6(\mathbb{Z}/D)^2$. Since $(\mathbb{Z}/D)^2/6(\mathbb{Z}/D)^2 \cong (\mathbb{Z}/e)^2$ and $SL_2(\mathbb{Z})$ maps onto $SL_2(\mathbb{Z}/e)$, the $G_{\mathbb{Z}}$ -orbits correspond to the $SL_2(\mathbb{Z}/e)$ -orbits on $(\mathbb{Z}/e)^2$, which are classified by the content $\gcd(v, e)$.

G has the same orbits. Since G is normal and $G_{\mathbb{Z}} = G\langle t_{\gamma} \rangle$ with t_{γ} central, it suffices to show $t_{\gamma}\lambda \in G\lambda$ for every λ ; if this holds at λ , it holds at $k\lambda$ for every $k \in G_{\mathbb{Z}}$, because $t_{\gamma}k\lambda = kt_{\gamma}\lambda \in kG\lambda = Gk\lambda$. Every point is $G_{\mathbb{Z}}$ -equivalent to one with $c = 0$, because $SL_2(\mathbb{Z})$ maps onto $SL_2(\mathbb{Z}/D)$, which moves every vector of $(\mathbb{Z}/D)^2$ to one of the form $(g, 0)$. At a point with $(b, c) = (g, 0)$, take $M = (T^{\top})^{-1}$ with $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then $M^{-\top} = T$ fixes $(g, 0)$, so L_M fixes λ ; and $M = STS^{-1}$ with $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, so $\Phi(L_M) = \text{ab}(T) = 1$. Hence $t_{\gamma}L_M^{-1}$ lies in G and moves λ exactly as t_{γ} does. $\square$

Corollary 5.2. *For $D = 2^k, k \geq 1$, there are exactly two orbits, of sizes $D^3/4$ and $3D^3/4$. For odd D there is one orbit if $3 \nmid D$, and there are two, of sizes $D^3/9$ and $8D^3/9$, if $3 \mid D$. For $6 \mid D$ there are four orbits; at $D = 6$ their sizes are 6, 18, 48 and 144, and at $D = 12$ they are 48, 144, 384 and 1152. In general the orbits at $D = 2^k m$ with m odd are the products of those at 2^k and at m .*

Proof. Count the vectors of $(\mathbb{Z}/e)^2$ by content. For $e = 2$ the contents 2 and 1 occur 1 and 3 times; for $e = 3$, 1 and 8 times; for $e = 6$, the contents 6, 3, 2, 1 occur 1, 3, 8, 24 times. Multiply by D^3/e^2 . The product statement is the Chinese remainder theorem, since $e = \gcd(2^k, 2) \gcd(m, 3)$. $\square$

At the levels $D = 2^k$ with $1 \leq k \leq 5$ the genera of the two components are $(0, 0)$, $(1, 3)$, $(17, 53)$, $(177, 537)$ and $(1569, 4721)$ (found by the referee pass of version 1 and recomputed here from cycle counts). By Theorem 5.1 the components of these covers at every level are indexed by the invariant $\gcd(b, c, \gcd(D, 6))$; their genera at all levels are Question 9.

6 The author’s notes, April–June 2026

At the author’s request this paper also treats the author’s notes on the equation, written from April to June 2026. There are two collections:

- one of twenty $\text{\TeX}$ notes and one 52-page paper;
- the Erdős–Straus part of a second collection, about fifty-five further notes, among them the source note of the terminal core.

They are cited by title and theorem number.

The notes were read in six independent passes by separate Claude instances. Each pass tabulated every numbered statement and checked it with its own programs. Every result proved in this section was then re-derived and re-checked with this paper’s own programs (Appendix B). The statements that rest only on a reading pass, on the referee of version 3, or on a cited source say so. Drafts that the author’s latest record supersedes are treated only for the results they contain that were verified.

What the notes contain. Their Erdős–Straus arithmetic consists of the shell calculus of Section 3 and the classical identity families of Rosati, Yamamoto, Elsholtz–Tao, Salez and Lopez. To it they attach structural layers, which are correct where they were checked:

- the Niemeier lattice $A_5^4 D_4$, which is also archive Theorem 25.8, and its umbral group $GL_2(3)$, the automorphism group of the glue code (checked by a reading pass);
- the Hauptmodul of $\Gamma_0(6)$ and the Monster class $6B$;
- Ogg’s primes;
- Lorentzian and Apollonian geometry;
- Cayley–Dickson algebras;
- the split-zero semiring, where one isomorphism needs an additional relation (Proposition 8.1);
- mock theta functions, where the notes’ identification with the minimal model $M(2, 7)$ holds in the form of Theorem 6.15.

None of these structures takes a prime, a shell or a certificate as input, so a statement about solvability drawn from them needs a typed map from the shell data to them; Section 8 states, for each, what is established about such a map.

What this section records.

- what the notes add to Sections 3 and 4: the exact form of the terminal core, a sieve of quadratic-residue conditions on a counterexample, and facts about single shells, identities and locks;
- corrections to the notes, each with its proof or counterexample;
- the mock theta functions of the notes, with the theorem that identifies their shadow with the modular representation of $M(2, 7)$.

6.1 The terminal core

Put $M_{23} = 840 \cdot 11 \cdot 19 \cdot 23 = 4,037,880$. A *base class* is a class $n \equiv r \pmod{840}$ with $r \in S_{840} = \{1, 121, 169, 289, 361, 529\}$, together with units modulo 11, 19 and 23. There are 23,760 base classes. The source note, *Finite central-cover reductions for residual Erdős–Straus shells*, calls (R, m, u) an *elementary $j = 1$ certificate* when:

- $R \equiv 3 \pmod{4}$ and $u \mid m^2$;
- $n \equiv -R \pmod{4m}$, $n \equiv -4u \pmod{R}$, and $\gcd(n, R) = 1$.

Then $m \mid a = (n + R)/4$, $u \mid a^2$ and $u \equiv -a \pmod{R}$, so the shell is occupied through the middle channel M_a . The primes that may divide m are those of $\{2, 3, 5, 7, 11, 19, 23\}$ not dividing R whose divisibility of a the class forces.

The note makes two computations:

- (i) The 32 shells $R \mid M_{23}$ with $R \equiv 3 \pmod{4}$ cover all but 4951 base classes. This is a covering statement.
- (ii) Over all R built from 3, 5, 7, 11, 19, 23, it tests the certificate congruence modulo $\gcd(M_{23}, R)$, so that a class counts as hit when some lift of it modulo $\text{lcm}(M_{23}, R)$ is certified. Exactly 2970 classes are then never hit, namely $S_{840} \times Q_{11} \times Q_{19} \times Q_{23}$, where Q_q is the group of non-zero squares modulo q .

The note calls this set the *support-zero core*. It uses the set to bound from below the residue of any obstruction built from such certificates.

Lemma 6.1. *Let (R, m, u) be an elementary $j = 1$ certificate with $\gcd(m, R) = 1$, and let n satisfy its congruences. Put $\varepsilon_\ell(n) = (n/\ell)$ for odd primes ℓ . Put $\varepsilon_2(n) = 1$ if $n \equiv 1 \pmod{8}$ and $\varepsilon_2(n) = -1$ if $n \equiv 5 \pmod{8}$. Then the Jacobi symbol satisfies*

$$\left(\frac{n}{R}\right) = - \prod_{\ell|u} \varepsilon_\ell(n)^{v_\ell(u)}.$$

In particular no n in the class is a unit square modulo $\text{lcm}(4m, R)$.

Proof. From $n \equiv -4u \pmod{R}$ we get $(n/R) = (-1/R)(4/R)(u/R) = -(u/R)$.

For an odd prime $\ell \mid m$, reciprocity with $R \equiv 3 \pmod{4}$ gives $(\ell/R) = (-R/\ell)$. This equals (n/ℓ) because $n \equiv -R \pmod{\ell}$.

If $2 \mid m$, then $n \equiv -R \pmod{8}$. So $(2/R) = 1$ exactly when $R \equiv 7 \pmod{8}$, that is, when $n \equiv 1 \pmod{8}$.

If n were a square modulo $\text{lcm}(4m, R)$, every $\varepsilon_\ell(n)$ with $\ell \mid m$ would be 1. Then $(n/R) = -1$, which is impossible for a square. $\square$

Theorem 6.2 (the terminal core). *(a) A base class has no elementary $j = 1$ certificate with R built from 3, 5, 7, 11, 19, 23 on any lift if and only if it is a square modulo 11, 19 and 23. So the support-zero core is exactly $S_{840} \times Q_{11} \times Q_{19} \times Q_{23}$, which has 2970 classes.*

(b) The base classes that no single identity among Salez’s seven modular equations [13] with modulus dividing M_{23} covers number 3520. They are the 2970 square classes and 550 classes that are non-squares modulo 11, 19 or 23.

(c) So the classes left open at level M_{23} strictly contain the core. For example the prime $3361 \equiv 1 \pmod{840}$, $\equiv 6 \pmod{11}$, lies outside the core, and none of the 64,149 families with modulus dividing M_{23} covers its class. The primes 18,481 and 2,840,041 behave in the same way.

Status. (a) The *if* direction is proved here. The *only if* direction is a finite computation: the source note’s script and an independent program of this paper both give 4951 and then 2970. (b) A finite computation, calibrated at Salez’s modulus $G_7 = 840 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23$, where it leaves 147,348 classes, the number Salez reports. Salez states that his seven equations form a complete set of modular equations. The primes in (c) are of course solvable, for instance $4/3361 = 1/841 + 1/110139970 + 1/950330$.

Proof. (a), *if*. Let n lie over a square class.

- S_{840} is the group of unit squares modulo 840, so n is a quadratic residue modulo every prime $\ell \in \{3, 5, 7, 11, 19, 23\}$.
- Also $n \equiv 1 \pmod{8}$.
- So $(n/R) = 1$ for every R built from these primes.
- The primes that may divide m lie in $\{2, 3, 5, 7, 11, 19, 23\}$, so Lemma 6.1 forces $(n/R) = -1$ for any certificate.

Hence no lift is certified. The remaining statements are the computations in the status line. $\square$

Remark 6.3. The notes’ later papers work with the core as the set of hard classes that remain: the 52-page *Terminal residual coordinates for the Erdős–Straus problem* (26 May 2026) and its June replacement *Finite support algebra, cubic middle geometry, and orientation descent*. Theorem 6.2 locates the core inside the set that single identities leave open. At level M_{23} that set has 3520 classes. The core is the part of it that quadratic reciprocity forces, and the 550 further classes are the part that remains from using one identity per class.

In the vocabulary of the split-zero semiring (Section 8.1):

- the 2970 core classes are *unsupported*;
- the 550 further classes are *supported but not covered*: some lift of each is certified, but no identity of Salez’s seven families holds on the whole class.

Salez’s statement that his seven equations are complete concerns identity families of his linear type; the archive’s audit of Salez (§13.2) records this scope. Whether combinations of identities cover the 550 classes at some finite level is Question 4.

By Dirichlet’s theorem the core carries $2970/\varphi(M_{23}) = 1/256$ of all primes, and the classes left open at level M_{23} carry $3520/760,320 = 1/216$.

6.2 Quadratic-residue conditions on a counterexample

Throughout this subsection, $p \equiv 1 \pmod{24}$ is prime. A certificate is a shell $R \equiv 3 \pmod{4}$ with $\gcd(R, pa) = 1$ and a divisor $u \mid a^2$ with $R \mid 4u + 1$ (channel E) or $R \mid u + a$ (channel M). Theorem 3.1 then gives the solution.

Theorem 6.4 (eight shifts). *Let q be an odd prime dividing*

$$F(p) = (p+1)(p+2)(p+3)(p+4)(p+7)(p+8)(2p+1)(3p+1)$$

with $(p/q) = -1$. Then the shell and divisor of Table 2 give a certificate. Equivalently, a counterexample $p \equiv 1 \pmod{24}$ is a quadratic residue modulo every odd prime factor of $F(p)$.

Table 2: The certificates of Theorem 6.4, with $a = (p + R)/4$.

shift	q with $(p/q) = -1$	shell R	u	channel
$p+1$	$q \equiv 3 \pmod{4}$	q	a	E
$p+4$	$q \equiv 3 \pmod{4}$	q	1	M
$p+2$	$q \equiv 5, 7 \pmod{8}$	q if $q \equiv 7$, $3q$ if $q \equiv 5 \pmod{8}$	$a/2$	E
$p+8$	$q \equiv 5, 7 \pmod{8}$	the same	2	M
$2p+1$	$q \equiv 5, 7 \pmod{8}$	the same	$2a$	E
$p+3$	$q \equiv 2 \pmod{3}$	3	q	E
$p+7$	$q \equiv 5, 6, 3 \pmod{7}$	7	$q, aq, 2aq$	E, M, M
$3p+1$	$q \equiv 2 \pmod{3}$	$(4q+1)/3$	q	E

Status. Proved here.

- *Recorded rows.* $p+1$ and $p+4$ are archive Theorem 24.50. $p+3$ and $p+7$ are Propositions 3.2 and 3.3, read through reciprocity. $3p+1$ is archive Theorem 24.171 with $k=3$.
- *Salez’s charts.* Each of the rows $p+1, p+2, 2p+1, p+4, p+8$ is the union, over the free divisor R , of one of Salez’s charts [13] with small fixed constants: (15b) with $(B, C) = (1, 1), (1, 2), (2, 1)$ and (14c) with $(B, D) = (1, 1), (1, 2)$.
- *Not in the archive.* The quadratic-residue conditions from $p+2$ and $2p+1$, and the stated condition from $p+8$, are not stated in the archive, in versions 1–2 of this paper, or in 43_/43a_. No literature search was made.
- *Checks.* Every certificate for every prime $p \equiv 1 \pmod{24}$ below 10^7 was built and verified: 780,700 of them, none bad. 247 primes survive all eight shifts.

Proof. The characters. For $q \mid p+k$ with $q \nmid k$ we have $(p/q) = (-k/q)$; similarly $(p/q) = (-2/q)$ for $q \mid 2p+1$ and $(p/q) = (-3/q)$ for $q \mid 3p+1$. These characters are -1 exactly on the classes of q listed in the table; for $p+7$ this uses $(-7/q) = (q/7)$.

Each row is a certificate. For each row the stated divisibility is an identity:

- $4a + 1 = p + R + 1$;
- $4(a + 1) = p + R + 4$;
- $2(2a + 1) = p + R + 2$;
- $4(a + 2) = p + R + 8$;
- $8a + 1 = 2(p + R) + 1$;
- $4q + 1 \equiv q + 1 \pmod{3}$;
- for $p + 7$: $4q + 1 \equiv 0$, $a(q + 1)$ and $a(2q + 1)$ modulo 7.

The rows $p + 2$, $p + 8$ and $2p + 1$. Since $p \equiv 1 \pmod{3}$, 3 divides each of these shifts, so R divides the shift. Here $q \neq 3$ because $(p/3) = 1$. Also $R \equiv 7 \pmod{8}$, so $8 \mid p + R$ and a is even.

The row $p + 7$. $8 \mid p + 7$, so a is even, and $q \mid a$.

The row $3p + 1$. Put $N_3 = (3p + 1)/4$, which is odd and $\equiv 1 \pmod{3}$. Then $h = N_3/q \equiv 2 \pmod{3}$. With $a = q(h + 1)/3$ one checks $4a - p = (4q + 1)/3 = R$.

Coprimality. A common prime of R and a divides $4a - R = p$, and $p \nmid R$ in every row. $\square$

At the strict record $p_* = 8,803,369$, whose least occupied shell is 107, two shifts give four certificates:

- $p_* + 2 = 3 \cdot 223 \cdot 13159$, where $223 \equiv 7$ and $13159 \equiv 7 \pmod{8}$, giving the shells 223 and 13,159;
- $2p_* + 1 = 3 \cdot 677 \cdot 8669$, where $677 \equiv 5$ and $8669 \equiv 5 \pmod{8}$, giving the shells 2031 and 26,007.

None of the other six shifts gives a certificate at p_* . The least primes certified by $p + 2$ alone, by $2p + 1$ alone and by $p + 8$ alone are 2521, 9601 and 33,961.

Theorem 6.5 ($p+5$). *Let $N = (p+5)/2$, and let $p \equiv 9 \pmod{40}$, or $p \equiv 121 \pmod{840}$, or $p \equiv 1,361 \pmod{840}$ with $p \equiv 4 \pmod{9}$. Suppose some prime $q \mid p + 5$ has $(p/q) = -1$. Then N has a divisor $d \equiv -p \pmod{20}$ or a divisor $h \equiv -1 \pmod{20}$, and either gives a certificate by Proposition 6.14 with $\mu = 5$.*

So a counterexample in these classes, which contain the hard classes 121, 169, 289 and 529, is a quadratic residue modulo every odd prime factor of $p + 5$.

Status. Proved here. The case $p \equiv 9 \pmod{40}$ is in the note *Full divisor locks, QR subtori, and survivor idempotents* in substance (its Theorem 3.7(ii), Corollary 3.8(ii) and Theorem 4.2). The extension to the classes 121, 1 and 361 was found by a reading pass. The theorem was checked on all 41,419 primes $p \equiv 9 \pmod{40}$ below 10^7 and on 5688 primes of the other classes. At $p = 12601 \equiv 1 \pmod{840}$, with $p \equiv 1 \pmod{9}$, it fails: $p + 5 = 2 \cdot 3 \cdot 11 \cdot 191$ has two non-residue factors, yet N has no divisor in either class.

Proof. For $q \mid N$ we have $(p/q) = (-5/q)$. This equals 1 exactly when $q \equiv 1, 3, 7, 9 \pmod{20}$, a subgroup H of index 2.

The case $p \equiv 9 \pmod{40}$. Here $N \equiv 7 \pmod{20}$, and the targets are 11 and 19. Let $q \notin H$ divide N :

- if $q \equiv 11$ or 19 , then q itself is the divisor;
- if $q \equiv 13$, then $N/q \equiv 19$;
- if $q \equiv 17$, then $N/q \equiv 11$.

The case $p \equiv 1 \pmod{40}$. Here $N \equiv 3 \pmod{20}$, and a divisor $\equiv 17$ or 19 suffices, because its cofactor is then $\equiv 19$. Let $q \notin H$ divide N :

- $3 \mid N$ always, so $q \equiv 13$ gives $3q \equiv 19$;
- $7 \mid N$ when $p \equiv 121 \pmod{840}$, so $q \equiv 11$ gives $7q \equiv 17$;

- $9 \mid N$ when $p \equiv 4 \pmod{9}$, so $q \equiv 11$ gives $9q \equiv 19$.

Conversely, the targets lie outside H , so a divisor in a target class has a prime factor outside H . $\square$

Theorem 6.6 (the visible-box principle). *Let $R \equiv 3 \pmod{4}$ be prime and c a class of hard primes modulo $10080 = 2^5 \cdot 3^2 \cdot 5 \cdot 7$. Let*

$$\text{Box}(c) = \left\{ \prod_{f \in \{2,3,5,7\}} f^{i_f} \pmod{R} : 0 \leq i_f \leq 2e_f \right\},$$

where e_f is the least value of $v_f((p+R)/4)$ that c forces. If $\text{Box}(c)$ contains every non-zero square modulo R , then for every prime $p \in c$ the shell R is occupied exactly when some odd prime $q \mid p+R$ has $(p/q) = -1$.

Status. Proved here; found by a reading pass. The archive's census refinements (Theorems 24.228 and 24.233) already give certificates in the shells $R = 31, 47$ and 71 on parts of these classes; the equivalence is not stated there. The search up to $R = 700$ is complete, because $|\text{Box}(c)| \leq 7 \cdot 5 \cdot 3 \cdot 3 = 315$. Among the 72 hard classes modulo 10080 it finds 72 good classes for $R = 3$ and $R = 7$, 48 for $R = 11$, 12 for $R = 19$, 48 for $R = 23$, and 30, 20, 4, 12, 1 for $R = 31, 47, 59, 71, 311$. The last five shells are new. All 90,897 tests with $p < 10^7$ agree, with 50,545 certificates verified.

Proof. Every element of $\text{Box}(c)$ is the residue of a divisor of a^2 .

If $q \mid a$ is a prime with $(q/R) = -1$, then $(p/q) = (-R/q) = (q/R) = -1$. Also $q \cdot \text{Box}(c)$ is the non-square coset, so the divisors reach every unit, including the target -4^{-1} of channel E .

The condition $(p/R) = -1$ needs no separate treatment. Let m be the odd part of $p+R$. Since $p \equiv 1 \pmod{8}$, reciprocity gives $(p/m) = (m/p) = (R/p)(2/p)^{-v_2(p+R)} = (p/R)$. So if $(p/R) = -1$, some prime factor q of m has $(p/q) = -1$.

Conversely, suppose that no odd prime $q \mid p+R$ has $(p/q) = -1$. For an odd prime $\ell \mid a$, reciprocity gives $(\ell/R) = (-R/\ell) = (p/\ell) = 1$; the prime 2 divides a only when $R \equiv 7 \pmod{8}$, and then $(2/R) = 1$. So every prime factor of a is a square modulo R , and Proposition 3.6 shows that the shell is empty. $\square$

Proposition 6.7 ($\mu = 17$). *Let $p \equiv 361$ or $529 \pmod{840}$, $p \equiv 1 \pmod{9}$ and $p \pmod{17} \in \{8, 15, 16\}$. Then $(p+17)/2$ has a divisor $d \equiv -p$ or $h \equiv -1 \pmod{68}$, giving a certificate by Proposition 6.14, if and only if some prime $q \mid p+17$ has $(p/q) = -1$.*

Status. Proved here; found by a referee pass. Checked on all 433 such primes below 10^7 .

Proof. Put $N = (p+17)/2$. Then $63 \mid N$. The character $(p/q) = (-1/q)(q/17)$ lives modulo 68, and both targets $-p$ and -1 lie outside its kernel. A finite check of the six classes shows that for every non-kernel class r the sets $r \cdot W$ and $N(rW)^{-1}$, with $W = \{1, 3, 7, 9, 21, 63\}$, meet the targets modulo 68. The converse holds as in Theorem 6.5. $\square$

The primes $p \equiv 1 \pmod{24}$ that satisfy the conditions of Theorems 6.4, 6.5 and 6.6 and Proposition 6.7 are counted in Table 3.

Table 3: Primes $p \equiv 1 \pmod{24}$ satisfying the conditions of Theorems 6.4, 6.5 and 6.6 and Proposition 6.7, cumulatively. The column 10^8 is from a referee pass.

conditions	below 10^6	below 10^7	below 10^8
hard classes	2370	20,513	179,468
and the eight shifts	54	247	1431

and $p + 5$ on its classes	29	146	816
and $\mu = 17$	29	145	806
and the visible-box shells	7	31	165

The first primes that satisfy all the conditions of the table are 43,201, 65,521 and 196,561.

The referee of version 3 found certificates of three further shapes, on parts of the hard classes. Let $R \equiv 3 \pmod{4}$ with $a = (p + R)/4$ and $\gcd(R, pa) = 1$, and let $s, t \geq 1$.

- $u = s$ is an M -certificate when $s \mid a^2$ and $R \mid p + 4s$, since $4(s + a) = p + 4s + R$.
- $u = a/t$ is an E -certificate when $t \mid a$, $\gcd(t, R) = 1$ and $R \mid p + t$, since $t(4a/t + 1) = p + R + t$.
- $u = ta$ is an E -certificate when $t \mid a$ and $R \mid tp + 1$, since $4ta + 1 = t(p + R) + 1$.

Applied to the shifts $p + 6$ ($t = 6$), $p + 12$, $p + 16$, $p + 20$, $p + 24$, $p + 36$ ($s = 3, 4, 5, 6, 9$), $5p + 1$ and $6p + 1$ ($t = 5, 6$), they give conditions of the same kind as those above. The referee verified 46,441 such certificates below 10^7 , and its script was re-run here. Below 10^7 , exactly 23 primes $p \equiv 1 \pmod{24}$ have no certificate of any of the kinds of this subsection. The prime 43,201, for example, has one from $p + 24$: $4/43201 = 1/10914 + 1/1036824 + 1/1885982856$.

The survivors are rare, and already the first condition removes almost all primes.

Proposition 6.8 (density zero). *The primes $p \equiv 1 \pmod{24}$ that satisfy the condition of the row $p + 1$ of Theorem 6.4 have relative density zero among the primes $p \equiv 1 \pmod{24}$. A fortiori, so do the primes that satisfy all the conditions of this subsection.*

Status. Proved here; new in version 4.

Proof. If a prime $q \equiv 3 \pmod{4}$ divides $p + 1$, then $(p/q) = (-1/q) = -1$, so the row $p + 1$ gives a certificate. Hence a prime satisfying the condition has $p \not\equiv -1 \pmod{q}$ for every prime $q \equiv 3 \pmod{4}$. (The prime $q = 3$ never divides $p + 1$ when $p \equiv 1 \pmod{24}$.) Let $q_1, \dots, q_k$ be the first k primes $\equiv 3 \pmod{4}$ greater than 3. By Dirichlet's theorem for the progressions modulo $24q_1 \cdots q_k$, the primes $p \equiv 1 \pmod{24}$ with $p \not\equiv -1 \pmod{q_i}$ for all $i \leq k$ have relative density $\prod_{i \leq k} (1 - 1/(q_i - 1))$ among the primes $p \equiv 1 \pmod{24}$. Since $\sum 1/q$ over the primes $q \equiv 3 \pmod{4}$ diverges, this product tends to 0 as k grows, and the relative upper density of the set in question is at most every such product. $\square$

Whether infinitely many primes survive all the conditions is Question 3. A negative answer would prove the conjecture at all but finitely many primes.

Remark 6.9 (the Collatz reading). For $p \equiv 1 \pmod{8}$ the number $(3p + 1)/4$ is the odd Collatz successor of p . So a counterexample $p \equiv 1 \pmod{24}$ has an odd Collatz successor built only from primes $\equiv 1 \pmod{3}$.

The odd Collatz predecessors of an odd n with $3 \nmid n$ satisfy $\Pi_{k+1} = 4\Pi_k + 1$, which is the notes' map $a \mapsto 4a + 1$. So $\Pi_k \equiv 5 \pmod{8}$ for $k \geq 1$. Every prime $P \equiv 5 \pmod{8}$ is solved by the shell $R = 3$ with $u = 2$. Hence only the first member of a predecessor chain can be a hard prime.

This is the classical Syracuse predecessor structure [7], and it links this workbench with the Collatz workbench [17]. The link uses single steps of the Collatz map.

6.3 Single shells

Proposition 6.10 (non-coprime shells). *Let p be prime and $R \equiv -p \pmod{4}$. Then $\gcd(R, pa) > 1$ exactly when $p \mid R$; write $R = kp$.*

- (a) If $p \nmid k$, the one-congruence criterion, that some $d \mid N^2$ has $d \equiv -N \pmod{R}$, is equivalent to occupancy, and every such d gives integral y and z .
- (b) If $p^2 \mid R$, it is not. At $p = 73$, $R = 7 \cdot 73^2$, the divisor $d = 4 \cdot 73^3$ satisfies the congruence and gives $y = 60$ but $z = 1920/73$, and the shell has no solution.

Status. Proved here; found by a reading pass. Four of the notes state the criterion without the coprimality hypothesis: the defect-completion article (Lemma 2.1), *Split-zero residual moonshine* (Theorem 3.1), the A_8 snowflake note (Theorem 2.1) and *Terminal-core lock covers* (Theorem 2.1). The error does not affect the notes' applications, which are at hard primes with $R < 3p^2$ (a reading pass): for $p \equiv 1 \pmod{4}$, $p^2 \mid R$ forces $R \geq 3p^2$. For $p \equiv 3 \pmod{4}$ it allows $R = p^2$.

Proof. (a) Put $m = (k + 1)/4$, so that $a = pm$ and $N = p^2m$. A divisor d with $d \equiv -N \pmod{kp}$ is divisible by p ; write $d = pd_1$, so that $k \mid d_1 + pm$.

- If $p^3 \nmid d_1$, then $d_1 \mid (pm)^2$ is a certificate for the reduced shell (k, pm) , and the formulas agree.
- If $d_1 = p^3w$ with $w \mid m^2$, then $k \mid 4p^2w + 1$, so $(w/k) = (-1/k) = -1$.
- But every prime $\ell \mid m$ has $(\ell/k) = 1$: for odd ℓ because $k \equiv -1 \pmod{\ell}$ and reciprocity applies; for $\ell = 2$ because $k \equiv 7 \pmod{8}$. So $(w/k) = 1$, a contradiction.

(b) Direct computation, including a search over all y . □

Proposition 6.11 (ramified shells). *Let (R, p) be a coprime shell.*

- (a) If $R \mid p - 1$, then $E_a = M_a$.
- (b) If $R \mid p^2 - 1$, then $u \mapsto a^2/u$ maps E_a to itself, and $|E_a|$ is odd exactly when $R \mid p + 1$.

Status. Proved here; checked on all 947 coprime shells with $R \mid p^2 - 1$ and $p \equiv 1 \pmod{4}$ below 2000.

Proof. (a) $4a \equiv p \equiv 1 \pmod{R}$, so $4(u + a) \equiv 4u + 1$.

(b) The complement sends the class -4^{-1} to $-4a^2 \equiv -p^2/4$, which is -4^{-1} again when $p^2 \equiv 1 \pmod{R}$. The only possible fixed point is $u = a$. Finally $a \in E_a$ exactly when $R \mid 4a + 1 = p + R + 1$. □

Proposition 6.12 (the pair-count series). *Let $A_R(n)$ be the number of triples (X, Y, Z) with $XYZ = n$, $\gcd(X, Y) = 1$ and $R \mid X + Y$. For a coprime shell, $A_R(pa)$ is the number of ordered certificates. For $R \geq 3$,*

$$\sum_{(n, R)=1} \frac{A_R(n)}{n^s} = \frac{1}{\varphi(R)} \sum_{\chi \bmod R} \chi(-1) \frac{\zeta_R(s) L(s, \chi) L(s, \bar{\chi})}{\zeta_R(2s)},$$

where ζ_R omits the primes dividing R .

Status. Proved here. This is Theorem 4.3 of the defect-completion article; 8757 coefficients were checked by a referee pass. No positivity statement follows.

Proof. Write $1_{X \equiv -Y} = \varphi(R)^{-1} \sum_{\chi} \chi(-1) \chi(X) \bar{\chi}(Y)$. The variable Z contributes $\zeta_R(s)$. Möbius inversion over the common divisors of X and Y contributes $L(s, \chi) L(s, \bar{\chi}) / L(2s, \chi \bar{\chi})$, and $\chi \bar{\chi}$ is principal. □

The divisor-pair note asks for finitely many central and edge templates covering S_{840} . As a congruence cover this is impossible, because no template class contains a square modulo its modulus.

- For the edge templates this is Proposition 4.6.

- A central template is the class $n \equiv -R \pmod{4c}$ with $s \mid c^2$, $\gcd(c, R) = 1$, $R \equiv 3 \pmod{4}$ and $R \mid s + c$. If $-R$ were a unit square modulo $4c$, then:

- every odd prime $\ell \mid c$ would have $(-R/\ell) = 1$, hence $(\ell/R) = 1$;
- if $2 \mid c$, then $-R \equiv 1 \pmod{8}$, so $(2/R) = 1$.

So $(c/R) = (s/R) = 1$. But $R \mid s + c$ gives $(s/R) = (-c/R) = -1$. (A reading pass found this argument.)

By Dirichlet's theorem, primes $p \equiv 1$ modulo the product of 840 and the template moduli exist, and they escape every template.

6.4 Identities, locks and types

Proposition 6.13 (Rosati's family). *Let $k \geq 1$, $c \mid k$ and $m = 4k - 1$. If $n \equiv -c \pmod{m}$, then*

$$\frac{4}{n} = \frac{1}{kn} + \frac{1}{k(n+c)/m} + \frac{1}{kn(n+c)/(mc)}.$$

Status. Proved here; Salez's chart (14a) with $(B, C, D) = (c, 1, k/c)$. The case $k = 22$, $c = 11$ solves every $n \equiv 76 \pmod{87}$. The two direct 29-channels of Terminal residual coordinates (Theorems 7.1–7.2), $n \equiv 917 \pmod{1276}$ and $n \equiv 1605 \pmod{2204}$, are correct. They are Salez's chart (15b) with $(B, C, F) = (1, 29, 11)$ and $(1, 29, 19)$. The first lies inside this family with $(k, c) = (80, 40)$, $m = 319$. The second lies inside chart (14a) with $(B, C, D) = (2, 23, 3)$, $m = 551$. The notes' claim that these are the only direct channels holds only for lock divisors that are single primes from $\{5, 11, 19, 23\}$. The lock with $\mu = 5$ and $d = 31$ solves $n \equiv 429 \pmod{620}$, a class containing the core prime 33,289.

Proof. Over the common denominator $kn(n+c)$ the three terms have numerators $n+c$, mn and mc , which add up to $(1+m)(n+c) = 4k(n+c)$. The denominators are integers because $m \mid n+c$ and $c \mid k$. $\square$

Proposition 6.14 (the full-divisor locks). *Let p be an odd prime, μ odd and $N = (p + \mu)/2$.*

- If $d \mid N$ is odd with $d \equiv -p \pmod{4\mu}$, then the shell $R = d$ carries the E-certificate $u = a/\mu$.*
- If $h \mid N$ with $h \equiv -1 \pmod{4\mu}$, put $e = N/h$ and $t = (h+1)/(4\mu)$. Then $a = 2\mu et$ and $R = \mu + 2e$ satisfy $4a - p = R$, and $u = \mu^2 t$ is an M-certificate.*

Status. Proved here. These are the locks of the notes, which a reading pass identifies with the faces $A = 1$ of Type I and $C = 1$ of Type II of [4]; lock (b) is Lopez's Type B [8]. In the notes' proof of (a) the numerator should read $ps + p + \mu$, not $ps + p + \mu s$.

Proof. (a) Here $\mu \mid a$, and $4u + 1 = (p + d + \mu)/\mu$ is divisible by d .

(b) $4a - p = 2e(h+1) - 2eh + \mu = R$ and $u + a = \mu tR$. $\square$

Computations on these certificates.

- Every hard prime below 10^7 has a lock certificate. A reading pass found the same below 10^8 , where the hard primes number 179,468. Not every prime has one: $409 \equiv 1 \pmod{24}$, which is not hard, has none for any odd μ , by an exhaustive search. It is the only prime $p \equiv 1 \pmod{4}$ below $3 \cdot 10^6$ without one (found by the referee of this version).
- Every prime $p \equiv 1 \pmod{24}$ below 10^7 has a Type II solution, that is, an M-certificate, in a shell $R \leq 107$. The record p_* is the only one that needs 107. A reading pass found the same below 10^8 .
- The complement-square certificates $u = Q^2$ with $Q \mid a$ (Lopez's Type C [8]) are not universal. The least prime $p \equiv 1 \pmod{24}$ without one is 97, and the least hard prime without one is 3361.

6.5 Corrections

Besides the scope of the one-congruence criterion (Proposition 6.10), the numerator in the notes' proof of lock (a) (Proposition 6.14) and the scope of the direct 29-channels (Proposition 6.13), the checked corrections are the following. Each is stated with its proof or counterexample.

- *Star–Kneser positivity* states a converse of Star–Kneser forcing. The archive records that it fails at p_* , $R = 107$, and retracts it (§15.3; located).
- Four notes identify the umbral group of $A_5^4 D_4$, the automorphism group of the glue code, with $GL_2(3)$, by an argument that does not exclude the binary octahedral group. The conclusion holds: $GL_2(3)$ has 13 involutions, while the binary octahedral group has exactly one element of order 2 (a reading pass, and `misc_checks.py`).
- A decomposition of the shell count over the six hard classes, of the kind the notes call a Coxeter–Fourier decomposition, needs data finer than the class of p modulo 840, because the count is not a function of that class. With A_R as in Proposition 6.12 and $a = (p + 11)/4$, $A_{11}(pa)$ takes the values 0, 4, 6, 8, 10 at the primes 2521, 5881, 4201, 15121 and 13441, all $\equiv 1 \pmod{840}$ (`misc_checks.py`).
- The map $(\mathbb{Z}/18)^\times \rightarrow S_{840}$ of the terminal papers is a bijection and not a homomorphism; the decomposition that uses it holds as stated (Proposition 8.2).
- One isomorphism of the split-zero dossier, its Theorem 7.2, needs the relation $[\tau] \sim 0$ (Proposition 8.1).
- The Kneser theorem used is that of [6], of 1955, not the density paper of 1953.

6.6 The mock theta functions of the notes and the minimal model $M(2, 7)$

The notes attach to the cubic middle of the shell problem an explicit rank-three modular object. They follow the framework of 3d modularity [2] and its revision [23]. There the $\widehat{Z}$ invariants of plumbed 3-manifolds are quantum modular, and their companions under orientation reversal are mock or mixed mock modular. For the Brieskorn sphere $\Sigma(2, 3, 7)$ the relevant objects are Ramanujan's order-7 mock theta functions [3, 23].

The note *An explicit mixed-mock Virasoro support realization* works with $m = 42$ and the Atkin–Lehner group $K = \{1, 6, 14, 21\}$. It computes the projector $P_7 = P^+(6)P^+(14)P^+(21)P^-(42)$ on $\mathbb{C}[\mathbb{Z}/84]$ and its image basis

$$\begin{aligned} v_1 &= e_1 - e_{13} - e_{29} + e_{41} - e_{43} + e_{55} + e_{71} - e_{83}, \\ v_5 &= e_5 + e_{19} + e_{23} + e_{37} - e_{47} - e_{61} - e_{65} - e_{79}, \\ v_{11} &= e_{11} + e_{17} + e_{25} + e_{31} - e_{53} - e_{59} - e_{67} - e_{73}, \end{aligned}$$

and writes regularised indefinite theta functions whose shadows lie on these three vectors.

First, the explicit mixed mock functions of the note are Ramanujan's order-7 mock theta functions, in Zagier's vector M_7 , whose components are $q^{-1/168}F_0$, $q^{-25/168}F_1$ and $q^{47/168}F_2$ up to signs [15]. The q -expansions agree through q^{40} , up to one sign (a reading pass). For $m = 30$ the same construction gives the order-5 functions. The first order-7 function is the invariant $q^{1/2}\widehat{Z}_0$ of the Brieskorn sphere $-\Sigma(2, 3, 7)$ [3, eq. (3.20)]. So the notes' construction lands exactly on the mock theta functions of 3-manifold invariants.

Second, the note identifies the shadow of this vector with the characters of the minimal model $M(2, 7)$, whose central charge is $c = -68/7$ and whose conformal weights are $h = 0, -\frac{2}{7}, -\frac{3}{7}$. The following theorem establishes this identification at the level of modular representations.

For $r \in \{1, 5, 11\}$ put

$$G_r(\tau) = \sum_{\ell \in \mathbb{Z}} v_r(\ell \bmod 84) \ell q^{\ell^2/168},$$

the weight- $\frac{3}{2}$ unary theta series on the three support vectors. For a vector F of weight k define matrices $\rho_F(S)$, $\rho_F(T)$ by $F(-1/\tau) = (-i\tau)^k \rho_F(S) F(\tau)$ and $F(\tau + 1) = \rho_F(T) F(\tau)$. Let $\rho = \rho_G$, and let $\rho_{2,7}$ be the representation of $SL_2(\mathbb{Z})$ on the characters $\chi_s = \chi_{1,s}^{(2,7)}$, $s = 1, 2, 3$, given by the Rocha-Caridi formula [32]. Let ε be the multiplier of Dedekind's η , so that $\varepsilon(S) = 1$ and $\varepsilon(T) = e(1/24)$ in this convention, and write $c_k = (2/\sqrt{7}) \sin(k\pi/7)$.

Theorem 6.15 (the order-7 shadow carries the $M(2, 7)$ representation). *In the basis (G_1, G_5, G_{11}) ,*

$$\rho(S) = \begin{pmatrix} -c_1 & c_2 & c_3 \\ c_2 & c_3 & -c_1 \\ c_3 & -c_1 & c_2 \end{pmatrix}, \quad \rho(T) = \text{diag}(e(\frac{1}{168}), e(\frac{25}{168}), e(\frac{121}{168})).$$

Let P be the signed permutation sending (G_1, G_5, G_{11}) to $(\chi_2, \chi_3, -\chi_1)$. Then

$$\overline{\rho(S)} = P\rho_{2,7}(S)P^{-1}, \quad \overline{\rho(T)}e(1/8) = P\rho_{2,7}(T)P^{-1}.$$

Hence $\bar{\rho} \otimes \varepsilon^3 \cong \rho_{2,7}$: the shadow of the order-7 mock theta vector, conjugated and twisted by the character of η^3 , transforms exactly as the characters of $M(2, 7)$.

Status. Proved here; new in version 4. The S -identity is verified exactly in $\mathbb{Q}(\zeta_{168})$ and numerically to 38 digits (`exact_shadow_reps.py`, `m7_vs_M27.py`). Whether this isomorphism is stated in the literature was not checked.

Proof. The S -matrix of ρ . The theta decomposition gives

$$\theta_{m,\mu}(-1/\tau, z/\tau) = \sqrt{\tau/2mi} e(mz^2/\tau) \sum_{\nu} e(-\mu\nu/2m) \theta_{m,\nu}(\tau, z)$$

[27, §5]. Differentiating in z at $z = 0$ gives

$$\theta_{m,\mu}^1(-1/\tau) = (-i\tau)^{3/2} \frac{i}{\sqrt{2m}} \sum_{\nu} e(-\mu\nu/2m) \theta_{m,\nu}^1(\tau)$$

for the derivative series $\theta_{m,\mu}^1$. The span of v_1, v_5, v_{11} is the image of the projector P_7 , whose factors are Atkin–Lehner involutions; these commute with the finite Fourier transform above [27, 24], so the transform of each G_a lies in the span of G_1, G_5, G_{11} , as the direct evaluation below also confirms. Since the vectors v_r are odd with disjoint supports, $\rho(S)_{ab} = (i/\sqrt{84}) \sum_r v_a(r) e(-rb/84)$. This formula agrees with a direct evaluation of the series at sample points to 29 digits.

The S -matrix of $\rho_{2,7}$. The Rocha-Caridi formula gives $\rho_{2,7}(S)_{s\sigma} = (2/\sqrt{7})(-1)^{s+\sigma} \sin(2\pi s\sigma/7)$.

The S -identity. Both sides lie in $\mathbb{Q}(\zeta_{168})$, using $\sqrt{7} = -i \sum_{k=1}^6 (\frac{k}{7}) \zeta_7^k$ and $\sqrt{3} = -i(\zeta_3 - \zeta_3^2)$. The identity was verified exactly, by reduction modulo the cyclotomic polynomial Φ_{168} .

The T -identity. It follows from the exponents. The values $-r^2/168 + 21/168$ for $r = 1, 5, 11$ are $20/168$, $-4/168$ and $68/168$ modulo 1, that is $5/42$, $-1/42$ and $17/42$. These are the values of $h - c/24$ for $s = 2, 3, 1$: $c/24 = -17/42$, and $h_{1,s} = 0, -\frac{2}{7}, -\frac{3}{7}$ for $s = 1, 2, 3$. $\square$

The note reaches the $M(2, 7)$ exponents by its cubic phase transport $T \mapsto e(-1/24)T^3$. That map sends the labels 1, 5, 11 to $-1/42$, $17/42$ and $5/42$: the correct set of exponents, but a different matching of labels to characters. With the note's matching, $1 \mapsto \chi_3$, $5 \mapsto \chi_1$ and $11 \mapsto \chi_2$, the diagonal S -entries differ in absolute value: $\rho(S)_{11} = -c_1$, while $\rho_{2,7}(S)_{33} = c_3$. So no signed permutation and scalar realise that matching, and the pair $(\rho(S), e(-1/24)\rho(T)^3)$ does not satisfy the relation $(ST)^3 = S^2$ (`m7_vs_M27.py`). Theorem 6.15 realises the identification through conjugation and the η^3 twist instead, and with it the note's claim that the explicit mixed mock functions carry a $c = -68/7$ character shadow holds at the level of modular representations, after

conjugation and the η^3 twist. The note’s third-order modular differential equation also checks: in the monic form $\vartheta^3 f + \mu_1 E_4 \vartheta f + \mu_2 E_6 f = 0$, with ϑ the Serre derivative, the indicial polynomial is $x(x - \frac{1}{6})(x - \frac{1}{3}) + \mu_1 x + \mu_2$, and the roots $\frac{17}{42}, \frac{5}{42}, -\frac{1}{42}$ force $\mu_1 = \frac{1}{28} - \frac{1}{18} = -\frac{5}{252}$ and $\mu_2 = \frac{85}{74088}$, the note’s coefficients.

Proposition 6.16 (the order-5 block carries the Fibonacci data). *Let $m = 30$, $K = \{1, 6, 10, 15\}$, and take the note’s vectors*

$$w_1 = e_1 + e_{11} + e_{19} + e_{29} - e_{31} - e_{41} - e_{49} - e_{59}, \quad w_7 = e_7 + e_{13} + e_{17} + e_{23} - e_{37} - e_{43} - e_{47} - e_{53}.$$

The weight- $\frac{3}{2}$ series $G_r(\tau) = \sum_{\ell} w_r(\ell \bmod 60) \ell q^{\ell^2/120}$, $r = 1, 7$, satisfy

$$\rho(S) = \frac{2}{\sqrt{5}} \begin{pmatrix} \sin \frac{\pi}{5} & \sin \frac{2\pi}{5} \\ \sin \frac{2\pi}{5} & -\sin \frac{\pi}{5} \end{pmatrix}, \quad \rho(T) e(-1/8) = \text{diag}(e(-\frac{7}{60}), e(\frac{17}{60})).$$

These are the Fibonacci modular data, the data of the level-one G_2 WZW model at $c = 14/5$ [33]. The Lee–Yang data of $M(2, 5)$ are not a twist of them by any signed permutation and scalar, since the absolute values of their diagonal S -entries are $(2/\sqrt{5}) \sin(2\pi/5)$, not $(2/\sqrt{5}) \sin(\pi/5)$.

Status. Proved here by the method of Theorem 6.15, with the exact verification in $\mathbb{Q}(\zeta_{120})$; new in version 4. Scripts: `exact_shadow_reps.py` and `m5_vs_M25.py`.

The rank-two note matches the Lee–Yang exponents $-\frac{1}{60}$ and $\frac{11}{60}$ through the cubic phase transport. By Proposition 6.16, the order-5 shadow carries the Fibonacci representation rather than the Lee–Yang one. Rank-two modular data are classified in [33], where Lee–Yang is related to Fibonacci by Galois conjugation; that relation was not verified separately here.

What these results add. Theorem 6.15 places Ramanujan’s order-7 mock theta functions, and with them the $\widehat{Z}$ invariant of $-\Sigma(2, 3, 7)$, inside the modular representation of the $(2, 7)$ minimal model: after conjugation and a twist by the character of η^3 , their shadow transforms exactly as the $M(2, 7)$ characters. The classical relation of this kind for order 5 is the mock theta conjectures, which express the order-5 functions through the Rogers–Ramanujan functions, the characters of $M(2, 5)$ [21, 29]. Proposition 6.16 shows that for order 5 the shadow’s own modular data are the Fibonacci data, a Galois companion of Lee–Yang.

My assessment: the two results together suggest a pattern for the Brieskorn spheres $\Sigma(2, 3, q)$, namely that the shadow representation of the associated mock vector is a twisted minimal-model representation or one of its Galois conjugates. I think so because the two cases $q = 5, 7$ fall under this description, and the pattern can be tested on further q with the same scripts (Question 10). For the Erdős–Straus side, the notes’ map from the cubic middle to the three support labels is the link that would carry this structure over. That map is stated in the note’s theorem *Coordinate realization of middle support* and is not verified here.

6.7 Further structural facts

Three further structural statements of the notes were checked by program in reading passes, and one of them again here.

- *The Niemeier lattice.* The Niemeier lattice with root system $A_5^4 D_4$ and glue of order 72 appears in the notes and as archive Theorem 25.8 [26]. Its umbral group, the automorphism group of the glue code, is $GL_2(3)$ [24]; as noted in Section 6.5, $GL_2(3)$ and the binary octahedral group are distinguished by their involutions.
- $\Gamma_0(6)$ and the class 6B. $T_6 = \eta(\tau)^5 \eta(3\tau) / (\eta(2\tau) \eta(6\tau)^5)$ is a Hauptmodul of $\Gamma_0(6)$, and $T_6 + 5 + 72/T_6$ is the McKay–Thompson series of the Monster class 6B [25].

- *Cayley–Dickson algebras*. The statements on sedenion zero divisors were checked in the conventions stated in the notes [31, 22].

The Busy-Beaver notes propose a pattern in residues modulo 107, the residual of p_* . A test in an earlier reading pass compared these residues with a null model fixed in advance and found their frequencies within it (held-out $p = 0.38$); the test was not re-run here, and it concerns that one proposed pattern only. The Lorentzian notes are treated in Section 8.4, where the relation between the modular flow and the Lorentz boost of their section on finite-dimensional modular flow is made precise (Proposition 8.3).

7 Negative results, with exact scope

Each row states a proposed implication that fails, with its counterexample or derivation, and the scope of the failure: what the counterexample does not show. Every certificate prime in the table is itself solvable. So these results limit specific mechanisms, not the conjecture, and several of them name the further input a mechanism needs.

Source	What fails	Certificate	What it does not show
bounded transport	every occupied shell has a solution with unary layer $A \leq 2$	$p = 37, a = 12$: only $u = 8$	the global cutoff ($a = 10$ works at 37)
first-two-shell sieve	occupancy of the $R = 7$ shell gives a first-two-layer hit	$p = 373, a = 95$	anything about other shells
item 2	every canonical closed component contains a local hit	$p = 2521$: a 7-cycle sink	2521 is solvable
item 3	strict canonical descent ends at a hit	a closed triangle at a 22-digit prime	that prime is solvable
item 6	the principal-only bound, summed over shells, is positive for large p	$\sum L_a \leq -\frac{1}{24}p \log p$	anything about the true counts
item 6	a nonnegative reweighting of it certifies occupancy	$p = 87481$: all 21,870 shells have $L_a \leq 0$	only this certificate family fails
item 14	same-grade middle repair from exterior states with $\alpha = -4t, t \geq 4$	$p = 7840561$	those primes are solvable
item 21	a same-word residual return for words $u \nmid 36$	3,041 of the 3,654 sources with $u \nmid 36$ at primes below 2600	endpoint solutions exist
item 26	a trace-to-solution map never increases a	$p = 67369$	67369 is solvable
record shells (archive §19.3)	the deficit-box conclusions without strict recordness	$p = 806,521$: full difference set, eight points missed	anything about strict records
author’s notes (Section 6)	the one-congruence shell criterion without coprimality	$p = 73, R = 7 \cdot 73^2$	anything about shells with $p^2 \nmid R$
author’s notes	every prime $p \equiv 1 \pmod{4}$ has a lock certificate	$p = 409$	anything about the hard classes
Jacobi barrier (archive §10)	its converse: a non-residue prime factor of a makes the shell occupied	$p = 5, R = 7, a = 3$	the barrier itself (Proposition 3.6)

Source	What fails	Certificate	What it does not show
no-two-shears theorem (supplement §5)	the theorem at the primes $p \equiv 2 \pmod{3}$ with $R(R+4) \mid p+4$ for a suitable residual R , for instance at every $p \equiv 131 \pmod{180}$; so no congruence hypothesis weaker than $p \not\equiv 2 \pmod{3}$ suffices (Propositions 2.1 and 2.2)	two-edge paths at $p = 131$ and $p = 1613$	anything at $p \not\equiv 2 \pmod{3}$, where the theorem holds; it holds at most primes $p \equiv 2 \pmod{3}$ (Remark 2.3); anything about solvability
carrier census (archive §24)	forcing every row of the census domain by fixed-divisor carriers	the 2,759,345,613,832,500 square rows (Corollary 4.7)	anything about the primes in those rows; carriers that use the factorisation of $(p+R)/4$

All of these are statements about methods or certificate families, and none is a counterexample to the conjecture. Where a mechanism is sharp, the table says so: $p \not\equiv 2 \pmod{3}$ is the weakest congruence hypothesis under which the no-two-shears theorem holds at every prime, and it holds at most primes $p \equiv 2 \pmod{3}$ as well.

The archive records further negative results of the same kind. They are located, not checked here:

- the withdrawn constructions of §5 (p. 22), each with its point of failure;
- the retracted converse of the Star–Kneser forcing theorem (§15.3, p. 122);
- the failed universal bridge of Dyachenko’s bounded identities (§12.2, p. 103);
- a finite counterexample to one projection step in López Bonilla’s 2026 preprint (§19, provenance, p. 148): in C_6 , the set $\{0, 1\}$ has trivial stabiliser, but its reduction modulo 2 is all of C_2 . The archive restricts this to that step and does not assess the whole paper.

8 Bridges to other areas, and the directions they open

The research behind this paper explored many directions quickly, and in the author’s view the bridges it built are among its most valuable outcomes. This section describes them. It does not claim that any of them is a complete research programme, and it does not treat a direction that is still incomplete as invalid: much of mathematics consists of programmes that are incomplete and are nevertheless valid mathematics. The audit did not have the compute to examine every direction to the depth needed to decide how promising it is. For each direction the section states what is established, with references to the proved results; what is not verified here; and my assessment.

The assessment addresses two separate questions. The first is the direction’s interest on its own merit, for example as a bridge between two areas. The second is its bearing on the Erdős–Straus problem. When a direction relates the equation to a structure that does not take primes as input, the second question is what typed map connects the two, and how far that map is established. The assessments are mine and are marked as such; they are views held at this stage and may change as the directions are explored further.

8.1 The split-zero support language

The notes describe shells, classes and certificates through the split-zero semiring $G(R) = R \sqcup \{\tau\}$. In it, τ is an additive identity distinct from the ring zero 0_R , and τ absorbs multiplication. The

same semiring is the coefficient algebra of the zeta programme [16], and it is archive §21. The note *A split-zero repair dossier* develops the algebra of $G(R)$: ideals, prime ideals, spectrum, congruences, localisation and semimodules. Its classification statements were checked in a reading pass. One theorem of the dossier needs an additional relation, and the correction keeps to the construction’s own convention of keeping τ and 0_R apart.

Proposition 8.1 (the dossier’s Theorem 7.2, with its relation). *Let $S = G(R)$, let M_S be its multiplicative monoid, and let $\mathcal{R}$ be the congruence on the free commutative semiring $\mathbb{N}[M_S]$ generated by $[a] + [b] \sim [a + b]$ for $a, b \in R$ and $[x] + [\tau] \sim [x]$ for $x \in S$.*

- (a) *The class of the empty sum 0 in $\mathbb{N}[M_S]/\mathcal{R}$ is $\{0\}$.*
- (b) *The natural map $\tilde{\Theta} : \mathbb{N}[M_S]/\mathcal{R} \rightarrow S$ sends both 0 and $[\tau]$ to τ , so it is not injective.*
- (c) *After adjoining the relation $[\tau] \sim 0$, $\tilde{\Theta}$ is an isomorphism.*

Status. Proved here; found by a reading pass of version 3, new as a proposition in version 4.

Proof. (a) Every generating relation relates two non-empty formal sums. Adding a sum, or multiplying by a non-zero sum, preserves non-emptiness, and multiplying by 0 gives $0 \sim 0$. So no chain of relations connects 0 to a non-empty sum. (b) A semiring homomorphism sends 0 to $0_S = \tau$, and $\tilde{\Theta}([\tau]) = \tau$. (c) With $[\tau] \sim 0$, every formal sum reduces to $[\tau]$ or to $[r]$ for some $r \in R$: the terms $[\tau]$ can be removed by $[x] + [\tau] \sim [x]$, the empty sum is $\sim [\tau]$, and the remaining terms $[r_1], \dots, [r_k]$ combine to $[r_1 + \dots + r_k]$. These representatives are pairwise inequivalent, because $\tilde{\Theta}$ sends them to the distinct elements τ and $r \in R$ of S (recall $\tau \neq 0_R$). So $\tilde{\Theta}$ is bijective, and it is a homomorphism by construction. $\square$

The support states also have a typed meaning in the shell arithmetic. For a shell a , the three states τ , 0 and $+$ correspond to three situations:

- neither -1 nor -4 lies in the group H_a , and the barriers of Section 3.4 apply;
- -1 or -4 lies in H_a , but no divisor of a^2 reaches a target;
- the shell is occupied.

By Proposition 3.8, under the exponent condition the middle state does not occur. Applied to classes of primes, as in Remark 6.3, the same vocabulary calls a class supported when some certificate applies on a lift of it.

My assessment: I think this vocabulary is useful for the problem, because it separates the two reasons for which a shell can be empty. One is group-theoretic, and the barriers handle it; the other is a matter of size, in which the exponents of a are too small to reach a target inside the group. They call for different tools. The algebra of $G(R)$ developed in the dossier is also an object of independent interest.

8.2 Mock and quantum modularity

What is established is in Section 6.6: the notes’ explicit mixed mock functions are Ramanujan’s order-7 mock theta functions, and the first is $q^{1/2}\widehat{Z}_0(-\Sigma(2, 3, 7))$; their shadow carries the $M(2, 7)$ representation after conjugation and the η^3 twist (Theorem 6.15); and the order-5 block carries the Fibonacci data (Proposition 6.16). Not verified here: the note’s support isomorphism from the cubic middle of the shell problem to the three support labels, its mass-gap coordinate for the cubic middle, and its Virasoro support-gap corollary.

My assessment: on its own merit, I think this is the most developed of the structural directions. It has produced a result, Theorem 6.15, that stands independently of the Erdős–Straus problem and connects Ramanujan’s order-7 functions, the 3-manifold invariant of $-\Sigma(2, 3, 7)$ and the $(2, 7)$

minimal model. For its bearing on the equation, the typed map needed runs from the shell data, which depend on p through the factorisation of $(p + R)/4$, to the modular object. The notes' map goes through the finite support labels. An analytic object in the notes whose coefficients depend on that factorisation is the pair-count series of Proposition 6.12. A typed map between the pair-count series and the mock vector would carry information in both directions, and constructing one is Question 11.

8.3 Moonshine, Niemeier lattices and Ogg's primes

Checked in reading passes (Section 6.7): the Niemeier lattice with root system $A_5^4 D_4$, its umbral group $GL_2(3)$, the $\Gamma_0(6)$ Hauptmodul and the series of the class 6B. A constructive-algebra preprint of the notes also connects the extension prime 29 with the Heegner discriminant 163. The factorisation it uses is classical and correct: $j((1 + \sqrt{-163})/2) = -640320^3$ with $640320 = 2^6 \cdot 3 \cdot 5 \cdot 23 \cdot 29$; by contrast, $5280 = 2^5 \cdot 3 \cdot 5 \cdot 11$ for the discriminant 67. Not verified here: the preprint's rank-15 Ogg module, its genus-zero and supersingular realisations, and its phase decomposition, apart from the map treated next.

Proposition 8.2 (the torsor map). *The map $\Theta : (\mathbb{Z}/18)^\times \rightarrow S_{840}$ of Terminal residual coordinates §10, given by $1 \mapsto 1, 5 \mapsto 289, 13 \mapsto 361, 17 \mapsto 169, 7 \mapsto 121, 11 \mapsto 529$, is a bijection but not a homomorphism. The map $5^k \mapsto 289^k$ is an isomorphism, and it differs from Θ exactly by exchanging the images of 7 and 13.*

Status. Proved here; new as a proposition in version 4.

Proof. Both groups are cyclic of order 6: 5 generates $(\mathbb{Z}/18)^\times$, with powers 5, 7, 17, 13, 11, 1, and 289 generates S_{840} , with powers 289, 361, 169, 121, 529, 1 modulo 840. So $5^k \mapsto 289^k$ sends 5, 7, 17, 13, 11, 1 to 289, 361, 169, 121, 529, 1. It agrees with Θ except at 7 and 13, whose images are exchanged. In particular $\Theta(5)^2 = 289^2 \equiv 361 \pmod{840}$, while $5^2 \equiv 7 \pmod{18}$ and $\Theta(7) = 121$. $\square$

The note's odd Ogg phase decomposition (its Theorem 10.1) uses only that Θ is a bijection, so it holds as stated; with the isomorphism in place of Θ , two of its sectors exchange their contents (a reading pass).

My assessment: the moonshine data organise the residue layer of the problem, meaning the six hard classes and the core, in a way that matches known structures: the umbral group of $A_5^4 D_4$, the $\Gamma_0(6)$ Hauptmodul and Ogg's primes. I think this match is of independent interest as an observation. Its bearing on the hard classes depends on the typed map that is used. What is proved is that polynomial identities and fixed-divisor certificates do not reach the square classes (Section 1, Proposition 4.6). I have not seen a map from the shell data to the moonshine data in the notes, and I do not have a view yet on whether one exists.

8.4 Lorentzian and Apollonian geometry

The Descartes form $F(x) = 2 \sum x_i^2 - (\sum x_i)^2$ has signature $(3, 1)$. The Apollonian group is a subgroup of $O_F(\mathbb{Z})$ of infinite index that is Zariski dense in O_F , a thin group, and it acts on hyperbolic 3-space [34, 28]. The notes identify a Lorentz form of their shell geometry with the Descartes form. The same Lorentzian geometry of circle configurations underlies Kocik's circle-geometric reading of the Koide relation [30].

The note *Lorentzian fixed fibers, Descartes sheets, and origin-resolved positivity in residual Erdős–Straus shells*, in its section *Finite-dimensional modular flow as a Lorentz boost*, states that the congruence action $X \mapsto A_\rho X A_\rho$, with $A_\rho = \text{diag}(e^{\rho/2}, e^{-\rho/2})$, is a Lorentz boost on the Hermitian 2×2 matrices with the form $\det$. This is correct. The relation between this boost and the modular flow of the section's title is the following.

Proposition 8.3 (modular flow and boost). *Let $H_r = \text{diag}(e^r, e^{-r})$ and $X = \begin{pmatrix} h+w & u+iv \\ u-iv & h-w \end{pmatrix}$, with the Minkowski form $\det X = h^2 - u^2 - v^2 - w^2$. For $z \in \mathbb{C}$ the map $X \mapsto H_r^z X H_r^{\bar{z}}$ lies in $SO^+(1, 3)$.*

- (a) *At $z = it$ it is the modular automorphism $\sigma_t = \text{Ad}(H_r^{it})$ of the state φ_{H_r} with density proportional to H_r . It acts as the rotation by the angle $2rt$ in the (u, v) -plane and fixes h and w .*
- (b) *At real $z = s$ it is the boost of rapidity $2rs$ in the (h, w) -plane and fixes u and v . The note's A_ρ is the case $s = \rho/(2r)$.*

So the modular flow and the note's boost are the compact and the non-compact real forms of one complex one-parameter subgroup $z \mapsto H_r^z$ of $SL_2(\mathbb{C})$, acting on Minkowski space by $X \mapsto gXg^$.*

Status. Proved here; new in version 4. The coordinate formulas are checked symbolically in `misc_checks.py`.

Proof. $\det H_r^z = 1$ and $(H_r^z)^* = H_r^{\bar{z}}$, so the map is the standard action of $SL_2(\mathbb{C})$ on Hermitian matrices, which lies in $SO^+(1, 3)$ because $SL_2(\mathbb{C})$ is connected. At $z = it$, $H_r^z = H_r^{-it}$, so the map is $\text{Ad}(H_r^{it})$, the modular group of φ_{H_r} ; it multiplies the off-diagonal entry $u + iv$ by e^{2irt} and fixes the diagonal, which gives $(h, u \cos 2rt - v \sin 2rt, u \sin 2rt + v \cos 2rt, w)$. At $z = s$ it multiplies the diagonal entries $h + w$ and $h - w$ by e^{2rs} and e^{-2rs} and fixes $u + iv$, which gives $(h \cosh 2rs + w \sinh 2rs, u, v, w \cosh 2rs + h \sinh 2rs)$. $\square$

My assessment: on its own merit, I think the Descartes setting is a natural place to ask how such Lorentz one-parameter families interact with the Apollonian thin group, and the question is one about thin groups in its own right. A companion note answers it [19]. The orientation-preserving subgroup A^+ of the Apollonian group meets every compact subgroup of $SO_F(\mathbb{R})$ only in the identity, so under any identification of the Descartes space with the Hermitian matrices no nontrivial element of the Apollonian group is a modular automorphism of a faithful state (the modular automorphisms lie in compact subgroups of $SO^+(1, 3)$, and the reflections S_i are not in SO_F). Boosts occur in the stabilisers of the circles of a packing, which are congruence groups (in the strip packing, the image of $\Gamma^0(4)$ in $PSL_2(\mathbb{Z})$), and also as boosts that fix no circle, such as $S_1 S_2 S_1 S_3 S_4 S_3$; every other loxodromic element of A^+ is a screw motion whose rotation angle is an irrational multiple of 2π . For the bearing on the equation, the notes' map from shell data to Descartes quadruples is not verified here. I have not identified a typed map from shell occupancy to the Apollonian group, and I have no view yet on whether one exists.

8.5 Additive combinatorics: Kneser's theorem

The notes apply Kneser's theorem on sumsets in abelian groups [6] to the residues of divisors, producing Star–Kneser forcing, which the archive states as a per-shell sufficient condition for occupancy (Theorems 15.6 and 20.7; located, not verified here). Its converse fails at p_* , $R = 107$, as the archive records (Proposition 15.7).

My assessment: I think additive combinatorics is a natural toolset for the second reason for which a shell can be empty (Section 8.1), where the target lies in H_a but may not be reached. In additive notation the residues of the divisors of a^2 form a sumset of arithmetic progressions, one for each prime $\ell \mid a$ with $2v_\ell(a) + 1$ terms, so sumset growth is exactly what decides whether they fill H_a .

8.6 Computability: the Busy-Beaver notes

The statement that $4/n = 1/x + 1/y + 1/z$ is solvable for every $n \geq 2$ is Π_1 . Each instance is decided by a bounded search: in a solution with $x \leq y \leq z$ one has $x \leq 3n/4$, then $y \leq 2/(4/n - 1/x)$, and z is determined. So there is a Turing machine that halts if and only if the conjecture fails, and if it has k states, the value of the Busy-Beaver function at k decides the conjecture. This is the standard bridge between Π_1 statements and the Busy-Beaver function; such machines have been constructed explicitly for Goldbach's conjecture and for the Riemann hypothesis [20]. The

archive uses Busy-Beaver values, quoted from the literature and not checked here ($S(4) = 107$, $\text{Space}(5) = 12289$), to select the shells of its §§3 and 20, and its §25 maps Busy-Beaver extrema to shell carriers (located). The proposed pattern in residues modulo 107 is discussed in Section 6.7. The Π_1 bridge is exact, and the least number of states of a machine that halts exactly when the conjecture fails is a well-posed target (Question 13).

8.7 Further directions in the same collections

The collections also contain notes on Cayley–Dickson algebras and sedenion zero divisors, on the Koide relation, on a mass gap and on anomalies, and a split-zero zeta note. The Cayley–Dickson statements were checked in a reading pass (Section 6.7). The Koide, mass-gap and anomaly notes lie outside the scope of this paper; they were not read for it and are not verified here.

8.8 The author’s other workbenches

Material crosses between this workbench and the author’s other workbenches. The audit note 47_ records every crossing that the audit found as a typed edge, of one of four types:

- *P*: a proof-level import;
- *D*: a dependency or citation;
- *I*: the same identity or object on both sides;
- *A*: an analogy, with no typed map yet.

The table lists the edges that touch this workbench, and adds three that this paper found in the archive (marked *new*).

Edge	What crosses	Type	Status
NS → ES	The auxiliary torus endomorphism $J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$ (determinant 14) of the Navier–Stokes manuscript. It enters 21 statements of the supplement (§8) as an operator only. The core identity is $\{(u, v) \in G^2 : u^3v = 1 = uv^5\} = \{(u, u^{-3}) : u^{14} = 1\}$	P	audited (21_ §1); no fluid theorem is used
$S^6 \rightarrow \text{ES}$	The integral monodromies A_1, A_2, A_∞ of the $(3, 4, \infty)$ period system, in ES-09	P	audited for odd levels; the orbits at every level are Theorem 5.1 (§5.2)
$\text{ES} \leftrightarrow S^6$	The Niemeier lattice with root system $A_5^4 D_4$ and glue index 72 (ES result R10)	I	ES’s glue code verified; the two codes not compared
Jacobian map	Alpöge’s map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with $\det DF = -2$ and the three-point fibre over $(-\frac{1}{4}, 0, 0)$ (archive §§4, 7). The same map is in the Yang–Mills workbench and the zeta programme. ES’s four-dimensional extension P ($\det DP = -2$) crosses to the zeta programme’s conductor	P	det DF , det DP and the fibre verified (zeta paper [16], Lemma 6.6; 21_); the conductor maps located
$\text{ES} \leftrightarrow \text{zeta}$	The split-zero semiring $G(R) = R \sqcup \{\tau\}$ with its Boolean support character (archive §21; ES R01)	I	audited (21_ §6)
$\text{ES} \leftrightarrow \text{zeta}$	A finite Weyl phase-space basis	I	ES side verified on $C_2 \times C_3$
$\text{zeta} \rightarrow \text{ES}$	NG20–NG21, used by item 20; the SplitZero packages of 13–16 September	D	located
$\text{ES} \rightarrow \text{zeta}$	The 488-page reader’s source, vendored into the zeta repository	D	located

Edge	What crosses	Type	Status
ES $\rightarrow$ zeta	The 124-page ES/Fable reader: solutions as binary quartics, then the Fable cover and a weighted conductor (§2.4)	P	the quartic relation proved here (Proposition 2.4); the rest located
ES $\leftrightarrow$ Collatz	$3(4n+1)+1=4(3n+1)$, i.e. $\theta T=4\theta$ (ES R05, diagrams by u/CivQ17)	I	audited
Collatz $\leftrightarrow$ ES $\leftrightarrow$ zeta	The homogenisation $(x, h) \mapsto (ax+bh, h)$ (ES X4)	I	ES’s diagram read
Busy Beaver (new)	Busy-Beaver extrema quoted by the archive ($S(4)=107$ and $\text{Space}(5)=12289$; for the five-state values it cites the formal enumeration of the bbchallenge collaboration) select the shells of archive §§3 and 20; §25 maps extrema to shell carriers. The zeta programme’s detectability argument [16] uses a different machine for a different purpose	A	located; the two use different machines for different purposes
Connes (new)	Supplement §6 reads Connes’ primitive intersections; archive §24 opens with Tomita–Takesaki transport. The zeta programme uses Connes’ summation map and Meyer’s quotient	A	located; no common statement was found
Fable (new)	<i>Fable</i> names three different objects across the repositories: the three-dimensional Keller map, the S^6 construction in older files, and a conductor cover (47a_ §10). Archive §5 withdraws one Fable construction	—	disambiguate at each use
ES $\leftrightarrow$ YM	A shared term, <i>gap</i> (ES X7), and a proposed success projector	A	located; no typed map is written

The crossings with typed content are the torus operator, the monodromy matrices, the Keller maps, the split-zero semiring and two identities; each is audited or recorded as located. The other connections are proposals awaiting typed maps, and the archive describes its late Fable, operator, Connes and moonshine material in the same way (p. 4): as a candidate interface, pending an explicit typed map and compatibility check.

9 Questions, and what was not checked

9.1 Questions

1. *The global statement.* Show that for every prime $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$ some shell is occupied. This is the conjecture itself, restated by Theorem 3.1, and the archive names it as the remaining obligation (p. 4). By Proposition 3.7, a proof must produce, for some R , a shell whose group H_a contains -1 or -4 ; when R is prime this means a prime factor of $(p+R)/4$ that is a non-residue modulo R . That condition alone does not suffice, and Proposition 3.8 says when it does.
2. *Type II universality* (Section 6.4). Does every prime $p \equiv 1 \pmod{24}$ have a solution with p dividing exactly two denominators? A positive answer proves the conjecture. Below 10^7 every such prime has one in a shell $R \leq 107$, and only $p_* = 8,803,369$ needs 107; a reading pass found the same below 10^8 .
3. *The quadratic-residue survivors* (Table 3). Are there infinitely many primes that satisfy every condition of Section 6.2? They have density zero (Proposition 6.8), and 23 of them lie below 10^7 . A negative answer would prove the conjecture at all but finitely many primes.

4. *The 550 supported classes* (Theorem 6.2). Is each of the 550 non-square classes left open modulo $840 \cdot 11 \cdot 19 \cdot 23$ covered by finitely many single identities at some finite level? If one is not, no identity sieve reduces the problem to the square core.
5. *The census survivors*. 2.394% of the final hard domain survives the carrier chain (§4.3), and the square rows among them can never be forced by its carriers (Corollary 4.7). Can the non-square survivor rows all be forced by carriers of the same kind, if more moduli are added? Which carriers, using the factorisation of $(p + R)/4$ and not only residues of p , force the square rows?
6. *The strict-record deficit-box conjecture* (§4.1; archive §19.2). It holds at the two strict records with non-empty deficit below 8,803,369, and strict recordness cannot be dropped from it ($p = 806,521$). A further test needs a strict record above 8,803,369, and there is none below $3 \cdot 10^9$ (Proposition 4.3).
7. *Growth of the least class*. Is h unbounded on the primes? Along the strict records up to $3 \cdot 10^9$ it takes the values 1, 2, 3, 6, 8, 15, 27; below $3 \cdot 10^9$ the next largest values are 21, 20 and 19, at nine primes; and every record after 73 lies in a hard class (Corollary 4.2). The workbench states no conjecture on the growth of $h(p)$, and none is stated here.
8. *The density of the shear paths*. What is the exact relative density, among the primes $p \equiv 2 \pmod{3}$, of the primes with a path of two shear edges? It is at least $1/24$, from the class 131 mod 180, and at most 0.335 (Remark 2.3); below 10^7 the proportion is 5.24%.
9. *Genera of the monodromy covers*. What are the genera of the components of the $(3, 4, \infty)$ covers at every level? Theorem 5.1 gives the orbits, and the genera follow from the cycle counts of A_1 , A_2 and A_∞ on each orbit (§5.2).
10. *Brieskorn spheres* $\Sigma(2, 3, q)$. For which q does the shadow representation of the associated mock vector coincide, after conjugation and a twist, with a minimal-model representation, as for $q = 7$ (Theorem 6.15), or with a Galois conjugate of one, as for $q = 5$ (Proposition 6.16)?
11. *A modular object for the shell counts*. Is there a modular, mock or quantum modular object whose coefficients are the shell counts $2|E_a| + |M_a|$, or a typed map from the pair-count series of Proposition 6.12 to the mock vector of Section 6.6?
12. *A typed map to the Descartes geometry*. Is there a typed map from shell occupancy, or from the notes' Descartes quadruples, to the Apollonian group? The companion note [19] determines how the rotations and boosts of Proposition 8.3 meet that group; the map from the shell data is what is missing.
13. *The least Π_1 machine*. What is the least number of states of a Turing machine that halts if and only if the conjecture fails?

9.2 What this paper did not check

- *The archive*. Apart from the parts marked proved or reproduced in §2.2, its 619 pages were mapped, not audited. In particular, the structural constructions (§§2, 4, 6–8, 9 after its opening, 21–23, the operator part of §24, the structural material of §24.4, and §25) were not audited, and the literature audits (§§11–14 and 17) were not compared with the papers they audit.
- *Lean*. The source archive contains 67 files with the extension `.lean` (seven of them `.axioms.lean`); they were not rebuilt. The archive's own account of what each Lean module checks is quoted where it matters (§4.3).
- *Section 6*. Several statements there rest on the reading passes and are marked so:
 - the lattice and modular facts of Section 6.7, apart from the involution count;
 - the comparison of the notes' q -expansions with Zagier's vector M_7 ;

- the Busy-Beaver residue test, and the Busy-Beaver values that the archive quotes from the literature;
- the Cayley–Dickson identities;
- the counts below 10^8 .

Salez’s completeness statement was not proved here. Drafts that the author’s latest record supersedes were not treated.

- *The census.* The enumerations of §4.3 were not re-run. Only the arithmetic consistency of the reported counts was checked, together with the base rows and the carrier sets used in Corollary 4.7.
- *The supplement.* The general proofs of the supplement’s Sections 1–7 were not re-derived, except that of the no-two-shears theorem of Section 5 (Proposition 2.1); their finite content was reproduced in part (§2.3). Section 8, the auxiliary torus, was not checked.
- *The 124-page reader.* Only Proposition 2.4 was checked.
- *The continuation.* Items 1, 5, 9–13, 15–20, 24, 25, 27 and 28 were located, not checked. So were item 4’s enumeration, the deduction in item 23, and ES-04 to ES-08 beyond the ES-08 witness.
- *Novelty.* No literature search for novelty was made here, except where a status line says so. The workbench claims no novelty for the classical coordinates and identities it uses. Several results were found by referee passes (Propositions 2.2, 3.7, 3.8, 4.4 and 4.6, and Corollaries 4.2 and 4.7); some of them are forms of statements of the archive or the notes, as their status lines say. The results marked new in version 4 are not in the workbench, the notes or the earlier versions of this paper, as far as these were read; whether they are in the literature was not checked, and none is claimed as new mathematics beyond that.

A Catalogue of statements

This catalogue lists every statement of the workbench that this paper states, proves, reproduces or locates. For each it gives where the statement is in the workbench and how far it was checked here, in the labels defined in the introduction. *WB* quotes the workbench’s own label where it adds something. Locators are as in the front matter: *archive* is the 619-page PDF, *supplement* the 86-page supplement, and *item n* the n -th entry of the README’s *Start reading* list.

Statement	Where in the workbench	Status here	Here
<i>The shell reduction</i>			
The ordered pairs completing a first denominator a number $2 E_a + M_a $; solvable iff some shell is occupied (ES-01, ES-02)	archive Theorems 9.1 (p. 38) and 9.64 (p. 90), table (394); supplement §1 (for $p = 12h + 1$)	Proved here; the bijection and the count are the archive’s, for every odd prime; this paper adds the swap analysis and (c)	Thm 3.1
The least denominator lies in $(p/4, p/2)$, except for one solution at $p \equiv 3 \pmod{4}$	a reduction step of item 22	Proved here, for every odd prime	Thm 3.1(a)
A counterexample prime has $E_a = M_a = \emptyset$ at every shell (the all-shell no-hit sieve)	supplement §1	Audited (43a_ §1.2); it is Thm 3.1(c)	§3
The residual-3 shell: criterion and count	supplement §1; archive Theorem 10.3	Proved here; count extended to $p \equiv 5 \pmod{12}$	Prop. 3.2

Statement	Where in the workbench	Status here	Here
The residual-7 shell: criterion	supplement §1; archive §10	Proved here, for every $p \equiv 1 \pmod{4}$	Prop. 3.3
Both first shells empty only if p lies in a hard class modulo 840	archive Corollary 10.7 (p. 95); the first pass observed the congruence modulo 24	Proved here	Prop. 3.4
Reduction to the six hard classes modulo 840	archive Theorem 10.6, p. 95 (with shells $R \leq 7$); classical (five identities)	Proved here	Props. 3.4, 3.5
Jacobi-symbol barrier, and the failure of its converse	archive Theorem 10.9 and Proposition 10.10, pp. 96–97	Proved here	Prop. 3.6
The group barrier $-1 \notin \langle 4, \ell : \ell \mid a \rangle$, and a converse under an exponent condition	implied by archive §10	Proved here (found by the referee pass)	Props. 3.7, 3.8
<i>Least classes, records and deficits</i>			
$h(p) = (R + 1)/4$ for the least occupied shell	archive §§17.4, 19	Proved here	Lemma 4.1
$h(p) \leq 2$ outside the six hard classes; every strict record after 73 is hard	implied by archive Theorem 10.6	Proved here (found by the referee pass); checked to $3 \cdot 10^9$	Cor. 4.2
The seven strict records up to 8,803,369; $1201 \in C_6 \setminus C_5$; $h(806521) = 15$	archive Proposition 17.6, pp. 131–132 (recovering Ionascu–Wilson’s printed list and correcting one label); §19.2, p. 152	Reproduced	Prop. 4.3
No strict record in $(8,803,369, 3 \cdot 10^9]$; the primes with $h = 19, 20, 21$ there	—	Reproduced (two independent programs)	Prop. 4.3
At $p_* = 8,803,369$ the least occupied shell has $R = 107$, and it carries one unordered solution	archive §3	Reproduced	§4.1
The mod-107 deficit progression, the covering, and the two middle fibres (WB: credited to u/CommonCareful3149)	archive §16; bench R06	Reproduced	§4.1
The deficits at 118,801 and 8,803,369; the boundary example 806,521	archive §19.3	Reproduced	§4.1
The strict-record deficit-box conjecture	archive §19.2 (WB: conjecture)	Open	§4.1, §9
The shell $a = 3075$ at $p = 12289$: counts $(3, 6, 3)$	archive §20	Counts reproduced (esr_referee_checks.py); the orbit and the quotient certificate located	§2.2
<i>Families and the census</i>			
Fixed-divisor families: a divisor $u = sh$ or $u = s$ with $s \mid c^2$ solves the equation on a class modulo $4cR$	archive Lemma 24.218, p. 422 (for primes)	Proved here, for all n (found by the referee pass)	Prop. 4.4

Statement	Where in the workbench	Status here	Here
Four families solving the equation on $248k + 209$, $248k + 89$, $17112k + 11137$, $17112k + 14113$	archive Theorem 24.219, p. 423; Proposition 24.222	Proved here; instances of Prop. 4.4; that they meet every hard class is shown here	Thm 4.5
The carrier census: 97.606% of the final hard domain forced	archive pp. 3–5 and §24 (WB: bounded sufficient-carrier census)	Located; the arithmetic of the counts checked	§4.3
No fixed-divisor carrier forces a square class; the census cannot force its 2,759,345,613,832,500 square rows	—	Proved here (found by the referee pass)	Prop. 4.6, Cor. 4.7
The residual-eleven support ideal; the paired $R = 7/R = 11$ affine theorem	archive §18	Located	§2.2
<i>The supplement and the 124-page reader</i>			
The finite content of Sections 1–7 (gluing, ratio transport, shears, successors, orchard partition, boundary law, unary box, incidence, raw scale)	supplement §§1–7	Reproduced in part	§2.3
No two shear edges compose	supplement §5 (stated for $p \equiv 1 \pmod{12}$)	Generalised here to $p \not\equiv 2 \pmod{3}$; no weaker congruence hypothesis suffices; fails only on a minority of the $p \equiv 2 \pmod{3}$	Props. 2.1, 2.2, Rem. 2.3
The Navier–Stokes auxiliary cover	supplement §8	Not checked	§2.3
The quartic of a solution: $p = -5u_4/u_3$ and the coefficient relation, and its converse	124-page reader; item 11	Proved here (also 21_); the converse found by the referee pass	Prop. 2.4
The Fable cover and the weighted conductor	124-page reader	Located	§2.4
<i>The September continuation</i>			
Item 2: a seven-cycle sink at $p = 2521$ (WB: method counterexample)	research/incoming/	The sink audited (certificate); the closure theorems located	§5, §7
Item 3: a closed triangle at a 22-digit prime	research/incoming/	The triangle audited (certificate), its primality proved here; the escape theorem located	§5
Item 4: six nonresidue primes within 25 factorisations (WB: computer-assisted)	research/incoming/	Reductions read; conditional on an enumeration not re-run	§5
Item 6: the principal-only bound is negative in sum	research/incoming/	Audited	§5
Item 7: least-nonresidue localisation	research/incoming/	Audited	§5
Item 8: the middle identity and its bounds	research/incoming/	Reproduced on 1,613 middle states; the atlas located	§5
Items 14 and 21: template bound; same-word returns	research/incoming/	Audited	§5
Item 22: the degree-8 invariant is negative on solutions	research/incoming/	Audited (certificate)	§5
Items 23 and 26	research/incoming/	Audited in part	§5

Statement	Where in the workbench	Status here	Here
Items 1, 5, 9–13, 15–20, 24, 25, 27, 28	research/incoming/	Located (item 11’s quartic proved here)	43a_ §2
ES-09: the $(3, 4, \infty)$ monodromy covers	received/es_s6_counterfactual/	Audited for odd levels; at even levels the statements do not hold as stated; the orbits at every level proved here (new in version 4)	§5.2
ES-08: a forced solution at $p = 1201$, $a = 310$, $R = 39$	12 September continuation	The witness checked; the forcing inequality not audited	43a_ §4
Results bench R01–R10, arguments X1–X7	RESULTS.md	Checked in 43a_ §3 (X1 not recomputed)	§5
<i>The author’s notes (Section 6)</i>			
The terminal core $S_{840} \times Q_{11} \times Q_{19} \times Q_{23}$	source note of the core; the terminal-coordinates papers	Proved here (one direction; the other a finite computation, reproduced); the covering count 3520 computed	Thm 6.2
The width-one barrier	pre-Niemeier note, Thm 3.1; archive Thm 15.5	Proved here	Prop. 3.7(c)
Eight quadratic-residue shifts; $p + 5$; the visible-box shells; $\mu = 17$	implied by the notes’ branches and locks	Proved here; $p + 2$, $2p + 1$, the extension of $p + 5$ and five shells new	§6.2
Non-coprime and ramified shells; the pair-count series	defect-completion article, Lemma 2.1 (which needs $p^2 \nmid R$) and Thm 4.3	Proved here	§6.3
The direct 29-channels; Rosati’s family; the full-divisor locks	terminal-coordinates paper, Thms 7.1–7.2; the locks note	Proved here; <i>only two channels</i> holds only for single-prime divisors	Props. 6.13, 6.14
The survivors of the row $p + 1$ have density zero	—	Proved here (new in version 4)	Prop. 6.8
The shadow of the order-7 mock theta vector and the characters of $M(2, 7)$	<i>An explicit mixed-mock Virasoro support realization</i>	Proved here in the form of Thm 6.15 (new in version 4); the note’s label matching does not realise it	Thm 6.15
The order-5 block and the Fibonacci modular data	the rank-two note	Proved here (new in version 4)	Prop. 6.16
The split-zero dossier’s Theorem 7.2	<i>A split-zero repair dossier</i>	Proved here with the relation $[\tau] \sim 0$	Prop. 8.1
The torsor map $(\mathbb{Z}/18)^\times \rightarrow S_{840}$	terminal-coordinates paper, §10	A bijection, not a homomorphism; the decomposition using it holds (a reading pass)	Prop. 8.2
Modular flow and Lorentz boost	the Lorentzian Descartes note	The note’s boost proposition correct; the relation proved here (new in version 4)	Prop. 8.3

Statement	Where in the workbench	Status here	Here
<i>The rest of the archive</i>			
Structural constructions: §§2, 4–8, 21–23, 25 and the operator part of §24	archive	Located; §5 read; the fibre of §7 verified in the zeta paper	§2.2, §8
Literature audits: §§11–14 and 17	archive	Located; the classes of §17 replayed	§2.2
The Star–Kneser forcing theorem and its retracted converse	archive §15	Located	§7

B How this paper was checked

Sources.

- The repository KokunoYumeto/erdos-strauss-foundation at commit d20b32e, read-only.
- The 619-page archive and its source archive, downloaded on 26 September 2026 from the Zenodo record 10.5281/zenodo.22678971:
00_ERDOS_STRAUSS_Project_Reader.pdf MD5 c1ee553967e0e6f12d4976e171c11b15
01_ERDOS_STRAUSS_Source_Verification_and_Machine_Memory.zip MD5 4129ec50d0c02ff7daf61fde6c589de4
The main source file of the archive (60,561 lines) and its fourteen satellite files were read where they are cited.
- The supplement’s source, in the repository folder research/expanded-2026-09-08/exact_bounded_transport_72/.
- For the literature statements of §1: the source of Elsholtz and Tao’s paper (arXiv 1107.1010v6), which the repository holds in its literature folders.

What was refereed earlier. Theorem 3.1 and Propositions 3.2–3.5, the audited items of Section 5, and the audit of the package ES-09 in Section 5.2 come from the Erdős–Straus section of the reader of four workbenches. There they went through five stages:

1. a first-pass audit by a separate Claude instance, which read the workbench and wrote independent checks (43a_);
2. a referee pass on the notes 43_–47_, by an instance that had written none of them; it found, among other things, that the orbit statements of Section 5.2 fail at even levels;
3. a referee pass on that reader;
4. two further passes that read it for understatement and missed generality as well as for overstatement.

These passes produced the range $x < p/2$ in Theorem 3.1, the first form of Proposition 3.4, Proposition 3.5, and the extension of the count of Proposition 3.2 to $p \equiv 5 \pmod{12}$. The checks are checks/workbenches/workbench_reader_claims_checks.py and checks/workbenches/generality_checks.py; the referee reports are in checks/workbenches/.

What version 1 added to the reader of four workbenches. *New* here means new relative to that earlier reader, not a claim of new mathematics. Where the archive already has a statement, this paper cites it.

- Proposition 3.6, which is the archive’s Theorem 10.9 and Proposition 10.10, re-proved;
- Lemma 4.1 and Proposition 4.3, with the reproduction of the deficit data in §4.1;
- Theorem 4.5, and the statement that each family meets every hard class;
- Proposition 2.1;
- Proposition 2.4(a), whose relation was also checked in 21_;
- the map of the archive (§2.2), the consistency of the census counts (§4.3), and the three new rows of the overlap table (Section 8).

These went through the referee pass described next.

The referee pass of version 1. An independent Claude instance (claude-opus-5-5), which wrote none of the text, refereed version 1 on 27 September 2026. It read in both directions, as the author asked: for statements stronger than their proofs or sources, and for understatement, missed generality and missed or implied results. It reran the five scripts of version 1, whose outputs were identical, and wrote its own checks. It found every proof correct as written and every recomputed number in agreement. It reported five major and ten minor findings, and seven results that the reader’s material implies. Every finding was verified here before it was applied: by reading the archive at the cited page, or by an independent script.

- *Attributions.* Theorem 3.1(b) is not a generalisation of the archive. The archive’s Theorems 9.1 and 9.64 already hold for every odd prime; the paper now says so. The check here also found that the count $2|E_a| + |M_a|$ is in the archive’s Theorem 9.64, which the referee had credited to version 1.
- *Understatements.* Proposition 3.4 now concludes that the survivors lie in the hard classes modulo 840, as the archive’s Corollary 10.7 does. The census domain is stated. The page map of §24 is split, and the page counts are made consistent.
- *Two misdescriptions.*
 - A remark of version 1, that the census carriers are not polynomial identities alone, was wrong. They are polynomial identities on their rows, and Proposition 4.6 and Corollary 4.7 replace it.
 - Proposition 2.1(b) had suggested that the theorem fails for every $p \equiv 2 \pmod{3}$. It fails only on a minority of them, and Proposition 2.2 and Remark 2.3 now describe that set.
- *Minor findings.* Among them:
 - the residual $R = 2$ at $p = 2$;
 - the archive’s correction of Ionascu and Wilson’s label for 1201;
 - the count of .lean files (67, not 84);
 - the title-page date (30 August);
 - Elsholtz and Tao’s definition of solvability by polynomials;
 - the full citation of Vaughan’s paper.
- *Results adopted.* After verification, these are Corollary 4.2, Propositions 3.7, 3.8, 2.2, 4.4 and 4.6, Corollary 4.7, Proposition 2.4(b), the records to $3 \cdot 10^9$, and the even levels of §5.2.
- *A correction to the referee.* It gave 3,533,141 as the prime with a shear path of length 5. A complete enumeration here shows that the least such prime is 1,183,451.
- *Not adopted.* The referee also reported an observation of Ionascu and Wilson on residues modulo 9240. It could not be checked against their paper here, and is omitted.

The referee’s report and its check programs are in `checks/es_reader/referee/`; the verification scripts of version 1 are in `checks/es_reader/referee_verify/`.

The scripts. They import no workbench code and recompute every number from the definitions. They are republished in `contrib/claude-ab/rewrite/checks/es/`; the folders `referee/` and `referee_verify/` are in the branch history (see below).

Script	What it checks	Time, result
<code>esr_records.py</code>	$h(p)$ for every prime up to 8,803,369 through Lemma 4.1 and Theorem 3.1; the strict records; $h(806521) = h(118801) = 15$ (for $p = 2$ the script prints $R = 4h - 1 = 3$; the text’s $R = 2$ is $4x - p$ with the least denominator $x = 1$)	3 s; as stated
<code>esr_lemma_checks.py</code>	Proposition 3.6 on all 47,210 shells meeting its hypothesis at odd primes below 3000 (none occupied), and the failure of its converse; Lemma 4.1 against a brute-force enumeration of all solutions at the odd primes below 700	25 s; all pass
<code>esr_deficits.py</code>	the deficit data of §4.1 at 73, 1129, 1201, 21169, 118,801, 8,803,369 and 806,521, their independence of the primitive root, the corollary of archive §16, and the counts $ E_a = 0$, $ M_a = 2$ at p_* (22 checks)	1 s; all pass
<code>esr_families.py</code>	the four identities of Theorem 4.5 as polynomial identities, exact fractions for $k \leq 200$, the hard classes met and the least prime in each	1 s; all pass
<code>esr_twshears.py</code>	Proposition 2.1: the full shear graph of every prime below 2000, and a complete search for two-edge paths up to 10^7	23 s; all pass
<code>referee_verify/</code> <code>shears_independent.py</code>	Proposition 2.2: degrees on the full graph below 1200; complete enumeration of paths of length 2 to 5 below 10^7 ; the class 131 mod 180	21 s; as stated
<code>referee_verify/</code> <code>shears_density_bound.py</code>	the bound $\sum_R \delta_R < 0.335$ of Remark 2.3	5 s; as stated
<code>referee_verify/</code> <code>hsegment.c,</code> <code>hsegment_A.c</code>	$h(p)$ for all primes below $3 \cdot 10^9$ by a segmented sieve; the records, the primes with $h \geq 19$, and Corollary 4.2	2 min; as stated
<code>esr_referee_checks.py</code>	Propositions 3.4, 3.7, 3.8; the census domain; the base rows and the square-row count of Corollary 4.7; the shell at $p = 12289$; the even levels of §5.2	2 min; all pass

A partial second pass. A second-pass audit of the supplement was started with a separate Claude instance and stopped before it wrote its report. Its scripts and outputs are kept in `checks/es_reader/second_pass_partial/`, and §2.3 reports what they show. The one reported mismatch was traced to the script’s own expectation (§2.3).

Version 3: the author’s notes. Section 6 was added in version 3, on 27 September 2026.

- Six Claude instances, each of which wrote none of this paper, read the notes in full in separate passes. Each tabulated every numbered statement and checked it with its own programs.
- `claude-ab` then re-derived every result proved in Section 6 and re-checked it with its own programs, which are in `checks/es50/`:
 - `es50_checks.py`: 25 items, all pass, in 49 s. Every certificate is turned into (x, y, z) and tested by $4xyz = p(xy + yz + zx)$.
 - `es50_support_core.py`: an independent reproduction of the counts 4951 and 2970 of Theorem 6.2(a), in 4 s.

- `es50_survivors.py`: Table 3.
- `salez/`: Theorem 6.2(b). The enumerator of Salez’s seven modular equations was written by one of the reading passes. It gives 3520 classes at $840 \cdot 11 \cdot 19 \cdot 23$, and 147,348 at Salez’s modulus, which is the number he reports.
- The full record, with the passes’ findings and the attributions, is note 50_.
- *The referee of version 3*. An independent Claude instance, which wrote none of the text, refereed note 50_ and Section 6 on 27 September 2026.
 - It re-ran every program, including the author’s own scripts, and recomputed the main numbers by independent code. It found every proof correct.
 - It reported three major findings, all applied: one error charged to the wrong note, an open question false as stated (the prime 409), and this section’s own verification status, which was stated too broadly.
 - It reported fifteen minor findings, applied after checking. Among them: Salez’s scope, attributions to the notes and the archive, and the reduced form of Theorem 6.6.
 - It found results that the material implies. The eight further shifts after Table 3 and the equivalence in Proposition 3.7(c) are adopted.

Its report and programs are in `checks/es50/referee/`.

Version 4: the new results. The results new in version 4 were proved by `claude-ab` and checked by programs written for them, which are in `contrib/claude-ab/rewrite/checks/es/`:

- `exact_shadow_reps.py`: Theorem 6.15 and Proposition 6.16, exactly in $\mathbb{Q}(\zeta_{168})$ and $\mathbb{Q}(\zeta_{120})$, together with the agreement of the S -matrix formula with a direct evaluation of the series (to 29 and 30 digits);
- `m7_vs_M27.py` and `m5_vs_M25.py`: the same S -matrices reproduced numerically from the q -series to 38 digits, the search over signed permutations, conjugation and twists, and the failure of $(ST)^3 = S^2$ for the pair of the note’s cubic transport;
- `misc_checks.py`: the values of $A_{11}(pa)$ on the primes $p \equiv 1 \pmod{840}$, the involutions of $GL_2(3)$, and the coordinate formulas of Proposition 8.3;
- `es09_even_levels.py` and the scripts of the companion S^6 paper [18] (in `checks/s6/`): Theorem 5.1 and Corollary 5.2.

Proposition 6.8, Proposition 8.1 and Proposition 8.2 are proved in the text.

What has not been done. The results adopted from the referee of version 1 were proved by that referee and checked by `claude-ab`, and they have had no further independent reading. Version 4, including its new results, was read by one further independent referee, a Claude instance that wrote none of the text; its findings were verified and applied. The parts of the workbench marked located were not checked, and §9 lists what else was left out.

Where everything is. This paper, its source and the scripts it rests on are on the branch `claude/claude-ab-grind-20260925` of `KokunoYumeto/zeta-function-research-reader`: the paper in `contrib/claude-ab/rewrite/papers/erdos_straus/`, the scripts in `contrib/claude-ab/rewrite/checks/es/`, and the condensed Markdown record of the same material in `contrib/claude-ab/rewrite/condensed/`. The earlier versions, the audit notes 21_, 43_, 43a_, 47_, 47a_ and 50_, and the folders `checks/es_reader/` and `checks/es50/` with the referee reports, are in the branch’s history at commit `a90d9e5b`, in `contrib/claude-ab/grind_20260925/`.

C Changes in version 4

Version 3 of this paper was withdrawn on 27 September 2026, together with the audit notes, because of its framing. Version 4 keeps its theorems, proofs and citations, corrects its framing, and adds new results. The changes are the following.

Framing.

- The title, abstract and introduction are rewritten. The introduction states the main results as Theorems I–V, explains what they mean, describes the bridges, credits the contributors, and sets out the four statuses and how the status labels map onto them.
- Section 6 is retitled *The author’s notes, April–June 2026*, and the notes are described in the same way throughout.
- Quotation marks around the terms of the workbench and the notes are replaced by italics.
- In the map of the archive (§2.2) the category *side* is renamed *structural* and defined through the archive’s own description of these sections as an interface still to be supplied by a typed map.
- Statements that structures which do not take primes as input cannot bear on solvability are replaced by what is established about the typed maps that would connect them (Sections 6 and 8). The negative results of Section 7 are stated with their scope, and the table’s closing sentence now says which mechanisms are sharp.
- The drafts that the author’s latest record supersedes are treated only for the results they contain that were verified: the paragraph on one such draft, its row in the catalogue, and the row of Section 7 on the terminal core as the residual locus are removed. The terminal core is now described by what Theorem 6.2 locates (Remark 6.3).
- Statements that rest only on a reading pass over the notes are marked (*a reading pass*) and count as not verified here.
- Section 8 is rebuilt as a section on the bridges to other areas, with what is established, what is not verified here, and my assessment for each; the table of edges to the author’s other workbenches is its last subsection.

Corrections of content.

- *The minimal model $M(2,7)$.* Version 3 said that the note’s identification of the mock theta shadow with the characters of $M(2,7)$ fails, that the note’s modified modular pair is not a representation of $SL_2(\mathbb{Z})$, and that no relabelling matches both S and T . Re-deriving the statement shows that the identification holds after conjugation and the twist by the character of η^3 (Theorem 6.15). What fails is the note’s matching of labels through its cubic phase transport: no signed permutation and scalar realise it, and the pair it produces does not satisfy $(ST)^3 = S^2$; version 3’s statements hold in that scope only.
- *The Lorentzian note.* Version 3 listed the note’s section on modular flow as an exception to the correctness of the notes’ structural links, stating that the modular flow is a rotation and not a Lorentz boost. That statement stands (Proposition 8.3(a)). What changes is the verdict on the note: its proposition, that the congruence action of A_ρ is a Lorentz boost, is correct, and Proposition 8.3 gives the exact relation between the two.
- *The sieve.* Version 3 said that a sieve bound of the form $x/(\log x)^\kappa$ follows from the quadratic-residue conditions and that the conditions do not approach a proof. The first statement was not proved there and is removed; Proposition 6.8 proves density zero, and Question 3 states what a negative answer would prove.

- *The shell count as a class function.* Version 3 said that $A_{11}(p)$ takes the values 0, 4, 6, 8, 10 on primes $p \equiv 1 \pmod{840}$. The count meant is $A_{11}(pa)$ with $a = (p + 11)/4$, the number of ordered certificates of the shell; $A_{11}(p)$ itself takes only the values 0 and 2 at primes. Section 6.5 now states it so, with the primes at which each value occurs.
- *The further certificates of the referee of version 3.* They are now stated with their proofs, and the 23 primes below 10^7 are described as the primes $p \equiv 1 \pmod{24}$ with no certificate of the kinds considered, rather than as survivors of conditions that version 3 did not state.
- *The Busy-Beaver notes.* Version 3's statement that the residues modulo 107 are at chance level is replaced by the description of the test and its scope (Section 6.7), and its statement that the only link with the Busy-Beaver function is the classical one by the exact Π_1 bridge (Section 8).

New results. Proposition 6.8 (density zero of the survivors of the row $p + 1$); Theorem 6.15 (the order-7 shadow carries the $M(2, 7)$ representation); Proposition 6.16 (the order-5 block carries the Fibonacci data); Propositions 8.1 and 8.2, which state as propositions, with proofs, two points that version 3 listed as errors; Proposition 8.3 (modular flow and boost); and Theorem 5.1 with Corollary 5.2 (the orbits of the $(3, 4, \infty)$ covers at every level), which answer version 3's question on the covers at even levels. The coefficients of the note's modular differential equation are checked in Section 6.6. Section 8 also states, from the companion note [19], how rotations and boosts meet the Apollonian group, and, from a reading pass, that the note's odd Ogg phase decomposition holds with the map of Proposition 8.2.

Questions. Section 9 is reordered. Version 3's question on the covers at even levels is answered and replaced by the question on their genera; Questions 10–13 are new.

Apart from the corrections listed above, no statement of version 3 changed its mathematical content.

References

- [1] *A compact complex threefold fibred by tori over the projective line, and the six-sphere*, manuscript circulated by L. Alpöge, 2026, <https://alpo.ge/s6.pdf>.
- [2] M. C. N. Cheng, S. Chun, F. Ferrari, S. Gukov, S. M. Harrison, *3d modularity*, J. High Energy Phys. **2019**, no. 10, 010.
- [3] M. C. N. Cheng, F. Ferrari, G. Sgroi, *Three-manifold quantum invariants and mock theta functions*, Phil. Trans. R. Soc. A **378** (2020), 20180439, arXiv:1912.07997.
- [4] C. Elsholtz, T. Tao, *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Aust. Math. Soc. **94** (2013), 50–105, doi:10.1017/S1446788712000468.
- [5] E. J. Ionascu, A. Wilson, *On the Erdős–Straus conjecture*, arXiv:1001.1100 (2010).
- [6] M. Kneser, *Ein Satz über abelsche Gruppen mit Anwendungen auf die Geometrie der Zahlen*, Math. Z. **61** (1955), 429–434.
- [7] J. C. Lagarias, *The $3x + 1$ problem and its generalizations*, Amer. Math. Monthly **92** (1985), 3–23.
- [8] M. A. Lopez, *A complete congruence system for the Erdős–Straus conjecture*, arXiv:2404.01508 (2024).
- [9] Spiridon Mihnea, Dumitru C. Bogdan (author names as listed on arXiv), *Further verification and empirical evidence for the Erdős–Straus conjecture*, arXiv:2509.00128 (2025).
- [10] H. L. Montgomery, R. C. Vaughan, *The large sieve*, Mathematika **20** (1973), 119–134.
- [11] L. J. Mordell, *Diophantine Equations*, Academic Press, London, 1969.

- [12] J. B. Rosser, L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), 64–94 (Theorem 15).
- [13] S. E. Salez, *The Erdős–Straus conjecture: new modular equations and checking up to $N = 10^{17}$* , arXiv:1406.6307 (2014).
- [14] R. C. Vaughan, *On a problem of Erdős, Straus and Schinzel*, Mathematika **17** (1970), 193–198, doi:10.1112/S0025579300002886.
- [15] D. Zagier, *Ramanujan’s mock theta functions and their applications [d’après Zwegers and Ono–Bringmann]*, Séminaire Bourbaki, exp. 986, Astérisque **326** (2009), 143–164.
- [16] Claude (Anthropic), *The split-zero programme around the Riemann zeta function: its results, with proofs, and its bridges*, version 3, 2026, in the Zenodo record *Split-Zero cohomology and the zeta-function research programme*, concept DOI 10.5281/zenodo.22678085; source on the branch `claudio/claude-ab-grind-20260925` of `KokunoYumeto/zeta-function-research-reader`, folder `contrib/claude-ab/rewrite/papers/zeta/` (Lemma 6.6 has the same number in versions 1–3).
- [17] Claude (Anthropic), *Four workbenches and where they meet: the Erdős–Straus, Collatz, Erdős Problem 817 and Yang–Mills workbenches*, 2026, in the same Zenodo record as [16]; source in `contrib/claude-ab/rewrite/papers/workbenches/` on the same branch.
- [18] Claude (Anthropic), *The S^6 record and the integral monodromy of its $(3, 4, \infty)$ period system*, 2026, in the Zenodo record 10.5281/zenodo.22678442 and its later versions; source in `contrib/claude-ab/rewrite/papers/s6/` on the same branch.
- [19] Claude (Anthropic), *Rotations, boosts and the Apollonian group*, 2026, a companion note to this paper; source in `contrib/claude-ab/rewrite/papers/apollonian/` on the same branch.
- [20] S. Aaronson, *The Busy Beaver frontier*, SIGACT News **51** (2020), no. 3, 32–54.
- [21] G. E. Andrews, F. G. Garvan, *Ramanujan’s “lost” notebook VI: the mock theta conjectures*, Adv. Math. **73** (1989), 242–255.
- [22] J. C. Baez, *The octonions*, Bull. Amer. Math. Soc. **39** (2002), 145–205.
- [23] M. C. N. Cheng, I. Coman, P. Kucharski, D. Passaro, G. Sgroi, *3d modularity revisited*, arXiv:2403.14920 (2024).
- [24] M. C. N. Cheng, J. F. R. Duncan, J. A. Harvey, *Umbral moonshine and the Niemeier lattices*, Res. Math. Sci. **1** (2014), Art. 3.
- [25] J. H. Conway, S. P. Norton, *Monstrous moonshine*, Bull. London Math. Soc. **11** (1979), 308–339.
- [26] J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999, ch. 16–18.
- [27] M. Eichler, D. Zagier, *The Theory of Jacobi Forms*, Birkhäuser, 1985.
- [28] R. L. Graham, J. C. Lagarias, C. L. Mallows, A. R. Wilks, C. H. Yan, *Apollonian circle packings: geometry and group theory I. The Apollonian group*, Discrete Comput. Geom. **34** (2005), 547–585.
- [29] D. Hickerson, *A proof of the mock theta conjectures*, Invent. Math. **94** (1988), 639–660.
- [30] J. Kocik, *The Koide lepton mass formula and geometry of circles*, arXiv:1201.2067 (2012).
- [31] G. Moreno, *The zero divisors of the Cayley–Dickson algebras over the real numbers*, Bol. Soc. Mat. Mexicana (3) **4** (1998), 13–28.
- [32] A. Rocha-Caridi, *Vacuum vector representations of the Virasoro algebra*, in: Vertex Operators in Mathematics and Physics, MSRI Publ. 3, Springer, 1985, 451–473.
- [33] E. Rowell, R. Stong, Z. Wang, *On classification of modular tensor categories*, Comm. Math. Phys. **292** (2009), 343–389.
- [34] P. Sarnak, *Letter to J. Lagarias about integral Apollonian packings*, June 2007.